

Bibliography

Keyboards and keying An annotated bibliography of the literature from 1878 to 1999

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Abstract. This bibliography covers the period from 1878 through 1999. It contains, in chronological order, a thorough sampling of the literature concerning the design and use of keyboards. The sources are selected and annotated to reflect the status of engineering and technology know-how, and knowledge about ergonomic aspects of the use of the keyboards with, first, mechanical typewriters, then electric typewriters and finally, from the 1960s on, computers. The bibliography illustrates the origin of Sholes' 1878 QWERTY keyboard and its continued use in spite of its many shortcomings, which may be – at least partially – the reason for cumulative trauma disorders in yesteryear's typists and today's keyboarders.

Keywords: Keyboards – Keyboarding – Typing – Ergonomics – Cumulative trauma disorders

1 Introduction

This bibliography covers the period from 1878, the year of Sholes' QWERTY-keyboard patent, through 1999. It is a chronological sampling of publications, in the German and English languages, that relate to keyboard design and to the activities of typing on mechanical type-writing machines, then of keying on electric typewriters and, finally, of keyboarding on electronic keyboards. I collected the information first on filing cards, and later in computer files, from my reviews of the literature, from information supplied by my colleagues and students, and from computerized searches. The vast majority of the references stem from scientific and occupational journals and books, but a few come from other sources such as trade publications. All references are

meant to highlight the conditions of technology, knowledge, and attitudes at a given time. Of course, one cannot claim any such collection to be exhaustive, but I have included all the references known to me that indicate the state-of-the-art at their time. This listing divides naturally into several sections according to the development of technology and ergonomics over a certain period of time.

1.1 The early years

In the early 1800s, the engineering challenge was to find a workable technical means for replacing manual penmanship by mechanisms that print information on paper. The tinkers and inventors of the time commonly followed the traditional practice of composing text from single letters, numerals, signs, and spaces. This led to the “one-key for one-character” design principle that, while following Western writing practice, poses high biomechanical loads on the typist's hand in terms of the large number of motions that the wrist and digits must perform. Unintentionally, this created an ergonomic problem with serious overexertion and disease outcomes, which could have been avoided had the printing of text been accomplished by some sort of chording or chunking of syllables, words, expressions, and sentences.

The typist's effort included the generation of all the energy needed to accelerate and move the masses of the type-writing machine's levers and rods to generate an imprint on the paper. This manual effort per keystroke was reduced in the 1960s through the introduction of auxiliary electric power in typewriters, then by the use of electric switches instead of mechanical parts, and finally by the employment of electronics – but the frequency of key presses remained as it was a century earlier: one stroke for every character, space, or control input.

1.2 The engineering problem

1.2.1 The origin of the QWERTY keyboard

It is appropriate to start the account of keyboarding on August 27, 1878, the day Christopher Latham Sholes patented his “Type-Writing Machine” (U.S. patent 207559). He was not the first person to invent a device that printed individual letters and numerals on paper, instead of scribes penning them by hand, but he was the first inventor of many others before and after 1878 to be successful in designing, producing, and marketing a practical type-writing machine.

Numerous inventors had already proposed a great variety of typographical devices with various keyboards. The main problems to overcome were of a biomechanical nature: how to make the manipulation of a mechanical input device (usually depressing a key of some sort) generate an imprint on paper that could be read as text. To this end, the early designers created a multitude of mechanical leverages as the innards of their typographical machines, as described in detail by [5]. Sholes uses nearly the entire text of his patent to describe the mechanical details.

Keyboard design Given the difficult mechanical problems of the type-writing machine itself, the designers were apparently little concerned about what kind of keys – or keyboards – to use. Many of their designs resembled the elongated black and white keys of the piano, whereas others had double or triple rows of button-like keys, some arranged in straight columns and rows, others in concave or convex circular segments [1, 2, 5]. Few inventors commented on why they chose their specific layouts of keys and their assignments to certain letters or numerals.

Starting in 1868, Sholes and his co-inventors obtained eight U.S. patents for “Type-Writers” of varying designs: in 1868, patent nos. 79265 and 79868 (both with Glidden and Soulé); in 1876, no. 182511 (with Schwalbach); in 1878, first no. 199382, then no. 200351 (with Glidden), followed by nos. 207557, 207558, and 207559.

Both 1868 patents show two rectilinear rows of keys: the first patent, no. 79265, has a total of 21 unmarked keys, described (on page 1) as “similar to the key-board of a piano”: 10 shorter keys atop 11 longer keys, akin to the white and black clavier keys.

The 36 keys in patent no. 79868 are also (as said on page 2) “similar to the keys of a piano or melodeon” but they all lie in one plane, side-by-side, and alternating in length. The keys on the left side, as the operator sees them, are numbered in increasing order, while the other keys, in the middle and on the right side, are lettered alphabetically. Neither patent contains any explanation for the choice of the respective key layout beyond the just quoted statements.

The following 1876 patent and the first four patents of 1878 all exhibit three straight rows of button-like keys affixed to lever-type bars: 32 keys in the 1876 patent,

and 21 keys in the 1878 patents. No labels with letters or numbers are on the buttons, and the texts of the patents provide no explanation or description of any kind.

Sholes’ last patent (no. 207559 of 1878) makes fourteen specific technical claims, but none refers to the key layout. One drawing in this patent shows a frontal view of four straight and horizontal rows of key tops. The rows are staggered at increasing heights so that the row closest to the operator is the lowest and the farthest row is the highest. Another drawing depicts a top view of the four straight rows, each with 11 round keys, for a total of 44 keys: see Fig. 1. The key tops are inscribed with numerals, letters, and signs.

Lacking any specific statements by Sholes (or his co-inventors of the earlier patents), one can only observe that the layout of the 1878 patent no. 207559 shows some remnants of an alphabetic arrangement. Sholes was a printer by trade, and so one can surmise similarities to the arrangement of the printer’s type case in which pieces presumably were assorted according to convenience of use instead of just alphabetically. Another possible reason for the arrangement of the characters and keys may have been the intent to prevent type bars (in the then-used mechanical leverages) from colliding or sticking together when neighboring bars are activated in a quick sequence. This may have led to a separation of certain bars and, hence, keys on the keyboard. However, there is no contemporaneous evidence for any of these speculations.

Sholes’ puzzling layout of the keyboard on his “Type-Writing Machine” per U.S. Patent 207559 is still commonly used, with only the positions of the *X* and *C* exchanged in English-language versions, and the *M* moved to the first row. The six leftmost keys on the third row, counted from the operator, carry the letters *QWERTY*. Today, the term “QWERTY keyboard” serves as a short name for any arrangement in which the letter keys (also

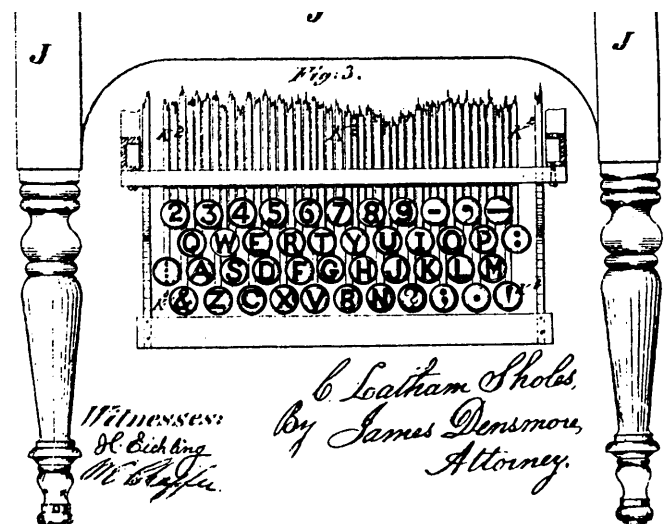


Fig. 1. The QWERTY keyboard as shown in Sholes’ 1878 U.S. patent 207,559

called “alpha” or “alphabetic” keys) essentially follow Sholes’ keyboard layout.

1.3 The USE problem

With the benefit of contemporary hindsight, it appears surprising that the inventors of the 1800s paid seemingly little attention to the usability of the input side, the “keys” and their respective arrangements, of their typing machines. A century earlier, in a book first published in 1700, then edited and enlarged in 1713, the Italian physician Bernardino Ramazzini [7] had reported on injuries and diseases that were specific to certain trades and kinds of work. He described overuse diseases that appeared in workers who did “violent and irregular motions” in “unnatural postures of the body” (page 43 in Wright’s 1993 translation from the 1713 Latin text). Of course, there were no typewriters or computers in 1713, but Ramazzini reported overuse diseases occurring among secretaries (*notaries*) and office clerks (*scribes*), saying (on Wright’s page 254) that their diseases were due to three causes: “First, constant sitting, secondly, the incessant movement of the hand and always in the same direction, thirdly, the strain on the mind...”. “[T]hey must stick to their writing all day long. ... [I]ncessant driving of the pen over paper causes intense fatigue of the hand and the whole arm because of the continuous and almost tonic strain on the muscles and tendons, which in course of time results in failure of power in the right hand.”

In 1872, the British physician Poore [6] referred to earlier work by Duchenne and Zuradelli who (following Ramazzini) had related the writers’ cramp to “muscular impotence” and “spasms” of frequently repeated muscle use. Poore stated that this health problem was not limited to writers, but also occurred in pianists, tailors, cobblers, fencing-masters, and others. Evidently, during the second half of the 1800s, repetitive strain disorders were well known to be associated with certain occupations such as telegraphists and musicians, particularly pianists (Ayoub and Wittels 1989, Burnette and Ayoub 1989, Hochberg 1983, Burry and Stoke 1985, Fry 1986, Lockwood 1989).

2 From 1878 to 1949

Annotated list of publications on keyboards and keyboarding from 1878 to 1949

Sholes, C.L. (1878). *Improvement in Type-Writing Machines*. Letter’s Patent 207 559. United States Patent Office. “The invention relates to that class of type-writing machines in which the successive depression and release of a series of keys, one after another, vibrate and throw a series of types against an inking substance and the substance to be written on ...” (page 1). Fourteen specific technical claims are made on page 4, none mentioning selection or layout of the keys. Figure 1 shows a frontal view of the invention with four horizontal

rows of 11 keys staggered in height to that the row farthest from the operator is the highest. Figure 2 shows a top view of the straight rows on which the third row (as counted from the operator’s position) of the keyboard starts, from the left, with the letters QWERTY. Neither this patent nor any of Sholes’ earlier patents explains the reasons for key selection or layout.

Poore, G.V. (1887). *Clinical Lecture on Certain Conditions of the Hand and Arm Which Interfere with the Performance of Professional Acts, Especially Piano-Playing*. The British Medical Journal, February 26 issue, 441–444.

After referring to previous communications on “writers’ cramp”, the author refers to 21 cases of a similar muscular disease seen in 21 pianists during his practice. He indicates that the cause is overuse, and the most appropriate and immediate treatment is rest.

Osler, W. (1892). *The Principles and Practice of Medicine, IX. Professional Spasms; Occupation Neuroses* (pp. 963–965). New York, NY: Appleton and Company.

“The continuous and excessive use of the muscles in performing a certain movement may be followed by an irregular, involuntary spasm or cramp, which may completely check [impede] the performance of the action. The condition is found most frequently in writers, hence the term writer’s cramp or scrivener’s palsy; but it is also common in piano and violin players and in telegraph operators. The spasms occur in many other persons, such as milkmaids, weavers, and cigarette-rollers (p. 963). ...The course of the disease is usually chronic. If taken in time and if the hand is allowed perfect rest, the condition may improve rapidly, but too often there is a strong tendency for recurrence. The patient may learn to write with the left hand, but this also may after a time be attacked” (p. 964).

Rowell, S.W. (1909). *Typewriter Keyboard*. Letter’s Patent 943 466. United States Patent Office.

The author groups the most commonly used letters in the middle of the keyboard.

Heidner, F. (1915). *Type-Writing Machine*. Letter’s Patent 1 138 474. United States Patent Office.

The author states that he “invented certain new and useful improvements” in keyboards “to enable the operator to obtain a better view of the keys and to write with greater ease, in a less cramped position than ordinarily. With this object in view, I divide the keyboard into halves and locate the two groups of keys thus formed in such manner that the forearms of the operator ...instead of converging, lie substantially parallel with each other.”... “[T]he hands have not to be twisted outward ... and there being thus much less strain upon the abducent muscles, writing is rendered considerably less fatiguing.” The keys are arranged in curved rows “in accordance with the natural form of the hand, that is to say, lengths of the fingers... [C]onvergence of the key groups further facilitates operation of the keys by the fingers in their natural position in the extended axis of the forearm.”

Banaji, F.M.M. (1920). *Keyboard for Typewriters*. Patent 1 336 122. United States Patent Office.

The author left ten letters of the QWERTY key set in their original locations and re-assigned the others. He assigned the letters Q and X to the index fingers contrary to the common belief that frequently used letters should be allotted to strong fingers.

Nelson, W.W. (1920). *The Improvements in Connection With Keyboards for Typewriters*. British Patent No. 155 446. The author proposes a new key arrangement, the "Combinational Keyboard," to suit the most frequent letter digrams occurring in English prose.

Wolcott, C. (1920). *Keyboards*. Letter's Patent No. 1 342 244. United States Patent Office.

The author proposes a new key arrangement. He assigned the letters J, Q, Z, and X to the index fingers.

Hoke, R.E. (1924). *Typewriter Keyboard Arrangement*. Letter's Patent 568 792. United States Patent Office.

The author proposes to split the keyboard into halves, locating the shift keys in the middle.

Book, W.F. (1924). *Voluntary Motor Ability of the World's Champion Typists*. *Journal of Applied Psychology*, 8, 283–308.

The author lists tests performed to determine whether certain individuals had the potential to become particularly good typists. These tests included various aspects, such as memory, intelligence, attention, but primarily tested motor activities, such as reaction time, rhythm capability, and the ability to perform quickly movements of the finger, hand, forearm, and upper arm. Such motor ability tests were given in state, district, and world competition in typewriting, and also to various students. The author stresses that such motor ability together with other characteristics, such as intelligence, are necessary components of high-performance typing.

Schroetter, H. (1925). *Knowledge of the Energy Consumption of Typewriting* (in German). *Pflueger's Archiv fuer die gesamte Physiologie des Menschen und der Tiere*, 207(4), 322–342.

Oxygen consumption and heart rate were recorded on two female typists before, during and after typing on mechanical typewriters. The heart rate increased by 13 beat per minute in one subject and by 17 bpm in the other typist during the work periods as compared to the respective resting levels while sitting. Respiration increased by 61 and 107%, with oxygen consumptions of 275 and 507 ccm/min, respectively. These data are said to compare well with other results published in the literature.

Klockenberg, E.A. (1926). *Rationalization of the Typewriter and its Operation* (in German). Berlin, Germany: Springer.

The "rational" functioning of a machine should not only be measured in terms of its energetic efficiency, but also with respect to the person who operates the machine. Accordingly, the book is divided into subsections:

Typing without visual control ("touch typing") and with visual control.

Static postural strains of the typist; back, forearm, upper arm, wrist and fingers.

Dynamic strain of the typist.

Possible performance of the hand of the typist, regarding the separate fingers (fatigue, reaction speed, duration of work, pressure and displacement), mobility of coordinating fingers, keying times, superiority of the left hand, vertical and horizontal mobility of the fingers, target-aiming of the fingers and errors.

Other required efforts, visual efforts, insertion of paper, making copies, etc.

The author describes a large number of experiments to assess the postural (static and dynamic), visual, and other strains of the typist. These experiments lead to a recommendation for various design changes in the typewriter, and the typist's workstation. Specifically, he recommends that the keyboard be divided into halves for the left and right hand, and that these halves be tilted downward to the sides. The improvements in design are expected to lead to less strain in the typist, and to better typing performance. On page 49 the author stresses that the (static postural) strains may be small if taken separately, but that they add up to a considerable magnitude over the course of many hours of work.

Zollinger, F. (1927). *A Few Remarks on the Question of Tubercular Tendovaginitis and Bursitis After an Accident* (in German). *Archiv fuer orthopaedische und Unfall-Chirurgie*, 24, 456–467.

The author indicates that many tendovaginitis and bursitis cases are not the result of traumas, but of "overuse". Yet, these efforts are often not unusual, extraordinary, or unexpected. Instead, overuse occurred because an already existing tendovaginitis became painful. "By far most cases are the result of a prolonged 'trauma' consisting of innumerable repetitive actions of specific groups of muscles; this is not a true effect of an accident, an occupational disease..." (p. 463). Tendovaginitis and bursitis in their early states, which often last for months, often cause no or only non-substantial problems that do not reduce the capability to perform the job (p. 466).

Gilbert, W.A. (1930). *The Orthographic Keyboards*. Spokane, WA: Shaw and Borden.

The authors methodologically reviewed the occurrence of letters and punctuation in the English language. He did not mention early related research but proposed a reformed key set, the Orthographic Keyboard, with F, P, B, and H placed at the center.

Marloth, K. (1930). *Improvements in Key-Boards for Typewriters and like Machines*. British Patent Specification 332476. His Majesty's Stationary Office.

The author splits the keyboard in left and right halves. On each, the keys are arranged in straight rows "at right angles to the natural positions of the fore-arms" resulting in a slant between the two key sets. "This arrangement allows the typist to occupy a perfectly natural position" for "comfort and health". Marloth also shows two parallel straight columns of keys assigned to each finger.

Conn, H.R. (1931). *Tenosynovitis*. *The Ohio State Medical Journal*, September, 713–716.

The author refers to an Ohio amendment to the Workmen's Compensation Act in which tenosynovitis of the flexor extensor muscles of the hand due to frequently repetitive motions or vibrations shall be considered occupational and compensable when contracted by an employee in the course of employment within 12 months prior to the date of disablement. Conn discusses medical aspects, particularly arguing that tenosynovitis does not necessarily result from a definite trauma, but that repetitive motion and vibration can be the primary causes.

Hammer, A.W. (1934). *Tenosynovitis*. *Medical Record (or: Review?)*. October 3, 353–355.

Referring to the Ohio Workmen's Compensation Act and the 1931 writing of Conn, the author states that the continuous use of special groups of muscles, such as in high-speed hand

operations, and often repeated manipulation, predispose to the development of tenosynovitis. This is put in contrast to an older classification that recognized acute, tuberculous, and purulent synovitis. "Where increased speed and the expenditure of muscular energy were demanded...tenosynovitis often developed when such laborious effort was persisted in, for many hours daily, over long periods of time" (p. 353). ... "In other words, human tendons will not tolerate more than 1500 to 2000 manipulations per hour" (p. 353). Hammer considers it conclusively proven that strain (and not direct injury) predisposes to the development of tenosynovitis (p. 354). This is most often found in the sheaths of the abductor and the extensor brevis pollicis.

Biegel, R.A. (1934). An Improved Typewriter Keyboard. The Human Factor, 8, 280–285.

The author points out several design defects in the common QWERTY keyboard, among them excessive stretches required from the ring and little fingers, and asymmetrical slant of key columns for the left and right hands.

Dvorak, A. (1936). Typewriter Keyboard. Patent No. 2040248. United States Patent Office.

The author proposed a new arrangement of the keys according to the frequency of use of letters in the English language and the abilities of the hand's digits to perform movements among the keys and to activate them. His intent was to obtain a keyboard with four rows of keys.

Flowerdew, R.E. and Bode, O.B. (1942). Tenosynovitis in Untrained Farm-Workers. Medical Memoranda, Barrain (?) Medical Journal. September 26, 362(?).

The authors describe a "large number" of cases of tenosynovitis crepitans among farm workers. The affected tendons were in all cases the extensor carpi radialis longus and the extensor digitorum communis. The patients were treated medically, and most improved rapidly and appeared to be cured within 10 days. Yet, in two cases, the onset of the disease was not noticed for about 10 days, and in these patients the cure took about three weeks after the beginning of treatment. Two patients who interrupted their treatment prematurely had a recurrence of the symptoms, which lengthened the forced period of inactivity. The authors believe that the cases occurred after work "involving repetition of stereotyped movements".

Dvorak, A. (1943). There is a Better Typewriter Keyboard. The National Business Education Quarterly, 12(2), 51–58, 66.

The author states six particular defects in the conventional keyboard:

1. Overloading of the weaker left hand.
2. Overworking certain fingers and not assigning enough work to others.
3. Too little typing done on the home row, too much on the upper and lower rows.
4. Fingers are required to execute an excessive amount of jumping back and forth from row to row.
5. Too few words can be typed exclusively on the home row.
6. Too many words must be typed by the left hand alone while the stronger and more agile right hand idles.

Dvorak then discusses the design characteristics of the keyboard proposed and patented by him in 1936, which he believes will eliminate these problems.

Griffith, R.T. (1949). The Minimotion Typewriter Keyboard. Journal of the Franklin Institute, 248, 399–436.

The author identifies five defects of the QWERTY layout: (1) an abundance of one-handed motions rather than the easier two-handed motions, and an enlarged number of reaches from the home row; (2) one-handed motions with "hurdles" over the home row or the row above; (3) overloading of the left hand; (5) "awkward fingerings" of one-handed motions that "loads the fingers inequitably". To overcome these problems, Griffith proposes a new keyboard, the "Minimotion Keyboard". He argues that such a keyboard contains the optimum arrangements of the letters for writing average English with the greatest possible ease (p. 404). The Minimotion Keyboard, proposed in this publication, is a successor of the "Random Keyboard", conceived by the author in 1942. The Minimotion Keyboard is claimed by the author to be superior to the "Simplified Keyboard" patented to Dvorak in 1936.

Soon after its invention in 1878, Sholes' typographic machine with its peculiar QWERTY keyboard swept the globe to become the predominant device to type text and numbers on paper. With thousands, and then millions of the mechanical typewriters employed in offices and privately, use problems inherent to the design became apparent. One issue was that the layouts of machine and workplace imposed a stressful body posture on the typist – see Fig. 2.

Well into the 1920s, even professional typists practiced a variety of techniques for striking the keys: either with a single finger, especially the strong middle finger, or using several fingers, or employing all ten digits, while looking at the keys or in "blind" ("touch") typing (Book 1924, Klockenberg 1926). Given the large displacement of the keys and their stiff resistance, and that they had to be struck thousands of times in the course of the work day, the work required of the typists' hands and arms when pounding on the keyboard of mechanical typewriters was tiresome. Furthermore, the arrangement of the keyboard



Fig. 2. Typical posture of a typist as depicted in 1926 by Klockenberg

itself, and of the keys on it as per Sholes' design, forced the arms into strong inward twist (pronation) and the hands into lateral bend (ulnar deviation) at the wrist, and it required complex motions between the ill-located keys. The unfortunate combination of bad posture and hard effort overloaded many typists' hands, wrists, arms, shoulders, and necks – as discussed by Book (1924), Schroetter (1925) and Klockenberg (1926).

"Myalgia" and related overexertions, reported earlier by Poore (1887) and Osler (1892) to occur in telegraphists and pianists, appeared in typists also (Klockenberg 1926). Tenosynovitis and other repetition-related diseases of the musculoskeletal system became a topic of theoretical and practical concern for physiologists and medical practitioners (Zollinger 1927, Conn 1931, Hammer 1934). That pathology was not yet generally related to typing, but would become so shortly.

In 1909, Rowell obtained U.S. patent 943 466 for a typewriter keyboard in which he placed the most commonly used letters in the middle. In the 1920s, several other patents for improved key locations were issued that

shifted the locations of several keys while keeping to Sholes' original layout. Gilbert (1930), Dvorak (1936/43), and Griffith (1949) proposed to substantially change the associations of letters with keys from the QWERTY layout, but adhered to the old layout of keys in bent columns and straight rows.

In contrast, Heidner's 1915 U.S. patent 1 138 474 shows substantial changes in the basic keyboard layout. These changes predate many of the ergonomic recommendations to re-appear in later years: see Fig. 3. In one of his designs, he divides the keyboard into left and right halves and locates and arranges the two sets of keys to make typing less fatiguing by reducing the operator's postural strains in the forearms and hands. Later proposals by Hoke (1924), Klockenberg (1926), and Marloth (1930) follow related ideas.

To summarize the development during the first seven decades, the use of mechanical typewriters, as designed by Sholes, becomes global, but often leads to physical overloading of the typists's musculoskeletal system. Many improved designs are proposed, but not put to use.

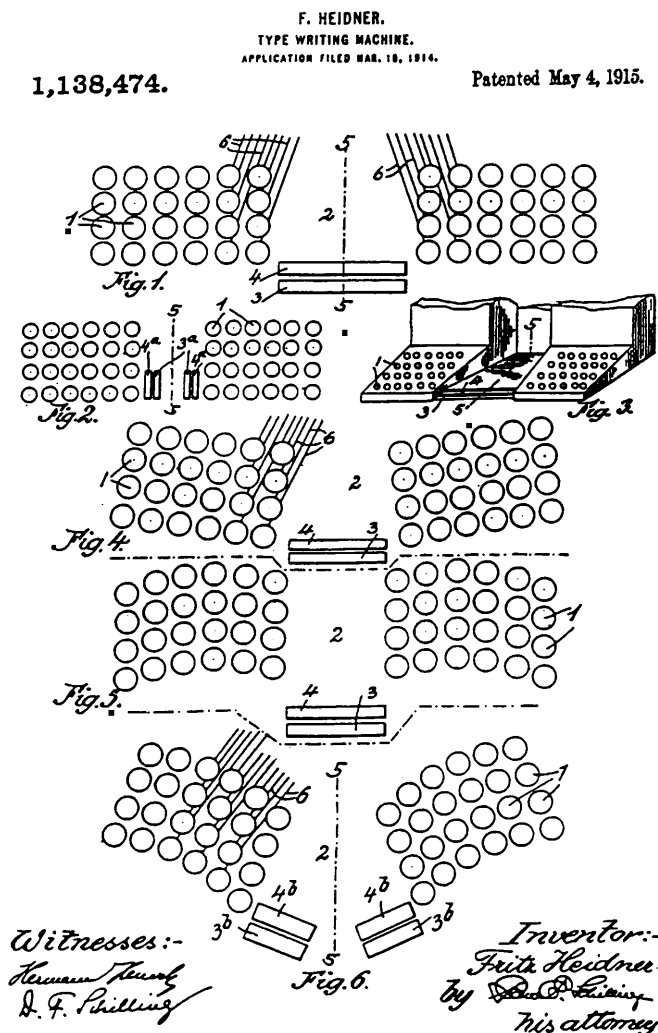


Fig. 3. Heidner's 1915 "ergonomic" keyboards as shown in his U.S. patent 1,138,474 (with permission from [3], all rights reserved)

3 From 1950 to 1969

Annotated list of publications on keyboards and keyboarding from 1950 to 1969

Hesh, F.H. (1950). *Electrically Operated Typewriter*. US Patent 2532 228. United States Patent Office.

The inventor indicates that the number of keys on a keyboard, at the time stated to be approximately 50, and their current arrangement on the keyboard, necessitate considerable shifting of the hands and movement of the fingers to strike all of the keys. As a result, such machines are fatiguing for their operators, the possibility of striking the wrong key is increased, and long practice is required to become a skilled operator (second paragraph). To alleviate these problems, a simplified keyboard is patented, in which the total number of keys is greatly reduced, and some keys are moveable on the keyboard.

Lundervold, A.J.S. (1951). *Electromyographic Investigations of Position and Manner of Working in Typewriting*. Acta Physiologica Scandinavica 24 (Supplement 84), 1–171.

This is an extensive description of electromyographic experiments performed between 1946 and 1949 to attain knowledge about the use of individual muscles while typewriting. Lundervold investigated 47 healthy persons, all beginning to accomplished typists, and 88 patients, most suffering from "occupation myalgia." This is described as "a disease which manifests itself by pains in muscles that have been overstrained during universed work, here mainly typewriting, and where no other cause can be found than the antecedent hyperfunction" (pp. 10, 11).

He used indwelling and surface electrodes. EMG signals were taken from the following muscles: ulnar group in the forearm, radial group in the forearm, biceps, deltoid, teres major, trapezius, latissimus dorsi, sacrospinalis, and pectoralis major. Four different typewriters were used: Remington Noiseless 6, Royal, Underwood Portable, and Olympia Plava Portable.

The subjects sat in three different chairs. All subjects were placed in approximately the same upright sitting posture, with the fingers placed in the same “starting position” on the keyboard.

Lundervold discovered that muscles were activated which were not expected to partake. He found great differences in the EMG records, both related to the fingers used and to the muscles observed. He also observed an increase in electromyographic activity in correspondence with the speed of key depression. The silent periods between key strokes became shorter with increasing keying speed, and the number of action potentials increased correspondingly (p. 95). Also, with more rapid key activation, the number of muscles increased from which action potentials were recorded (p. 96). The appearance of electromyographic events changed with increasing fatigue of the subjects.

Lundervold then discusses design features of office equipment, namely the chair, and footrests. He gives first an overview of past and present recommendations, and then continues by discussing electromyographic measurements taken in different sitting postures. In these experiments, he recorded EMGs on 47 persons, 18 female and 29 male. Sitting in a “tense upright position” showed larger electromyographic activities in back muscles than sitting in a “relaxed, well-balanced position.” Increased sitting duration also resulted in increased muscular activities. Leaning against a solid backrest resulted in smaller muscular activities than using a flexible backrest. Raising the forearm from a horizontal position while typing resulted in increased electrical activity, while lowering the forearm below horizontal when typing resulted in lowered electromyographic activity (pp. 133–135).

Scales, E.M. and Chapanis, A. (1954). The Effect on Performance of Tilting the Toll-Operator's Keyset. Journal of Applied Psychology, 38(6), 452–456.

An experiment was performed to investigate whether tilting the key set used by long-distance operators affects keying performance. The key set had three columns of keys, with the left column having six keys and the middle and right column five keys each, with keys of two different top sizes, and somewhat varying distances. This set was mounted with hinges, so that it could be inclined at eight different slope angles, namely 0, 5, 10, 15, 20, 25, 30, and 40 degrees. The stimuli for the keying task were ten-digit number and letter combinations. The experiment was performed in two parts, practice and test session, each extending over eight days. A Latin Square procedure was used. Sixteen subjects took part, eight males and eight females, no one having previous experience on this key set. Performance was not affected by inclination of the keyboards, but spontaneous comments from all subjects indicated that they preferred an inclined key set to a horizontal one. About half of the subjects expressed preference for key set slopes between 15 and 25 degrees.

Hagen, J. and Peters, T. (1955). About Occupational Diseases of Tendon Sheaths, Attachments of Tendons and Muscles (in German). Zentralblatt Arbeitsmedizin Arbeitsschutz, 5, 84–87.

The authors state that “chronic diseases of the tendon sheaths, of attachments of tendons and muscles through overexertion” (p. 84) are now officially listed as occupational diseases. Based on a table of reported occupational diseases, the authors state that the by far largest number of cases re-

ported by females were among typists, while males reported such incidences after various kinds of physical work. The authors conclude that the reason for this occupational disease is not so much the “heaviness” of the work, but rather the unaccustomed, monotonous, stereotypical “light” activities.

Strong, E.P. (1956). A Comparative Experiment and Simplified Keyboard Retraining and Standard Keyboard Supplementary Training. Washington, DC: General Services Administration.

A comparative experiment was conducted with two groups of subjects. An experimental group typed on the Simplified Keyboard (as per Dvorak 1936) and a control group used a standard keyboard. Each group contained ten subjects. There was no matching in terms of performance between the participants in each group. Random sampling methods were used to assign subjects to the keyboards. Each of the ten subjects in the experimental group was trained in the use of the Simplified Keyboard (SK) until the speed of typing on the standard keyboard prior to the beginning of the training was reached. Members of the control groups were injected into the experiment after the experimental subjects had reached the original speed.

The results showed that, on average, it took at least 100 hours of instruction and drill to regain the original speed, with large variations among individuals. Still, more errors were made on the SK than on the customary keyboard. Furthermore, subjects on the SK showed less improvement in speed for the total period of the experiment. They also showed less accuracy. Therefore, Strong wrote, a recommendation for the adaptation of the SK for use by the Federal Government can not be justified.

Buckup, H. (1957). Pocketbook of Occupational Medicine (in German). Stuttgart, Germany: Thieme.

In this compilation of the current knowledge in Occupational Medicine, a number of entries refer specifically to repetition-related occupational diseases. These are described (on page 72) as diseases that occur more often in certain occupations than in the population in general, resulting from occupation-related conditions or occupational activities. The symptoms are not necessarily different from those of other diseases, but present themselves in clinically uncharacteristic appearance, based on causal complexes that consist of expositional and dispositional factors. Occupational factors play a causal or contributory role, which needs to be demonstrated based on a comprehensive investigation of the history, the conditions at work, the materials handled, and the working methods employed.

Tendosynovitis is discussed at length and in detail (on pages 224 to 227). It is distinguished from other diseases of the tendons and muscle attachments that are often found in the lower arm area which, in turn, may be related to the cervicobrachial syndrome. Tendosynovitis is said to be probably caused by single, repeated, or long-endured motions, and overexertion. It is often observed in all sections of the working population, in all occupations, both in people performing heavy work, but also in craft persons, or in operators of office machines or typewriters. It appears primarily in the area of the wrist, but also at the elbow or the shoulder.

Klemmer, E.T. (1958). A Ten-Key Typewriter. (Research Memorandum RC-65). Yorktown Heights, NY: IBM Research Center.

A typewriter is described that has only 10 keys, one for each finger. The character inputs are achieved by pressing two or more keys simultaneously. Coding was chosen so that the eight most frequent letters in English had single key codes and the other 18 letters, two-key codes.

Performance of two subjects on the 10-key set was similar to the performance shown in learning to type on a conventional keyboard.

Lundervold, A. (1958). Electromyographic Investigations During Typewriting. Ergonomics, 1, 226–233.

Referring to his 1951 report (listed above), the author reports on electromyographic measurements on typists (the number and specifics of the persons are not reported). In experiments with five women and 10 men who had never previously written on a typewriter, the same measurements as on the typists were taken. Their recordings indicated that they used their muscles in fundamentally the same manner as the skilled typists, but each individual muscle was contracted more forcibly and for a longer time (pp. 228–229). As in the earlier experiments, back muscle activities were also recorded with the subject sitting in a tense upright position or in a relaxed, well-balanced position with flattened or even kyphotic lumbar spine. Similar to the description published in 1951, arm and back activities were found to be lower when the forearms were sloped below the horizontal as compared to forearms sloped upward above the horizontal position.

Hettinger, T. (1958). A Modification of the Skin Temperature Test to Determine the Disposition for Inflammations of Tendon Sheaths (in German). Int. Z. angewandte Physiologie Einschliesslich Arbeitsphysiologie, 17, 271–275.

The author describes the modification of a previously (1957) described test that relies on changes in skin temperature to determine the individual disposition for tendon sheath inflammations. Subsequently, the author describes the results of experiments with 24 female subjects. He concludes that the new instrument allows distinguishing between persons with higher and lesser propensity for the inflammations.

Tanzer, R.C. (1959). The Carpal-Tunnel Syndrome – A Clinical and Anatomical Study. Journal of Bone and Joint Surgery, 41-A(4), 626–634.

This paper describes the carpal tunnel syndrome. The problem has been investigated in depth during the past 10 years by several cited articles. Symptoms are often produced or aggravated by manual activity. Both flexion and extension of the wrist produce pressure on the median nerve.

Lockhead, G.R. and Klemmer, E.T. (1959). An Evaluation of an Eight-Key Word-Writing Typewriter (Research Report RC-180). Yorktown Heights, NY: IBM Research Center.

This report describes a typewriter which uses only eight keys, and types entire words by simultaneous depression of combinations of keys. This is in essence a design previously used and described (Klemmer, 1958), with a word-writing mode added: 100 multiple-finger patterns were coded to type entire words by a simultaneous depression of appropriate keys.

The authors conclude that "...the ability of people to learn and their willingness to use multiple key patterns is encouraging. Speed and error data for the present system do not imply an improvement over the standard typewriter for usual applications. The data do suggest that general rules can be formulated relating the number of keys simultaneously pressed to the maximum rate of pressing, and perhaps to rate of errors" (p. 21).

Deininger, R.L. (1960). Human Factors Engineering Studies of the Design and Use of Pushbutton Telephone Sets. The Bell System Technical Journal, July issue, 995–1012.

This paper reports on experiments to evaluate four categories of design features for telephone pushbuttons: key arrangement, force-displacement characteristics, button-top design, and general office factors.

The subjects were samples of 10 to 15 employees at Bell Telephone Laboratories, drawn randomly. They used two to five pushbutton telephone sets that differed in design characteristics, keying 10 to 15 standard telephone numbers on a set each day. The force required to depress the button was varied while its displacement was constant. Two different resistances were used but most of the subjects did not detect a difference. Out of the nine subjects who noticed a difference, eight preferred the key with less resistance. When the force was held constant and different displacements were used, between 1/32 and 3/16 in, there were no significant differences in performance or in preference.

Creamer, L.R. and Trumbo, D.A. (1960). Multifinger Tapping Performance as a Function of the Direction of Tapping Movements. Journal of Applied Psychology, 44(6), 376–380.

The authors report on experiments to determine how the direction of movements would affect tapping performance. Previous work seemed to have indicated that horizontal tapping movements were faster and less fatiguing than vertical movements. A typewriter keyboard was cut through its center from front to back, generating one keyboard for the left hand and one for the right hand. These halves were hinged so that the angle between them could be varied from 0 to 180 degrees. The tapping rates were greatest at intermediate positions, i.e., with keyboards tilted sideways at 22, 44, and 66 degrees below horizontal. Most errors were committed in the conventional keyboard arrangement (0 degrees), which was also reported by all five subjects to be the most fatiguing.

Baader, E.W. (1960). Occupational Diseases (in German). Munich, Germany: Urban und Schwarzenberg.

This is a comprehensive description of the clinical bases of 64 reportable occupational diseases. Among these diseases are "pressure paralyses of nerves" described on page 257 to 259. The author says that they should better be called nervous injuries and lists, among other occupations at risk, telephonists, telegraphists, and typists.

More than 13 pages are devoted to a discussion of diseases of tendon sheaths, tendons, and muscle attachments (pp. 260–273). Inflammation of tendon sheaths is said to be a very frequent occupational wear-and-tear effect due to continual repetitive motion with friction of the tendon. Serious acute inflammation of the tendon sheaths are found with light physical work, leading to the observation that the maximal number of movements for untrained workers should not be more than 1,500 to 2,000 per hour.

Among the many afflicted occupations listed by the author, are typists: "In typing, flexors and extensors of the fingers are fatigued by the repetitive fast work, and every key stroke generates a minimal impact, the summation of which is more injurious to fatigued muscle than to one not fatigued" (p. 264). Reference is made to a case of a large number of long distance operators who, after using a keyboard-like device, fell victim to the disease in large numbers.

Droege, R.C. and Hill, B.M. (1961). *Comparison of Performance on Manual and Electric Typewriters*. Journal of Applied Psychology, 45(4), 268–270.

The purpose of the experiment was to determine whether the performance of experienced typists on manual typewriters could predict their performance on electric typewriters. For this, more than 500 individuals took a standardized typing test, first on an electric typewriter, then on a manual typewriter. The results showed a substantial relationship between the scores obtained on manual and electrical typewriters, with more words per minute typed on the electrical typewriters, and fewer errors than on mechanical typewriters.

Peres, N.J.C. (1961). *Process Work Without Strain*. Australian Factory, July 1 issue, 1–12.

Peres discusses the effects of repetitive work on muscles and tendons in the hand and wrist. "It has been fairly well established, by experimental research overseas and our own experience in local industry, that the continuous use of the same body movement and sets of muscles responsible for that movement during the normal working shift (not withstanding the presence of rest breaks), can lead to the onset initially of fatigue, and ultimately of immediate or cumulative muscular strain in the local body area" (p. 2). "Cumulative muscle strain in industry is probably the reason for more injuries than has been generally thought" (p. 10). "It is sometimes difficult to see why experienced people, after working satisfactorily for, say, 15 years at a given job, suddenly develop pains and strains. In some cases these are due to degenerative arthritic changes and/or traumatic injury of the bones of the wrist or other joints involved. In other cases, the cause seems to be compression of a nerve in the particular vicinity, as for example, compression of the median nerve in carpal tunnel syndrome. However, it may well be that many more are due to cumulative muscle strain arising from wrong methods of working learnt earlier" (p. 11).

Ratz, H.C. and Ritchie, D.K. (1961). *Operator Performance on a Chord Keyboard*. Journal of Applied Psychology, 45(5), 303–308.

Experiments were performed concerning whether or not a multiple-finger or "chord" keyboard may be a useful alternative to the regular keyboard in which one key depression generates one character. The results indicated that single finger activations (no chord) were associated with the fastest response times. The slowest times were found with patterns where one or two fingers had to be kept still while their neighbors on each side activated keys. Considering also the information rate associated with the different chords, the authors conclude that motor system constraints are predominant over choice reaction in determining speed of activation on a chord keyboard.

Diehl, M.J. and Seibel, R. (1962). *The Relative Importance of Visual and Auditory Feedback in Speed Typewriting*. Journal of Applied Psychology, 46(5), 365–369.

A test was performed with an electric typewriter to determine whether depriving typists, in a speed-typing situation, of auditory or visual cues, or both, affected the number of net words typed per minute. Sixteen subjects underwent standard five-minute speed typing tests either under normal conditions, where they could both see and hear the typing occurring on the platen, or with the visual or auditory cues, or both, removed. Preventing auditory feedback reduced the performance.

Klemmer, E.T. and Lockhead, G.R. (1962). *Productivity and Errors in Two Keying Tasks: A Field Study*. Journal of Applied Psychology, 46(6), 401–408.

The authors estimate, conservatively, that two million people spend a large part of each day using a keyboard machine, such as typewriter, keypunch, adding machine, teletype, bank proof machine, and linotype. The authors conducted a statistical analysis of recorded keystrokes from IBM card punch machines or IBM bank proof machines. The numbers of key strokes per day, per operator, ranged from 56 000 to 83 000, of which 1600 to 4300 strokes were errors (undetected and uncorrected by the operator). The least accurate operators were found to make ten times as many errors as the most accurate operators. Fast operators tended to make fewer errors.

Seibel, R. (1962). *Performance on a Five-Finger Chord Keyboard*. Journal of Applied Psychology, 46(3), 165–169.

The experimental setup was similar to the one by Ratz and Ritchie, but five lights were assigned to provide the activation stimulus. Four subjects participated, and the daily experimental sessions took approximately 30 minutes. Each of the 31 patterns possible with the five lights and keys was presented 17 times per session. The results indicated that, after the first practice session, the average response time was less than 0.40 seconds for every session. The response time decreased during at least the first eight sessions.

Seibel R. and Rochester, N. (1962). *Communication Device*. Patent 3022878. United States Patent Office. (Patent signed to IBM Corporation, New York.)

The authors state that previous keyboards were not designed with a shape related to the form of the relaxed human hand, keys were not arranged so as to reduce hand and arm movements to a minimum, and keyboards were not designed to support the operator's hand, although it is desirable that the operator's effort be reduced as much as possible. The authors describe an invention in which the keys ("transducers") are associated with the individual digits of the hand, and can be moved by digit movements only to reduce physical movement. For this, the design can be arranged around the hand like a glove.

Yllo, A. (1962). *The Bio-technology of Card Punching*. Ergonomics, 5, 75–79.

The author describes a modification of IBM card punches. He says that poor health of punch machine operators has been reported due to the fast and monotonous character of their work. The author attributes the reasons for the physiological overexertion to the keyboard, located too high in relation to the working arm. Furthermore, he says that the keyboard position forced twisting of the forearm to an extreme position towards the thumb side, and that the keyboard was positioned so as to cause an unnatural wrist position. Finally, the seeing distance to the moving card in the machine was too far. The keyboard was lowered, angled to the outside and adjustable for each operator. Transition from the old to the new working conditions is claimed to be very easy. The operators were satisfied and none was willing to work with the equipment as originally delivered by the manufacturer. Electromyograms taken in the old and new working position indicated a reduction of fatigue symptoms.

IBM Industrial Design Laboratories Germany (1962). Keyboard Project 840400 (Report 1962, Industrial Design Studies). Boeblingen, Germany: Industrial Design Department, IBM Laboratories Germany.

This report describes studies of positioning of the hands and finger motions at current keyboards. It is argued that because of the straight-line assignment of the keys, the hands have to be turned about 30 degrees with respect to the forearm, bringing about ulnar wrist deviation. Because of the horizontal arrangement of the keys, fingers as well as the whole arm have to be moved to touch the keys.

New designs of keyboards are proposed to overcome human factors disadvantages of the traditional arrangement. This includes the arrangement of keys on a curved surface adapted to the natural shape of the fingers, and to the direction of the motion curves of the fingers, which are in a "ray" fashion, meaning that in extended fingers the tips are further apart than in curved fingers. Figure 8 shows a keyboard in which the keyboard sections for the left and right half are slanted with respect to each other, in the top view, to avoid the ulnar deviation referred to earlier. Figures 9, 10, 11, 14, 15 show various arrangements of the keys to accommodate the natural position and motion capabilities of the human hand. Figure 12 is a drawing of a keyboard in which the keys are arranged to facilitate finger motions. Figure 13 depicts a detailed drawing of another model, showing different ways to accommodate natural motions of the hand.

Robbins, H. (1963). Anatomical Study of the Median Nerve in the Carpal Tunnel and Etiologies of the Carpal-Tunnel Syndrome. Journal of Bone and Joint Surgery, 45A(5), 953–966. This paper describes the carpal tunnel and median nerve in their anatomical features, particularly the passage of the tendons of muscles moving the fingers. The opening of the tunnel is reduced proportionally to the degree of extension or flexion of the wrist, as compared to the straight position.

Cornog, J.R., Hockman, J.F. and Craig, J.C. (1963). Address Encoding – A Study of the Double-Binary Keyboard as a Link in the Machine-Sorting of Mail. ASME Publication of the Winter Annual Meeting, Philadelphia, PA: November 17–22, 1963. New York, NY: The American Society of Mechanical Engineers.

The authors describe the development and experiments with their double-binary keyboard, which has an armrest bar, a panel of lights, and a foot switch. These devices could be arranged and adjusted in various manners by the subjects. They tested two groups of 14 subjects each. The main measurements concerned the speeds achieved and the error rates. Each subject was scheduled to work two hours a day with the new equipment. At reporting time, the performance figures were not yet available. However, the authors observed no significant differences in speed or accuracy among persons using either different postal codes, with or without speed reading training, and of different age or gender. The authors believe that the experiments demonstrated the practicality of the keyboard.

Seibel, R. (1963). Discrimination Reaction Time for a 1023-Alternative Task. Journal of Experimental Psychology, 66(3), 215–226.

The author arranged ten lights in a similar curved fashion as ten keys (push buttons) which were arranged, for each hand, in a curved form resembling the positions of the digit tips when the hand is relaxed. When one or several lights came

on, the corresponding key or keys had to be pressed, and the reaction time was measured. Given the large number of experiments needed (there were 1,023 unique alternatives), the author and two assistants were the subjects. The reaction time increased by 20 to 30 milliseconds when 8.5 to 10 bits of information processing were required instead of 4 to 5 bits.

Morgan, C.T., Cook, J.S., Chapanis, A. and Lund, M.W. (Eds.) (1963). Human Engineering Guide to Equipment Design. New York, NY: McGraw-Hill.

In providing guidelines for the design of controls on equipment, the authors state on page 17 that "...the characteristics of the controls to be used by each operator should be analyzed, taking into account the following factors...:

1. The distribution of tasks among the various limbs. Consider whether any one limb is overloaded... .
 2. The placement of controls. Consider whether all controls are within easy reach, ...and are placed according to their importance.
 3. The proper type of control. Consider whether the proper type of control has been provided"
- On page 36: "In this role [as controller], man also has his capabilities and his limitations. He can push buttons and pedals, turn knobs, throw switches, and operate various other kinds of control, provided such controls do not impose tasks that exceed his capabilities. Some of these can be listed as follows:
1. The forces that human operators can exert are limited.
 2. Human operators are relatively slow in moving controls
 3. Man is also limited in the kinds of movement he can make
 4. The precision with which man can apply a given force to a control is also limited. ...
 5. The time during which man can apply a force is limited.
 6. Man, as a controller, is easily overloaded."

Kroemer, K.H.E. (1964a). On the Effect of the Spatial Position of Keyboards on Performance with Typewriters (in German). Int. Zeitschrift Angewandte Physiologie Einschliesslich Arbeitsphysiologie, 20, 240–251.

In the introduction it is stated that: "Steno-typists and other persons working with keyboards are often afflicted with disorders of tendons, tendon sheaths, and synovial tissues of tendons, and of the tendon and muscle attachments. ...At this moment it is unknown what causes or aggravates these disorders. Possible sources may be, for example, the force needed to operate the keys, the displacement of the keys, or the frequency of operation. Hettinger (1957) attributes particular importance to the frequency. Practical experiences give reason to assume that 'electrical' typewriters are advantageous over 'mechanical' ones. With electrical machines, the typing frequency is certainly not lower, but the key displacement and the operational force are smaller. The body posture is indicted in several publications. Inappropriate posture and extensive muscle tension of the arms are mentioned... , and the working position and direction of arms and hands are indicated... . The disadvantageous posture of arms and hands appears to be an important but generally little considered attribute of the work with typewriters and other keyboard machines as indicated already in 1926 by Klockenberg and in 1931 by the Allgemeiner Freier Angestelltenbund" (p. 240).

In the first section, experiments are described in which subjects held their forearm horizontal and turned, with closed eyes, the hands and forearms in a twisting motion until they stopped in the most comfortable position. The angle of the palm against the horizon was recorded. The results are simi-

lar to those reported in 1926 by Klockenberg (listed in Table 1): with upper arms hanging, the most comfortable angles of hand tilt against the horizon are in the neighborhood of 30 to 45 degrees. The subsequent section describes experiments with a new keyboard ("K-keyboard") – see Kroemer 1964b.

Kroemer, K.H.E. (1964b). *Hand and Arm Postures with Typewriters: Comparison Experiments on a Traditional and a Gable-Like "K-Keyboard"* (in German). In Proceedings, 2nd International Ergonomics Association Congress (pp. 367–370). London: Taylor & Francis.

This paper reports on two sets of experiments that had been described in separate publications (Kroemer, 1964a, 1965). Both used a K-keyboard on which the keys are arranged on curves instead of straight rows. The keyboard is divided into halves for the left and right hand. The sections are separated, and each can be tilted laterally. Each keyboard half has a curved space bar. Only two keys were used per finger, in a sequence of operations determined by a sentence so selected that each finger struck keys about as often as when typing a common German text. That sequence was continuously keyed by the subjects.

In the first set of experiments, the K-keyboard sections were arranged so that the keyboards were either both horizontal (0 degrees) or each declined laterally at 30, 60, and 90 degrees, respectively. The following pairings were compared: 0 with 60, 30 with 90, and 30 with 60 degrees. Thirty-two experienced typists participated as subjects, as did 33 persons who could neither type nor play piano.

The results are separately displayed for experienced and novice typists. The number of subjects in each trial was not the same, and many of the main effects, and of the interactions, were not significantly different. Yet, the results are interpreted in such that the keyboards tilted 30 or 60 degrees were superior in performance and preference. The tilts of 30 and 60 degrees were judged to be more comfortable than the common horizontal arrangement, which was associated relatively often with complaints about discomfort and pain in hands, wrists, and forearms.

In the second set of experiments, a K-keyboard was tilted at 45 degrees to the side, and compared with a regular keyboard (at 0 degrees, i.e. horizontal). Twelve non-typist subjects participated. They used the same finger sequence as in the first experiment, and were instructed to continue the keying without break as long as they could.

The results indicated no statistically significant differences in terms of endured keying time, number of keystrokes executed, keystrokes per second, and heart rate. Yet, "aches and pains" in arms, wrists, and fingers, as well as in the back were more often voiced as reasons for terminating the trials with the standard arrangement than with the tilted K-keyboard. Pains in shoulders and impossibility of further concentration on the task were the main reasons reported for terminating the trials with the K-keyboard.

Seibel, R. (1964). *Levels of Practice, Learning Curves, and System Values for Human Performance on Complex Tasks*. Human Factors, 4, 293–297.

The author summarizes previous experiments, of his own and of other authors, regarding the validity of either using literature data, or the fidelity of research to shed light on specific questions. The author concludes that there are no reliable shortcuts to determine final performance on a complex task, such as keyboarding. Thus, it appears necessary that the subjects practice until their performance curves flatten out.

Fox, J.G. and Stansfield, R.G. (1964). *Digram Keying Times for Typists*. Ergonomics, 7, 317–320.

A standard typewriter was outfitted with a photocell and lamp to record the time interval between key strokes. Five subjects participated who could type about 80 words per minute. They were required to type a text of 27 lines, which used 23 letters of the alphabet and contained 115 different two-letter sequences (digrams). The results indicated that digrams keyed with alternate hands required shorter times than digrams keyed with the same hand. The authors conclude that a keyboard that is designed to allow many digrams to be keyed with alternate hands, will allow an increased keying rate.

Bowen, H.M. and Guinness, G.V. (1965). *Preliminary Experiments on Keyboard Design for Semiautomatic Mail Sorting*. Journal of Applied Psychology, 49(3), 194–198.

The experimenters intended to determine keyboard designs that can be used favorably for mail encoding. In the first experiment, three experienced typists used a conventional sequential keyboard with 29 keys arranged in three rows. The typists could strike any two of these keys for encoding, or they could employ three of the eight keys of the home row. Performance on these two key sets was approximately the same. Then the same sequential keyboard of 29 keys was used to make comparison experiments with two chord keyboards, one with a single arrangement of keys, the other with a double arrangement of keys. The chord keyboards were found to be associated with higher input speeds than the sequential keyboard. Of the two chord keyboards, the larger showed the smaller number of errors. A comparison between extraction and memory encoding showed no difference in performance on those three keyboards. Yet, altogether, the differences in performance were small, "provided that the keyboards are designed rationally and in keeping with good human engineering practice" (p. 198).

Conrad, R. and Longman, D.J.A. (1965). *Standard Typewriter versus Chord Keyboard – An Experimental Comparison*. Ergonomics, 8, 77–88.

To compare performance of postal employees, the authors designed a chord keyboard with ten "home keys". The comparison keyboard was that of a standard typewriter. The results showed that the people using the chord keyboard became "operational" about two weeks earlier than the persons using the regular keyboard. From then on, improvement rates and data input were almost parallel. Regarding possible improvements, the authors believe that provision of feedback would be particularly beneficial in the early learning stages.

Galitz, W.O. (1965). *CRT Keyboard Human Factors Evaluation*. (Report, March 19, 1965). Roseville: Systems Application Engineering.

Preliminary investigations of keying performance indicated that persons experienced with electrical typewriters might have difficulty in adapting to the Univac computer console. Twelve experienced typists were employed to input text abstracted from an article that contained 24 paragraphs, each containing between 45 and 55 words. Keyboard slopes of 9, 21, and 33 degrees were selected. The subjects preferred mostly the 21 degree keyboard slope, with 33 degrees least preferred. Some subjects complained that the lack of sound associated with the pressing of the keys meant a lack of feedback about key activation.

The recommendations drawn from the experiment include that the keyboard should be designed to approximate the touch characteristics of an electric typewriter, meaning activation forces of about 2.25 ounces and a total travel of one-eighth inch. The key activation should be at three quarters of the total travel distance. The keyboard slope should be adjustable between 10 and 35 degrees. Yet, in spite of the complaints of some typists, the author recommends that feedback should not be included in the key characteristics.

Kroemer, K.H.E. (1965). Comparison of the Keyboard of a Standard Typewriter with a "K-Keyboard" (in German). Internationale Zeitschrift Angewandte Physiologie Einschliesslich Arbeitsphysiologie, 20, 453-464.

This is a report on experiments regarding the spatial arrangement of a K-keyboard, comparing a keyboard tilted at 45 degrees with the usual horizontal typewriter keyboard. Twelve subjects, none used to typing or piano playing, participated. They were trained in depressing the key in the sequence of 60 key strokes used earlier. They were instructed to "type" the sequence rapidly and as long as they could without break, with speed more important than avoiding mistakes.

The reasons given by the subjects for terminating each trial showed a preponderance of "aches and complaints" in arms, wrists, and fingers with the normally arranged keyboard, with complaints in the shoulder region more frequent after use of the K-keyboard. The statement "can not concentrate any longer" on the task (instead of reasons related to aches and pains) was reported as a reason much more often with the K-keyboard than with the regular arrangement. The total numbers of key strokes per trial, the durations of trials, and the heart rates during the trials, were not statistically significant for the two keyboard arrangements, but more keying errors were committed with the conventional keyboard.

Mayzner, M.S. and Tresselt, M.E. (1965). Tables of Single-Letter and Digram Frequency Counts for Various Word-Length and Letter-Position Combinations. Psychonomic Monograph Supplements, 1(2), 13-32.

This is a tabulation showing single-letter and double-letter (digram) frequencies, based on 20000 English words. The tables are set up to show word-length and letter-position combinations for words from three to seven letters in length.

Galitz, W.O. (1966a). CRT Keyboard Human Factors Evaluation: Study II. (Report, February 28, 1966). Roseville: Systems Application Engineering.

This study, the second in a series of three, concerned the design of the keys of a Univac computer console. Seven female typists who had been using electric typewriters were employed. The results showed that performance on the modified keyboard was less than the one achieved on the customary electrical typewriter, but the modified Univac keyboard showed much better performance than the previously employed keyboard. Subjects complained, however, that the force required to depress the keys was too high even on the better keyboard. The statements of the subjects indicated that key displacement, force, and slope were quite important factors in terms of performance, fatigue, and satisfaction with the design.

Galitz, W.O. (1966b). CRT Keyboard Human Factors Evaluation: Study III. (Report, June 8, 1966). Roseville: Systems Application Engineering.

In this set of experiments, the keys on the Univac console were replaced by new switches with a displacement of 5/32 inch and an activation force of 2.25 ounces. Six experienced electrical typewriter typists participated, as in the previous set of experiments, over a period of three days. The results indicated that with these keys, the performance of the subjects was similar to their performance on the customary electrical typewriters. The subjects all found that the "touch of this keyboard was lighter" than on their typewriters. Four of the six subjects expressed concern about missing feedback regarding whether or not they had actually activated a key when depressed. Thus, the author concludes that, in contrast to his earlier statements, feedback may be an important feature.

Phalen, G.S. (1966). The Carpal-Tunnel Syndrome - Seventeen Years' Experience in Diagnosis and Treatment of Six-Hundred Fifty-Four Hands. Journal of Bone and Joint Surgery, 48-A(2), 211-228.

Phalen states that carpal tunnel syndrome is the most common cause of burning pain, numbness, and tingling in the digits and the lateral half of the palm. He refers to studies in 1958 by Paget, in 1930 by Marie and Foix, and to other studies in the 30s and 40s. He also indicates that he reported on carpal tunnel syndrome in 1947, 1949, and 1956. He states, on page 219: "Occupations that require active finger flexion with the wrist flexed should certainly predispose to a carpal-tunnel syndrome..." His findings are based on the investigations of "754 hands of 439 persons during the last 17 years" (p. 212).

Hymovich, L. and Lindholm, M. (1966). Hand, Wrist, and Forearm Injuries - The Result of Repetitive Motions. Journal of Occupational Medicine, 8(11), 573-577.

"It has long been known that repetitive motion of the hand can cause injury, especially when some sort of tool or instrument must be held in the grasp. Such injuries are encountered most frequently among workers employed in small-assembly and bench-work activities in plants engaged in the kinds of light manufacturing typified by the electronics industry" (page 573.) While many injuries are said not be serious, and responding to simple therapeutic measures, this report presents disabling and expensive cases observed over a period of six years within a group of workers manufacturing small electronic components. No type of work or tool could be indicated as a predominant factor with respect to frequency or seriousness of injury. Four injuries occurred in offices. A program is described to prevent these diseases on the job, including an analysis of hand motions performed by the injured workers to determine whether other ways of doing the same work could be developed.

Tichauer, E.R. (1966). Some Aspects of Stress on Forearm and Hand in Industry. Journal of Occupational Medicine, 8(2), 63-71.

The author states that, in modern industry, the total workload is much reduced but often concentrated on small elements of the worker's anatomy. Instead of traumatic injury, a new kind of impairment must be considered which is "caused insidiously over lengthy periods of time by gradual, cumulative, and often imperceptible overstrain of minute body elements". The author describes a biomechanical model of the arm designed to predict the nature and magnitude of stress and strain caused by industrial activities.

Max-Planck Gesellschaft (1967). Patentschrift. (West) German Patent 1255 117. (Inventor: Kroemer).

The patent describes a keyboard that is split into left and right halves. In the plane view, the keyboard halves are slanted against each other, with the slant angle open toward the operator. In the frontal view, the keyboard halves are tilted against each other, with the tilt angle open downward. Slant and tilt angles are adjustable. On each keyboard half, sets of keys are oriented in columns that spread ray-like so that they are more apart away from the operator. The rows of keys are bent to accommodate the fingers of different lengths. Key top surfaces form a concave arc along the columns, and a convex arc along the rows. These design specifics are chosen to improve keyboard usability, to bring about more natural hand and arm posture, and to avoid so-called typist diseases such as tendonitis.

Lewin, M.H. (1967). Portable Electronic Keyboard for Computer Input by Telephone. IEEE Transactions on Electronic Computers, EC-16, 332–334.

To allow access to a computer by telephone, a special input device, called “electronic keyboard” was developed. It consists of a circuit box that displays all characters, a speaker attachment, and a “selection pen”. To input messages, character by character, one points the selection pen at the selected characters.

Smith, W.A. (1967). Accuracy of Manual Entries in Data-Collection Devices. Journal of Applied Psychology, 51(4), 362–368.

The investigation concerned the accuracy of entries made manually in data collection devices, mostly modified telephone sets, by workers at their place of work. It was found that most errors were immediately detected and corrected by the operators. About every second mistake was caused by single-digit substitution. About every fifth mistake was omission of a digit.

West, L.J. (1967). Vision and Kinesthesia in the Acquisition of Typewriting Skill. Journal of Applied Psychology, 51(2), 161–166.

The purpose of the reported experiment was primarily to determine whether deprivation of visual feedback concerning what had just been typed, would affect performance.

Two hundred sixty-six typists, with skill ranges from 9 to 108 words per minute, participated in experiments where they typed a regular text when they could either see the typed text, and the typewriter keyboard, or could see neither. The results indicated that deprivation of visual feedback reduced the performance only in terms of errors, which were much increased.

Ulich, E. (1967). Some Experiments on the Function of Mental Training in the Acquisition of Motor Skills. Ergonomics, 10, 411–419.

Different methods of learning motor performance are reported. They included the Finger-Dexterity Test. Among other things, the studies led to the conclusion that motor skills can be learned in different ways, including mental training. Myographic measurements demonstrated changes in muscular activities during training. Some subjects reported pains or symptoms of fatigue.

Lhose, E. (Ed.) (1968). Proposed USA Standard - General Purpose Alphanumeric Keyboard Arrangement for Information Interchange. Communications of the ACM, 11(2), 126–131.

This is a USA standard proposal for general purpose typewriter-like alphanumeric keyboards. The keyboard arrangement was developed from a study of keyboards already in use by millions of trained operators in the Taylor printer, typewriter, and related fields. Also included in this study were the USA standard typewriter keyboards (USAS X4.7-1966) and applicable standards work carried out internationally.

The standard presents the arrangement of keys claimed to be “the most practical and economical for user and manufacturer”, with a study performed on “human engineering considerations and de facto practices” (p. 129). Forty-four keys are defined. In the second part of the document (pages 130, 131), a “dual case keyboard arrangement” is described which is applicable to both electric and manual machines. Forty-four printing keys are defined for 57 characters.

Adams, J.A. (1968). Response Feedback and Learning. Psychological Bulletin, 70(6), 486–504.

The author discusses the history and theory of the effects of sensory feedback about the results of an action to govern later segments of behavior. While the author’s interest is mostly in the relationship between feedback and learning, he discusses underlying principles, such as the derivation of the terms “kinesthetic” and “proprioceptive” feedback information. In his examples discussing the closed-loop theory, he refers to motor behavior like key tapping, which is said to be controlled by multiple loops. For example, a subject can see the key, hear its tapping, and “feel” it through proprioceptive and tactile organs. He refers to several articles in the early 1960s that investigated the influence of various loops on key tapping in which reductions in auditory and proprioceptive feedback produced lowered performance. If all modalities of feedback were degraded together, some subjects reported no knowledge of tapping the key.

Keele, S.W. (1968). Movement Control in Skilled Motor Performance. Psychological Bulletin, 70, No. 6 (Part 1), 387–403.

The author discusses various theories of movement control from their beginnings in the late 1800s. Skilled performance of motor activities (movement such as walking, driving, typing, etc.) is theoretically explained by the relations between speed, accuracy, and distance of the movement and the time required to process feedback. In single movements, corrections may be made according to Fitts’ Law of 1954. For repetitive movements, the control may be shifted from feedback to a motor program which, however, is probably acquired from a sequence of single movements. Also, the motor program deteriorates with time, but is increased in accuracy by the number of repetitions.

Kinthead, R.D. and Gonzalez, B.K. (1969). Human Factors Design Recommendations for Touch-Operated Keyboards (Final Report 12091-FR, Prepared for Micro Switch Division). S & RC Research: Honeywell, Inc.

Four experiments were conducted to determine the best experimental design for keyboard performance study, the optimal levels of key resisting force and displacement, and the importance of tactile feedback of the keystroke to the operator.

In the first set of experiments it was found that performance on the experimental keyboards approached the speeds of performance on the subjects’ own electrical typewriters after the sixth day. In the second experiment, the experienced typists employed in the experiment did not show a trade-off between

speed and errors. In the third experiment, the keys with a lower force requirement fared better than those with higher force. If the key force is low, then displacement should be low as well. If forces are high, then displacement should also be high. High force with low displacement was found to be the most undesirable condition. In the fourth experiment, the keys with the “snap action in the middle of the stroke” showed a higher error rate than keys that did not have such snap action.

The operators were questioned about their preferences at the end of each experiment. For about half of the subjects, their opinions about preferred conditions reflected their performance, but “only one or two of the subjects, however, could give these accurate opinions consistently. This lack of correspondence between preference and performance has been noted several times before” (p. 27).

Remington, R.J. and Rogers, M. (1969). Keyboard Literature Survey Phase I: Bibliography (TR 29.0042). Research Triangle Park, NC: IBM Systems Development Division.

This report contains a listing of more than 300 previous papers, most in the open literature, regarding keyboard entry devices. Three categories of reports are listed. The first category concerns reports on behavioral data associated with the use of keyboards and related data entry devices. This listing contains approximately 120 reports, some published in the 1920s, many in the 1930s, and most in the 1950s and 1960s. The second category includes documents concerning keyboard devices by themselves, and contains about 60 references, mostly from the 1950s and 1960s, but some going back decades more.

The third category contains “other” pertinent references. The listing contains about 130 entries, again with most of them from the 1950s and 1960s, but some going back even into the 1930s.

West, L.J. (1969). Fatigue and Performance Variability Among Typists. Journal of Applied Psychology, 53(1), 80–86. This experiment was performed to shed light on the relationship between fatigue and typewriting. Two hundred thirty-four typists, ranging in skill from 5 through 108 words per minute, participated. They performed 30 minutes of continuous typing of standard text. The data were analyzed for number of strokes and number of errors per minute. The results indicated a slight increase in error (attributed to fatigue) over the 30-minute work periods, while overall performance showed no trend over the work period. As skill was improved, measured by words per minute, the typing speed became more stable, but the number of errors showed no such trend.

The mechanical nature of the typewriter had made it impractical to change its basic lever system that transforms a hand-caused key displacement into a typed imprint. Similarly, it was also impractical to change the design of the keyboard. But, around 1950, the use of electric auxiliary power to help transform a key activation into a printed character became feasible (Hesh 1950, Droegge and Hill 1961). This reduced the amount of energy that the operator had to apply in each keystroke (but did not diminish the frequency of keystrokes).

Use of electricity opened the door for new designs, both of the basic machine and of its keyboard. The spatial

configuration of the keyboard with respect to the operator became a matter of experimental research (Scales and Chapanis 1954, Creamer and Trumbo 1960, Yllo 1962). The use of chording instead of the one-key/one-character design was intensively investigated (Klemmer 1958, Lockheed and Klemmer 1959, Ratz and Ritchie 1961, Seibel 1962, Seibel and Rochester 1962, Conrad and Longman 1965).

The terms “ergonomics”, “human factors”, or “human engineering” were emerging at this time, and the importance of the underlying concept was widely accepted: “Fitting the task and equipment to the human”, by reconfiguring the keyboard to suit the size and mobility of the hand, was seriously considered (IBM 1962, Kroemer 1964/5, Galitz 1965/6). On the emerging electronic computers, the mechanical key, a button at the operators’ side of a leverage system, was replaced with an electrical switch. Human-factor design recommendations for “CRT” keyboards (Cathode-Ray Terminals, Galitz 1965/6) and “touch-operated keyboards” (Kinkead and Gonzalez 1969) were published. Yet, the “Proposed USA Standard” for a general-purpose keyboard for information interchange (Lhose 1968) still incorporated the QWERTY keyboard, but more keys were added on its left and right sides so that the total number of keys was roughly doubled. This standardization of the keyboard perpetuated the convention of keys arranged in straight horizontal rows with zigzag columns on the QWERTY part but straight columns in all other key sets.

In 1951, Lundervold published the first report of his groundbreaking electromyographic (EMG) experiments to attain knowledge about the use of individual muscles while typewriting. Lundervold investigated 47 healthy typists and 88 patients, most of the latter suffering from “occupation myalgia”, a disease in muscles that have been overstrained during repetitive work, such as typewriting, “where no other cause can be found than the antecedent hyperfunction” (pages 10, 11). Lundervold discovered that, in typing, muscles were activated which were not expected to partake. He found that changes in the EMG records reflected the actual muscle activity and the status of muscle fatigue. Lundervold’s experiments provided clinical analytical evidence for the long-held opinion that repetitive typing can lead to a cumulative overexertion injury even when a single keying activity is not harmful by itself. After decades of observations and practical experiences, which had pointed in that direction, Lundervold’s findings supported the recognition of ailments such as tendinitis, tenosynovitis, and tendovaginitis of the upper extremities as “occupational diseases” of typists (Hagen and Peters 1955, Buckup 1957, Hettinger 1958, Baader 1960). The occurrence and causes of Carpal Tunnel Syndrome (CTS) associated with repetitive keyboarding activities was becoming a topic of medical and engineering concern (Tanzer 1959, Peres 1961, Phalen 1966, Hymovich and Lindholm 1966, Tichauer 1966).

To summarize the development during the 1960s and 1970s, musculoskeletal overexertion of typists evolves as a medical problem and a concern to human factors engineers who propose more ergonomic keyboard designs and uses. The QWERTY layout is maintained even in enlarged keyboards. Mechanical typewriters begin to be replaced by electrically powered machines which reduce the energy input required from the typist for each keystroke, but not the stroke frequency.

4 From 1970 to 1979

Annotated list of publications on keyboards and keyboarding from 1970 to 1979

Caldwell, L.S. (1970). Decrement and Recovery with Repetitive Maximal Muscular Exertions. Human Factors, 12, 547–552.

Sixty U.S. Army enlisted men, medically fit, but not selected with respect to size or strength, participated in experiments in which they pulled ten times with the hand as strongly as they could on a handle mounted at shoulder height. During the pulls, the elbow angle was 150 degrees. The pull time was, in each case, 12.5 seconds, but the ten trials were separated by rest periods of 12.5, 25, 50, 100, or 200 seconds.

The results indicated a clear dependency of the maximal strength that could be exerted in each subsequent trial on the rest periods between trials. The largest force could be exerted with the longest rest periods, and the smallest forces with the shortest rest periods. From trial one to trial ten, the possible exertion was reduced consistently from trial to trial. The smallest reductions occurred with the longest rest periods, the largest reductions with the shortest rest periods.

Galitz, W.O. and Laska, T.J. (1970). The Computer Operator and His Environment. Human Factors, 12, 563–573.

Operator performance data were collected at six computer customer centers. Questionnaires were sent to all customer personnel using computers. Thirteen categories of activities were established, based on which activity the operator was performing with the hands. This included the handling of tapes, cards, papers, instructions, writing, keying, telephone operation, etc. It was found that for almost one-half of the work day the operator was not engaged in any manual activity, while about one-third of the work day was spent at the computer console, and another 25 percent at manual input/output devices. Keying itself took more time than telephoning.

Hirsch, R.S. (1970). Effects of Standard Versus Alphabetical Keyboard Formats on Typing Performance. Journal of Applied Psychology, 54(6), 484–490.

The author performed an experiment in which he compared the “standard” key arrangement on a QWERTY keyboard with an arrangement in alphabetical order. The subjects performed significantly better on the “standard” keyboard than on the “alphabetic” keyboard. The author argues that, with limitations, the standard keyboard is better “human engineered” than the alphabetical one. For example, the most frequently used letters are clustered in the center of the keyboard.

Garrett, J.W. (1971). The Adult Human Hand: Some Anthropometric and Biomechanical Considerations. Human Factors, 13, 117–131.

This is a summary of recent studies on the anthropometry and on selected biomechanical characteristics of the human hand. The data concern: (1) conventional anthropometry of male and female hands, (2) anthropometry of the relaxed hand, (3) comparison of selected anthropometric and performance parameters, and (4) the ability to retain grips on selected handles.

Klemmer, E.T. (1971). Keyboard Entry. Applied Ergonomics, 2, 2–6.

This article summarizes about 35 papers on keyboard entry that had been published recently. The effects of feedback, keyboard design, and other factors on performance is discussed, together with physiological measurements. Among other statements, Klemmer indicates that suitable chording of keystrokes may allow higher input performance than single keystroke procedures. Regarding the stroke, it is said that a light, short stroke results in higher speed and less fatigue than other strokes. Concerning feedback, Klemmer believes that auditory feedback may be of little importance, while visual feedback might be more important. “Elemental muscle action” (kinesthetic) feedback can produce effects, in fact is required for learning. Such feedback can be attempted through key displacement, force, or spacing. (Yet, he quotes no evidence for his conclusion that kinesthetic advantage is more than offset by the added time and work involved.) The author also stresses that physiological measurements, particularly EMG recordings, can be used to evaluate keyboards, their uses, and training procedures.

Komoike, Y. and Horiguchi, S. (1971). Fatigue Assessment on Key Punch Operators, Typists and Others. Ergonomics, 14, 101–109.

The authors describe a study done on 233 female office workers, 53 of whom were telephone operators, 49 were typists, and 26 were keypunchers. They usually worked six days a week, and more than 40 hours a week. They often showed symptoms such as stiff shoulders, myalgia, and arthralgia, especially in the right arm, and finger tremor. In general, a significant increase in the number of subjective fatigue symptoms after work was observed. No significant differences were observed between the subjects of different job classifications.

Phillips, S.A. and Kincaid, J.P. (1971). The Human Engineering of the Typewriter Keyboard: A Case in Point. Human Factors Society Bulletin, July issue, 7–8.

This is an overview of the evolution of the keyboard from 1872 to the time of publication. In particular, human factors engineering aspects are discussed, such as the arrangement of the keys on the keyboard.

“Judged from the standpoint of the touch typist, the arrangement of the [key] alphabet is madly inconvenient. In the first place it is left-handed. Most typists are right-handed, and their right hands are more agile than their left. But the standard order...gives the right hand 56 percent of the total number of strokes. It also gives weak fingers, particularly the little finger of the left hand too much work, including the two toughest keys, the shift-lock and the back-spacer. In the third place it demands an entirely unnecessary amount of awkward

reaching because it ignores the frequency with which the letters of the alphabet are encountered. ... A rational keyboard would try to make most of what's easiest, which is hitting a letter with one hand and the next with the other hand. ... Among one-handed motions, hurdles - in which the finger has to jump a key to get from here to there - are the most difficult. ... Almost any reasonable man, upon realizing that the typewriter keyboard is silly, would ask, 'Why doesn't somebody do something about it?' ... Mechanically speaking, any of the typewriter firms could arrange the keys in any order quite easily. ... It remains a case in point which illustrates a sad fact known by human engineers everywhere: an ounce of human engineering prevention is worth a pound of human engineering cure" (p. 8).

Michaels, S.E. (1971). Qwerty Vs. Alphabetic Keyboards as a Function of Typing Skill. Human Factors, 13, 419-426.

Performance on two keyboards was compared. The keyboards were a three-row alphabetic key arrangement and the standard QWERTY array. Thirty subjects ranging from little to good typing skills operated the keyboards for 10 half-hour sessions, with half of the subjects starting on the alpha and the other half starting on the QWERTY keyboards, switching to the other keyboard after five sessions. For skilled and semi-skilled typists, the QWERTY keyboard resulted in better inputs, while unskilled operators performed equally on the two keyboards. The author compares his study to that performed by Hirsch (1970) and states that the QWERTY keyboard is probably better than an alphabetic keyboard for the pragmatic reason that skill in using the QWERTY array is so widespread.

Smith, S.L. and Goodwin, N.C. (1971). Alphabetic Data Entry via the Touch-Tone Pad: A Comment. Human Factors, 13, 189-190.

The authors refer to their earlier article and argue that a telephone-pad arrangement can be used to enter alphabetic information because a computer can be used to sort out the possible ambiguity of entering three different letters with the same key.

Galloway, G.R. (1972). Keyboards and Data Measures. Unpublished Master thesis, Department of Information Sciences, Graduate School of Arts and Sciences, University of Dayton, OH.

A study is described in which an engineering group had to develop a new line of alphanumeric touch operation keyboards, such as found on typewriters, calculators, and business machines. One of the goals of the design effort was to optimize the physical operating characteristics of the keyboards for the user.

For the experiment, key forces of 100, 200, and 300 grams were selected. The results indicated that the lower force keyboards were associated with higher input rates, but that there were fewer errors with the higher key force levels. The operators liked the low force-low displacement keys best, and the high force-high displacement keys least. Their preferences are said to be in some disagreement with regard to what, from a "management point of view", might be preferable in terms of force and displacement vs. performance and errors.

Kroemer, K.H.E. (1972). Human Engineering the Keyboard. Human Factors, 14, 51-63.

The author points out that at the standard keyboard the typist must twist the forearms so that the palms of the hand are

about horizontal, i.e., pronated close to the anatomical limits. This requires continual muscle tension. To relieve the muscles, the operator tends to lift the elbows laterally and forward, which also occurs if the keyboard is located too high. This, again, leads to continual muscle tension, both in the arms and across the shoulders. Some of the muscles continually strained to maintain the arm posture are also involved in moving the fingers to do the typing. Removal of unnecessary strain (by tilting key sets laterally) is said to possibly reduce the susceptibility to ailments of muscles, tendons, joints, and tissue often found among card punchers and typists. The author's experiments, reported in German in 1964/1965, are summarized.

Phalen, G.S. (1972). The Carpal-Tunnel Syndrome. Clinical Evaluation of 598 Hands. Clinical Orthopaedics and Related Research, 83, March-April Issue, 29-40.

The carpal-tunnel syndrome is said to be the most common and the most important of all nerve entrapment syndromes. This paper is based on the examination of a total of 1252 hands with carpal-tunnel syndrome. Phalen reiterates the "wrist flexion test" and the "tourniquet test", which by the time had become classical diagnosis procedures. He then continues to describe surgical procedures for relief, and their follow-up period.

Samuel, R.A. (1972). Electric Typewriter Key and Keyboard Arrangement. Patent 3633 724. United States Patent Office.

The patent describes a novel typewriter key, and keyboard, which can be used with conventional electric or standard typewriters. A particular reason for the invention was to avoid that a particular digit of the hand must move from the home position to a desired key. To achieve this, 8 alphanumeric keys are used which have moveable sections. These can be operated by simple finger pivoting and pressing on the different sections of the keys.

Seibel, R. (1972). Data Entry Devices and Procedures. Chapter 7 in H.P. Van Cott and R.G. Kinkade (Eds.), Human Engineering Guide to Equipment Design (pp. 311-344). Washington, DC: US Government Printing Office.

This chapter discusses current issues of data entry devices. Seibel states (p. 337) that neither visual nor auditory feedback appear to be necessary for the highly skilled data operator performing under speed-test conditions. Yet, he cites other studies that showed that removing visual feedback does have a detrimental effect on the number of errors made. Thus, he concludes that the major source of feedback for the highly skilled operator is "kinesthetic-proprioceptive-tactual feedback which the operator gets from actually making the movement and striking the key" (p. 337).

Regarding the force-displacement characteristics of the key, Seibel states that little information is available but then says "[M]ost keyboards exhibit an increasing amount of force required as the key is depressed, with a marked drop-off in necessary force as the entry (typing) mechanism is tripped, and then a relatively sharp further increase in force requirement as the key is 'bottomed'. Most electrically operated keyboards call for a relatively light touch in order to trip the entry mechanism, frequently in the range to 2 to 3 oz. There are no performance data to indicate the effects of increasing, or decreasing, this force requirement; nor are there data to indicate the effects of changing the characteristics of the force-displacement function, or its variations under dynamic conditions" (p. 338).

Smith, H.A. (1972). *Function Key Design Alternatives*. In: Proceedings of the 16th Annual Conference of the Human Factors Society (p. 422). Santa Monica, CA: Human Factors Society.

Research done at NCR [National Cash Register] indicates that it is no longer necessary to have a single key for each function. Rather, recent developments in technology have made it possible and practical to have other than single or fixed function keys by virtue of the fact that the machine can be made smart enough to know where the operator is in the operation of the device.

Alden, D.G., Daniels, R.W. and Kanarick, A.F. (1972). *Keyboard Design and Operation: A Review of the Major Issues*. Human Factors, 14, 275–293.

The authors start with the statement that the musculature of the hand places limitations on the ability of the fingers in a keyboard task. The tapping rates are different for the fingers. The five digits of the hand have different resistance to fatigue. The authors indicate means to measure performance. These are primarily input speeds and error rates, in general with a trade-off between them.

Regarding design characteristics, the authors state that no single industry or military standard for key characteristics (size, operating force, displacement) had been formally established. They find that the results of the few studies available indicate that keying force and displacement have, within certain limits, little effect on keying performance of experienced operators. They cite various studies to indicate the inconclusiveness regarding the effects of actuation feedback of a key (by either feel, hearing, or sight) on keying performance.

They argue that the standard QWERTY layout has become the de facto standard layout for communications and computer interface keyboards, in spite of suggestions that the arrangement is less than efficient (p. 282).

Tichauer, E.R. (1973). *Ergonomic Aspects of Biomechanics*. Chapter 32 in NIOSH (ed.), *The Industrial Environment – Its Evaluation and Control* (pp. 431–492). Cincinnati, OH: National Institutes of Occupational Safety and Health.

In the introduction to his chapter, Tichauer quotes the 1713 edition of the book “De Morbis Articulorum” by the “father of occupational medicine” Bernardino Ramazzini: “Manifold is the harvest of diseases reaped by certain workers from the crafts and trades that they pursue; all the profit they get is injury to their health.” The author then provides an overview of earlier biomechanical and physiological research that leads to an understanding of why working conditions, work tools and practices should be designed to suit the capabilities and limitations of the human body.

Yoder, T.A., Lucas, R.L. and Botzum, G.D. (1973). *The Marriage of Human Factors and Safety in Industry*. Human Factors, 15, 197–205.

The authors state that social concern as well as governmental legislation place great emphasis on the need for safe workplaces. They give examples of the efforts of the Eli Lilly and Company's safety and human factors department to achieve this purpose. One example is a study using electromyography in assembly tasks for which there were complaints from operators of forearm muscle fatigue and wrist tenosynovitis. The experimental results showed that eight of 21 subjects had increased energy levels when using the standard hand tool. In addition to proposals for the use of different tools, it was found that a mere correction of the hand orientation during the task would be sufficient to alleviate the problem.

Duncan, J. and Ferguson, D. (1974). *Keyboard Operating Posture and Symptoms in Operating*. Ergonomics, 17, 651–662.

In keyboard operators, “occupational cramps” (a functional incoordination of an accustomed repetitive skill) and “occupational myalgia” (recurrent muscle aching associated with repetitive tasks) were found to occur, either singly or combined. The occurrence of symptoms was related to work load, which was influenced by keyboard design and body postures. Among the postural phenomena, the following were found: shoulder depression, shoulder abduction, shoulder flexion, ulnar deviation of the hand, wrist extension, finger extension, mixed extension and flexion of fingers, finger flexion, little finger abduction. Some of these postural deficiencies were found to be correlated. Out of 90 patients, 69 cases of symptoms were found in the arm-shoulder-neck area, and 185 cases in the hand area. Adverse operating postures were observed more often in subjects with symptoms of cramp and myalgia than in the 45 unaffected telegraphists.

Ferguson, D. and Duncan, J. (1974). *Keyboard Design and Operating Posture*. Ergonomics, 17, 731–744.

The authors had found in an earlier study cases of occupational cramps and myalgia had been found in telegraphists. The layout of the standard QWERTY keyboard not only brings about an uneven distribution of load on the fingers, but also affects operating posture. The authors specifically recommend the rearrangement of the keys according to character occurrence and finger abilities, and the division of the single keyboard into two symmetrical keyboards, one for each hand, each tilted sideways.

Kassab, S.J. (1974). *Research Note: Reduction of Muscular Fatigue Associated with Repetitive Light Assembly Operations by Means of Arm Counterbalancing*. Human Factors, 16, 323–326.

In a light assembly task, the weight of the forearms of the worker was partly counterbalanced by a sling to which a cable was attached, which ran over a pulley, so that a weight attached to the cable helped lift the arms. At a work pace of 35 cycles per minute, electromyography indicated that the deltoid muscles showed a 50 percent reduction in effort with this arrangement.

Herndon, J.H., Eaton, R.G. and Littler, J.W. (1974). *Carpal-Tunnel Syndrome*. Journal of Bone and Joint Surgery, 56A(8), 1715–1718.

This is a report on a rare case of carpal-tunnel syndrome, caused by a strike to the wrist in 1971, and subsequently treated. The authors refer to twelve earlier publications, with the oldest dated 1935, and particularly to Phalen's report of 1966.

Maeda, K. (1974). *The Variety of Occupations Which Induce Neck-Shoulder-Arm Disorders, and Some Problems in Taking Measures to Meet the Situation*. Sumitomo Sangy Eisei (Japanese, Abstract in English), 135–143.

During the 20 years preceding the date of publication, neck-shoulder-arm syndrome disorders were noticed among business machine operators. Disorders developed in writers, stampers, belt conveyor workers, sewing machine operators, and others. Harmful factors commonly noticed in postures and motions of those workers where: (a) restrained posture, (b) unnatural posture, (c) repeated use of the upper extremities, and (d) static sustaining of the upper extremities. Administrative actions to prevent such disorders had been centered on business machine operators, with less regard to other workers.

Showel, M. (1974). *A Comparison of Alternative Media for Teaching Beginning Typists*. Journal of Educational Research, 67(6), 279–285.

Several kinds of instructions were applied to male military personnel who were able to type only ten or fewer words per minute in a screening test given prior to the start of training. The subjects received up to 33 hours of typing practice over a nine-day period. Progress measurements administered after every three three-quarter hours of practice indicate that the most effective programs are those which devoted relatively little time to detailed instruction, but emphasized speed by forced pace.

Sidorsky, R.C. (1974). *Alpha-Dot: A New Approach to Direct Computer Entry of Battlefield Data* (Report AD774841). Arlington, VA: U.S. Army Research Institute for Behavioral and Social Sciences.

The purpose of this study was to determine the suitability of a five-key alpha-dot keyboard. Ten enlisted soldiers were instructed to enter both free-form and formatted versions of a data input scenario. Learning time as well as rate and accuracy of data entry using the alpha-dot keyboard were compared to operator performance on a standard typewriter. The results indicate that no special skill is required to operate the device. All trainees were able to input messages satisfactorily after two or three minutes of instruction. Thus, learning time was very short. Nine of the ten trainees memorized the character set within one and one-half hours of practice and were then able to transmit messages without reference to the guide chart. Rate of data entry compared favorably with the standard keyboard: after less than five hours of practice, free text messages were entered at 60 percent of each trainee's standard keyboard rate. Yet, formatted messages were transmitted at a rate equal to, or exceeding, the rate achieved when using a standard keyboard.

Goodwin, N.C. (1975). *Cursor Positioning on an Electronic Display Using Lightpen, Lightgun, or Keyboard for Three Basic Tasks*. Human Factors, 17, 289–295.

In an experiment, six persons tried to position a cursor in an arbitrary location, to position the cursor in sequential operations, and to do check readings. They used a (“poorly designed”) keyboard whose keys are tapped to move the cursor, or a common light pen, or a pistol-shaped light gun. Each task was repeated ten times. The positioning task via keyboard took longest. Yet, the author states that the poor performance with the keyboard does not mean that cursor positioning keys should be eliminated. Rather, the results confirm that a light pen or light gun can be a useful addition.

Ayoub, M.M., Martz, H.F. and Wu, C.H. (1975). *Measurement of Muscle Fatigue Using Electromyography*. In Proceedings, 19th Annual Conference of the Human Factors Society (pp. 403–407). Santa Monica, CA: Human Factors Society.

A large number of reports is said to exist in the literature since 1923 regarding use of muscle action potentials to measure local muscle fatigue. A study was done to investigate the signal characteristics of muscle action potential as an indicator of fatigue of that muscle. Both dynamic and static muscular activities were investigated. Eight subjects participated in both the static and dynamic tests. The results indicated that changes in the low frequencies of the EMG can be used as a measure for muscle fatigue.

Hanes, L.F. (1975). *Human Factors in International Keyboard Arrangement*. Chapter 12 in A. Chapanis (Ed.), Ethnic

Variables in Human Factors Engineering (pp. 189–206). Baltimore, MD: Johns Hopkins University Press.

In this chapter, the author presents a number of guidelines for arranging the keys on a keyboard. These include:

1. Arrange the keys according to the frequency of their occurrence and to operator characteristics.
2. Follow established standards.
3. Group frequently used keys in the touch area. (The touch-operated area is described as the normal “home” or resting hand position, where keys may be easily operated with minimum fatigue and strain.)
4. Group common functions together.
5. Group keys logically and according to sequence of use.

Einbinder, H. (1975). *Input Keyboards*. Patent 3929216. United States Patent Office.

This invention concerns a keyboard in which the most common vowels are placed directly under the fingers of the left hand, the five most common consonants directly under the fingers of the right hand. The overall design is to minimize finger motions and to maximize entry rates. The keyboards are split into halves and contain curved key rows and slanted key tops at variable height to follow the anthropometry of the hand.

Engel, S.E. and Granda, R.E. (1975). *Guidelines for Man/Display Interfaces* (IBM Technical Report TR 00. 2720). IBM Poughkeepsie Laboratory.

The following controls are covered in the guidelines: light pen, selector pen, joystick, bowling ball, mouse, thumb wheel, stylus, and keyboards. Each of these is discussed with respect to their advantages and disadvantages, and it is concluded that, from an overall human factors viewpoint, the joy stick appears to be the best (p. 29).

Birkbeck, M.Q. and Beer, T.C. (1975). *Occupation in Relation to the Carpal Tunnel Syndrome*. Rheumatology and Rehabilitation, 14, 218–221.

The authors refer to earlier publications, starting with Brouwer in 1920, who stressed that “There is little doubt that occupation is a causal factor” in compression of the median nerve in the carpal tunnel.

They then report on a survey of the work and hobbies of 658 patients with carpal tunnel syndrome. Out of 516 employed patients, 401 (78%) were required to use “light, highly repetitive wrist and finger motions”: 71 patients were boot- and shoemakers, 44 were cashiers/secretaries, and 39 were clothing machinists (page 219). Out of 72 housewife patients “with no other work”, 37 (51%) reported knitting as their main hobby. The authors conclude (on page 220) “...that under some conditions the CTS [carpal tunnel syndrome] should be considered an occupational disease. These conditions appear to be fulfilled most readily by work or hobbies involving light, highly repetitive movement of the fingers and wrists.”

Clare, C.R. (1976). *Human Factors: A Most Important Ingredient in Keyboard Designs*. EDN Electronic Design News, 21(8), 99–102.

The author shows two key force-displacement graphs, one representing a “bad” design and the other a “good” design. The first is illustrated as one having linearly increasing resistance over the large portion of the key displacement, ending in a very large resistance when the key is pushed fully. The “good” example is one where key resistance lightly increases during “pre-travel”, then abruptly increases to a higher value from

which it increases again only slowly during "actuation", and then reduces again after the actuation has taken place. If that displacement is overshoot, then the resistance progressively increases until it stops the action. This procedure generates, according to Clare, a feel of actuation. The author then discusses preferred actuation directions and indicates that they should be tangential to the curve described by the fingertip as the finger moves from forward-extended to flexed. The author also recommends that the keyboard be so designed as to provide resting areas for the hand.

Wood, J. (1976). *The Application of Ergonomics in the Design of A Computer Console for Emergency Services: A Case Study*. In Proceedings, 20th Annual Meeting of the Human Factors Society (pp. 255-259). Santa Monica, CA: The Human Factors Society.

This is a detailed description of the ergonomic procedure used in the design and development of a series of eight consoles for the central control room of a police department. The needs of the operator were researched first. This concerned especially the location of controls that must be easily reached and displays which must be in the field of view. For this, an activity breakdown was done of the sequences of actions. Accordingly, six alternative configurations of the workstations were generated. These were evaluated by the design team and the two final candidates were then tested by police controllers who, sitting at the mock-up consoles, enacted various operational scenarios. Based on these procedures, one prototype was developed which was tested for its technical feasibility as well as for acceptance by the operators. Based on the prototype trials, modifications were made and incorporated into the production units.

Davies, B.T. and Pratt, M.C. (1976). *The Effect of Static Muscle Work on a Subsequent Fine Manipulative Task*. In Proceedings, 19th Annual Meeting of the Human Factors Society (pp. 531-533). Santa Monica, CA: Human Factors Society.

This report describes an experiment to assess the effect of static muscle contraction upon a subsequent fine manipulative task. The maximal grip strength of six subjects was established. Then the subjects were asked to maintain 15 percent of that force for either three minutes or for 90 percent of their maximal holding time. The effect of these two exertions on a following fine manipulative task (Purdue Pegboard) was investigated. It was found that performance on the manipulative task was reduced immediately after the static muscle contraction, but not after five or 30 minutes.

Ferguson, D. and Duncan, J. (1976). *A Trial of Physiotherapy for Symptoms in Keyboard Operating*. The Australian Journal of Physiotherapy, 22, 62-72.

The authors incorporated in their experimental treatment 29 of those telegraphists on which they had reported in 1974 (see listing above). All these persons had symptoms of muscular dysfunctions (occupational cramp, myalgia). The treatment consisted of three different kinds of exercises: active exercises to affect hand and forearm, together with ultrasonic radiation; a form of resisted exercises of the whole affected upper limb, preceded by ice pack application to hand and forearm; and massage of the affected hand. All three types of physiotherapy resulted in improvements of the operators' tested performance, i.e., in grip strength, tapping speed, speed, and accuracy of operating. The authors state that certain operating postures are associated with symptoms.

Posch, J.L. and Marcotte, D.R. (1976). *Carpal Tunnel Syndrome - An Analysis of 1201 Cases*. Orthopaedic Review, 5, 24-35.

The authors acknowledge the contributions of Dr. Phalen in the 1940s and 1950s, but indicate that carpal tunnel syndrome was discussed since 1865 repeatedly in the literature. Their findings are derived from examining 1,201 hands. They found that, of these cases, 36 percent were related to work. They argue that carpal tunnel syndrome can now be readily diagnosed and treated after the experiences of the last 30 years.

Einbinder, H. (1976). *Orthogonal Input Keyboards*. Patent 3945482. United States Patent Office.

Similar to the inventor's 1975 invention (see the listing above) this patent employs two separate keyboards with the keys arranged to maximize entry rates and minimize finger motions.

Tichauer, E.R. (1976). *Biomechanics Sustains Occupational Safety and Health*. Industrial Engineering, 2, 46-56.

The author describes a biomechanical model that can be constructed to establish a cause-and-effect relationship between the muscular effort involved in wrist extension and the pattern and quality of the motion produced. Using electromyography, changes related to tenosynovitis can be observed with ease. Fatigue changes the recorded profile.

Van Nes, F.L. (1976). *Analysis of Keying Errors*. Ergonomics, 19, 165-174.

The author analyzed keying errors and found that about 25 percent were due to the operator's misinterpretation of input data. At least 40 percent of the keying errors could be traced to underlying errors in finger movement control. About 20 percent of the errors could have been caused by hitting a key adjacent to the correct key.

Hadler, N.M. (1977). *Industrial Rheumatology - Clinical Investigations Into the Influence of the Pattern of Usage on the Pattern of Regional Musculoskeletal Disease*. Arthritis and Rheumatism, 20(4), 1019-1025.

In discussing carpal tunnel syndrome (as one of degenerative joint diseases) the author states that the vast majority of cases is not caused by underlying systematic illness. He conveys the thought that according to some observers, but not all, this syndrome is precipitated if not caused by stereotyped usage of the hand. He states, generally, that "there is little formal proof that pattern of usage, aside from overt trauma, causes or precipitates any regional musculoskeletal disease" (p. 1023).

LeCocq, A.D. (1977). *The Design of a CRT Computer Terminal - A Human Factors Dream or Nightmare?* In Proceedings of the 21st Annual Meeting of the Human Factors Society (pp. 164-166). Santa Monica, CA: Human Factors Society.

The evaluation and development of a computer CRT (Cathode Ray Terminal) design is described. The first step is the determination of the operating needs of the users. Basically, the authors consider the need for operating "easily, efficiently, effectively" (p. 164). For this, the users of terminals were interviewed with respect to the particular task they performed. Then, a videotape was made that displayed a mock-up of the proposed terminal and some of the operational procedures. Again, comments and suggestions were obtained from the operators. The use of this information established an overview of the real operating needs of the future users of the terminal.

Next, operators were given mock-ups of the keyboard and monitor. The operators simulated actual operation and rated the relative ease or difficulty to perform tasks. Furthermore, they were encouraged to make comments. This information was used to improve the design of software and hardware.

Bequaert, F.C. and Rochester, N. (1977). Teaching Typing on a Chord Keyboard (Technical Report 00.2918). Cambridge, MA: SPD Cambridge Systems Group (IBM Poughkeepsie Laboratory).

This report describes an automated system for the teaching touch-typing on a one-hand chord keyboard. To teach the students to use the keyboard, lines of text are displayed on a screen. The student is assigned the task to type the displayed line, and the characters that are typed appear directly below the displayed text. Upon completion of the line, the newly typed text is compared with the original and errors made are displayed in error codes beneath the typed line. Five subjects were employed, four college students and one IBM staff member. Their conventional typing speed was between 37 and 75 words per minute. Overall, the results indicate that the subjects were able to learn to use the system in a touch-typing mode as rapidly as they could learn to use a conventional typewriter keyboard in a classroom situation.

Rodbard, S. and Weiss, B. (1977). Performance and Discomfort During Ischaemic Accustomed Exercise (Typing). *Ergonomics*, 20, 535–541

Experiments performed with 50 female typists, 17 to 65 years of age, are described. A tourniquet was applied to one upper arm and blood flow cut off. The subjects typed nevertheless as long as possible. When the typing was terminated, the pressure cuff was removed and the subject was asked to describe the sensations felt in the arm. It was found that typing was normal at the onset of the test, but within two or three minutes, discomfort became evident, and the subjects began to lose voluntary control over their fingers. The total duration of typing endured ranged from 35 seconds to seven minutes, with most of the subjects being able to type between 1.5 and 4.5 minutes. The authors then discuss the effects of insufficient blood supply to working muscles, particularly to the intrinsic muscles of the hand that move the fingers in typing.

Roth, J.T., Ayoub, M.M. and Halcomb, C.G. (1977). Seating, Console and Work Place Design: Seated Operator Reach Profiles. In *Proceedings of the 21st Annual Meeting of the Human Factors Society* (pp. 83–87). Santa Monica, CA: Human Factors Society.

The authors describe a procedure to determine the reach envelopes of restrained and unrestrained, male and female seated operators. These data are to be used for the design of consoles.

Dorris, A.L. and Purswell, J.L. (1978). Human Factors in the Design of Effective Product Warnings. In *Proceedings of the 22nd Annual Meeting of the Human Factors Society* (pp. 343–346). Santa Monica, CA: Human Factors Society.

The authors discuss the need to provide warnings for product use to establish safety for the user and to fulfill the legal duty to warn on the part of the manufacturers.

Card, S.K., English, W.K. and Burr, B.J. (1978). Evaluation of Mouse, Rate-Controlled Isometric Joystick, Step Keys, and Text Keys for Text Selection on a CRT. *Ergonomics*, 21, 601–613.

The authors report on the evaluation of several input devices to select text shown on a CRT. They found that use of the mouse is by far the fastest. Use of keys is slower, depending on the number of key strokes that must be typed.

Rochester, N., Bequaert, F.C. and Sharp, E.M. (1978a). Chord Keyboard Architecture (Technical Report 00.2934-1). Cambridge, MA: IBM Cambridge Systems Group.

This report provides a description of the physical dimensions, logical operations, and user performance on a compact chord keyboard. The chord keyboard is operated with one hand and provides a larger character set than a conventional keyboard. In essence, the keyboard consists of adjacent keys which, when depressed singly, generate a designated character. However, if neighboring keys are depressed simultaneously, the combined key activation generates a new designated character. Where four keys meet in a “common corner”, the fingertip may overlap the adjacent corners of all four keys. Thus, depression of the fingertip activates four keys, and generates a new letter, for example V. The authors conclude that, given the design features of the keyboard and their experiments with subjects, high speed typing is possible with this keyboard.

Rochester, N., Bequaert, F.C. and Sharp, E.M. (1978b). The Chord Keyboard. *Computer*, December Issue, 57–63.

This report describes experiments performed with the keyboard described previously in the authors' technical report, 1978a, of the IBM Corporation (listed above). Four students participated, reaching high typing speeds in apparently rather short periods of time. The authors caution that a powerful automated teaching system was used and that the students were of above average intelligence and highly motivated. The keyboard is said to have advantages over a conventional keyboard, such as one-handed operation, small size, low cost, and a facility for touch-typing large alphabets, such as Japanese or Arabic.

Whitaker, R.O. (1978). Combinational Keyboard with Moveable Keys and Adapted for One-Handed Keying of Numerical Information. Patent 4067431. United States Patent Office.

This patent concerns the moveable location of keys for a keyboard that, with ten keys arranged in a natural pattern to follow the location of the fingertips, can be re-arranged to accommodate the hand configurations of various operators.

Zapp, W. (1978). Keyboard Actuable with the Aid of the Fingers of at Least One Hand. Patent 4081068. United States Patent Office.

This patent concerns a keyboard on which the keys can be moved out of an initial position by the tips of the fingers. Furthermore, the keyboard is equipped with a rest device on which the pad of the thumb or the wrist can be continuously supported during key actuation. Furthermore, the keys are arranged on a curve corresponding to the natural position of the fingertips. Finally, each key has a rest position on which the tip of the finger is able to rest without actuating the key.

Rothfleisch, S. and Sherman, D. (1978). Carpal Tunnel Syndrome-Biomechanical Aspects of Occupational Occurrence and Implications Regarding Surgical Management. *Orthopaedic Review*, 7, 107–109.

The authors report on a study with 16 workers in an automotive assembly plant, with a total of 25 hands showing CTS symptoms. The authors concluded that wrist position and vibration are etiologic in the development of CTS in industrial workers. From this, they derive that the workplace conditions should be changed to reduce the risk of developing CTS. These include: avoiding vibration generated in tools, re-designing tools to provide a biomechanical proper hand grip, decreasing exposure time by job rotation, and increasing awareness and use of ergonomic and biomechanical techniques.

Tichauer, E.R. and Gage, H. (1978). Ergonomic Principles Basic to Hand Tool Design. (Ergonomics Guides). Akron, OH: American Industrial Hygiene Association.

This ergonomic guide provides a survey of features of hand tools relevant to the physical interaction with the human operator. Concepts regarding the optimization of forces are developed, followed by a description of some common physiological problems in musculoskeletal complaints associated with improper hand tool design and usage. Among the most frequently observed complaints from overuse of the hand, tenosynovitis and carpal tunnel syndrome are mentioned. Trauma are said to be often induced through repetitive maneuvers and cumulative work stress rather than by one-time overexertion (page 2). Also, ganglia, trigger finger, and epicondylitis are discussed.

Armstrong, T.J. and Chaffin, D.B. (1978). An Investigation of the Relationship between Displacements of the Finger and Wrist Joints and the Extrinsic Finger Flexor Tendons. Journal Biomechanics, 11, 119–128.

This article describes several biomechanical models of bony structures of the hand, its joints, and the tendons. Experiments with cadaver hands showed that Landsmeer's Model I best described the interdigit tendon-joint mechanics, especially in regards to tendon displacements.

Cakir, A., Hart, D.J. and Stewart, T.F.M. (1979). Visual Display Terminals (In German). Darmstadt, Germany: Inca-Fiej.

The original German version of the 1980 English translation, listed in Sect. 5.

Armstrong, T.J. and Chaffin, D.B. (1979a). Carpal Tunnel Syndrome and Selected Personal Attributes. Journal of Occupational Medicine, 21, 481–486.

“Why do some persons develop occupational carpal tunnel syndrome while others do not?” One explanation may be that hand and wrist size are a factor. The other possibility is that some persons use work methods that are more stressful to the wrist tissues than others. To test these hypotheses, two groups of female workers were studied. The results indicated no effects of hand or wrist size, but use of forceful exertions and deviated wrists and pinch hand positions, particularly during forceful exertions, were found to be associated with carpal tunnel syndrome.

Armstrong, T.J. and Chaffin, D.B. (1979b). Some Biomechanical Aspects of the Carpal Tunnel. Journal Biomechanics, 12, 567–570.

The authors start by stating that 12 cited publications “have suggested that carpal tunnel syndrome can be caused, precipitated or aggravated by exertions with certain hand and wrist positions. ... There is general agreement among investigators

that the incidence of carpal tunnel syndrome is related to forces on the median nerve inside the carpal tunnel. The purpose of this paper is to present biomechanical argument that show quantitatively how forces inside flexed and extended wrists are related wrist size, hand force, and hand position.” (p. 567). The authors then indicate, both by anatomical and mechanical reasoning, that indeed hand and wrist positions, as well as wrist sizes, partly explain occurrences of carpal tunnel syndrome in male and female populations.

Gopher, D. and Eilam, Z. (1979). Development of the Letter-Shape Keyboard: A New Approach to the Design of Data Entry Devices. In Proceedings of the 23rd Annual Meeting of the Human Factors Society (pp. 40–44). Santa Monica, CA: Human Factors Society.

The authors describe the development of a chord keyboard, to be operated with one hand, in which Hebrew characters are typed by two successive strokes, with every stroke a specific combination of three keys. These three keys are struck in such a way that they represent points on a three times two matrix that, when connected, resemble the Hebrew character. The experiments performed with prototypes and relatively few subjects encouraged the authors, as they say, to continue research with the letter-shape keyboard.

Kuorinka, I. and Koskinen, P. (1979). Occupational Rheumatic Diseases and Upper Limb Strain in Manual Jobs in a Light Mechanical Industry. Scandinavian Journal of Work Environment and Health, 5 (Supplement 3), 39–47.

Ninety-three workers, almost all women, involved in scissor-making, were examined by a physiotherapist. The recorded results were later diagnosed against a set of predetermined criteria. Also, the jobs performed by the workers were analyzed. The work history of the previous year was collected and the actual work methods for each workstation and product were analyzed. Recorded symptoms were found to be concentrated in two anatomic regions, the neck and the wrist. No relation between syndromes and age of the workers was found. No inflammatory diseases were present. The only clear relation found was that the number of symptoms increased as the number of work pieces handled per year increased.

Luopajarvi, T., Kuorinka, I., Virolainen, M., and Holmberg, M. (1979). Prevalence of Tenosynovitis and Other Injuries of the Upper Extremities in Repetitive Work. Scandinavian Journal of Work Environment and Health, 5 (Supplement 3), 48–55.

The authors investigated the prevalence of tenosynovitis and other repetitive injuries of the upper extremities in two groups of workers. One group of assembly line packers in a food production factory consisted of 152 women with a mean age of 39 years and an average of six years of employment on the assembly line of. The comparison sample of shop assistants in a department store did not include cashiers. It consisted of 133 women, also with mean age of 39 years, and an average time as shop assistant of 13 years.

While many symptoms were similar in the two groups, tenosynovitis and other neck and shoulder disorders were significantly higher for the assembly workers. Their work consisted of static muscle tension in the arms, grasping and maximal extensions of the fingers, and lateral deviations of the wrist. These movements numbered up to 25 000 per day. The authors therefore assume that a relationship between workload factors and disorders of the upper extremities is probable.

Waris, P., Kuorinka, I., Kurppa, K., Luopajarvi, T., Virolainen, M., Pesonen, K., Nummi, J. and Kukkonen, R. (1979). *Epidemiologic Screening of Occupational Neck and Upper Limb Disorders – Methods and Criteria*. Scandinavian Journal of Work Environment and Health, 5 (Supplement 3), 25–38.

The authors describe a method to screen for occupational neck and upper limb disorders, most of which include CTDs. The examination consists of functional tests, measurements, and an interview. With the exception of grip force, all test results were labeled as either normal or abnormal. The recorded results were analyzed later and the number and type of symptoms that were either normal or abnormal were checked and compared to a predetermined set of criteria to ascertain the presence of a disorder. This separation of examination and diagnosis is deemed necessary to avoid a possible subjective confounding component in the clinical examination.

Around 1970, the proper design of the “computer keyboard” had become a hotly debated topic among ergonomists/human-factor engineers. The debate concerned the layout of the set of keys (the “keyboard”) in general, as well as the specific design of the keys, and their use. The discussion concerning the design of the keyboard addressed, among other issues, the size and spatial orientation of the board(s) and the geometric layout of the keys on them, including the arrangement of keys in use patterns. Figure 4 explains some of the technical terms that describe design features of the keyboard and its keys. Key displacement (travel) and resistance (force) were of particular interest.

Assessments of the effects of occupational or experimental conditions on keyboarders included fairly so-

phisticated procedures that went beyond the traditional recordings of keying performance, usually in terms of throughput and errors. The measurement routines now incorporated the biomechanical and physiological strain of the operator, such as EMGs (electromyograms), and psychological judgements (Galitz and Laska 1970, Klemmer 1971, Tichauer 1976, Armstrong and Chaffin 1978, LeCocq 1977).

Reports from Japan indicated that key operators suffered from overexertion injuries (Komoike and Horiguchi 1971, Maeda 1974, Nishiyama 1987). Concurrently, considerable medical research on, and experience with, cumulative trauma disorders (CTDs; *see the discussion of terminology below*), and especially the carpal tunnel syndrome, was going on (Phalen 1972, Birkbeck and Beer 1975, Posch and Marcotte 1976, Rodbard and Weiss 1977, Rothfleisch and Sherman 1978, Armstrong and Chaffin 1978/9).

Ergonomists were concerned that the (enlarged) QWERTY keyboard of the typewriter was becoming the de facto standard layout for communications and computer interface keyboards in spite of its human engineering shortcomings (Phillips and Kincaid 1971, Kroemer 1972, Seibel 1972, Hanes 1975). The ergonomic guidelines by Cakir, Hart, and Stewart, first published in 1979 in German, then immediately translated into English (1980), caused a great impetus toward human factors considerations for keyboarding.

To summarize the development during the 1970s decade, the concern about musculoskeletal overexertions in relation to the repetitive work of keyboarding becomes both a medical and an ergonomic problem. The new electric/electronic nature of key and keyboard permits and encourages the proposal of a variety of more ergonomic designs.

5 From 1980 to 1989

Annotated list of publications on keyboards and keyboarding from 1980 to 1989

Huening, H., Grandjean, E. and Maeda, K. (1980). *Constrained Postures in Accounting Machine Operators*. Applied Ergonomics, 11, 145–149.

The authors observed operators of accounting machines and found that they were frequently affected by impairments in hand and arms, and that the disorders appeared related to working posture and the operation of keys.

Maeda, K., Huening, W. and Grandjean, E. (1980). *Localized Fatigue in Accounting-Machine Operators*. Journal of Occupational Medicine, 22, 810–816.

The authors first list a series of publications, starting in 1959 and running through 1974, in which health problems in business machine operators related to repetitive exertions and posture have been reported to occur in many different countries, including Russia, Australia, Japan, and Europe. A survey was performed on 119 accounting-machine operators and 57 sales women. Only young persons (average about 21 years)

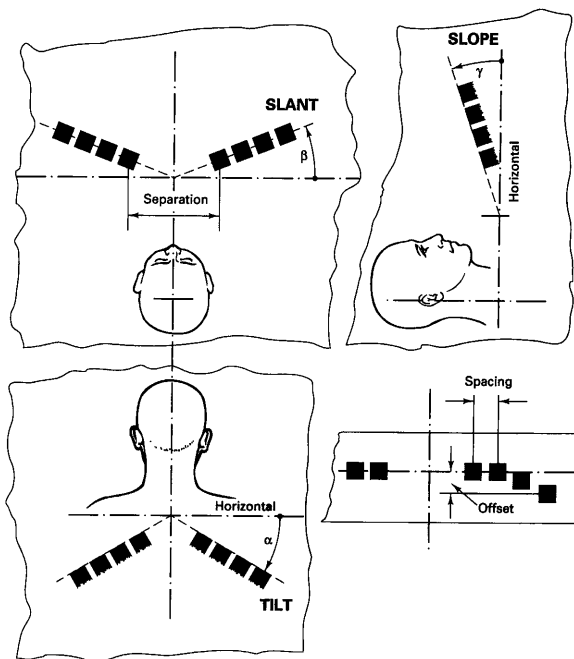


Fig. 4. Terminology to describe design features of the keyboard and its keys (with permission from [4], all rights reserved)

were included to remove from the data the effect of age related diseases in neck, shoulders, and low back. Each subject completed a questionnaire regarding symptoms, similar to that proposed by the Japanese Association of Industrial Health. Major differences in the reported symptoms were found between operators and clerks. In particular, tiredness, pain, stiffness, cramps, trauma, and numbness in the hand were important symptoms of operators, and are related to the repeated use of arms and hand in tapping the keyboard. Other symptoms in torso and neck were related to the sitting posture of operators. The authors recommend a number of ergonomic improvements of the operators' work places, particularly regarding the height arrangement of the keyboard, and support to the arms while tapping.

Cakir, A., Hart, D.J. and Stewart, T.F.M. (1980). Visual Display Terminals. Chichester, UK: Wiley.

This manual was originally published in German by the Inca-Fiej Research Association, Darmstadt, West Germany, in 1979. The manual covers, in considerable detail, ergonomic and health aspects of all components of VDT workstations, including keyboards and other input devices, displays, work place design, and task organization. For key force, 0.25 to 1.5 N is recommended (p. 129). A snap-action (force-drop) after key actuation is recommended for feedback to the operator to indicate that the keystroke has successfully activated the system. Such feedback is claimed to be an "extremely important keyboard characteristic" (p. 129).

Laeubli, T., Nakaseko, M. and Huening, W. (1980). Work-Related Cervicobrachial Complaints During Office Work (in German). *Sozial- und Praeventivmedizin, Medecine sociale et preventive*, 25, 407-412.

The authors report that many activities using modern office machines are associated with enforced postures in the neck, shoulder, and arms. Related complaints appear similar to the cervicobrachial syndrome previously reported in Japan. This occupational disease is frequent with various activities on keyboards. The authors investigated a Swiss sample consisting of nearly 300 persons working with data entry terminals. Questionnaires were used and medical examinations were performed. The results indicated that in Switzerland, similarly to Japan, cervicobrachial complaints are frequent with the work at keyboards.

Zapp, W. (1980). Keyboard Actuatable with the Aid of the Fingers of at Least One Hand. U. S. Patent 4201489. United States Patent Office.

This patent describes a key that can be moved out of its initial inoperative position by being tilted with the tip of the finger. This tilting action, either forth or back, or left or right, results in an actuation signal. However, since no actuation occurs when the key is simply pressed downward, the finger can rest on the key. This is expected to be reducing the strain of muscles.

Cannon, L.J., Bernacki, E.J. and Walter, S.D. (1981). Personal and Occupational Factors Associated with Carpal Tunnel Syndrome. *Journal of Occupational Medicine*, 23, 255-258.

The authors describe a controlled study in which they compared 30 patients with carpal tunnel syndrome and 90 matched controls. The results indicate that job-related events, particularly the use of vibrating hand tools, as well as gynecological surgery are strongly associated with the onset of carpal tunnel syndrome.

Grandjean, E., Nakaseko, M., Huening, W. and Laeubli, T. (1981). Ergonomic Studies to Develop a New Keyboard for Office Machines (in German). *Z. Arbeitswissenschaft*, 35, 221-226.

The authors report on three experiments with a split keyboard. The keyboard halves could be arranged in the conventional mode with both halves on the same plane, or separated. When separated, the key sets could be slanted (in the top view) with angles of 15, 25, and 35 degrees between the halves; and it could be tilted sideways-downward at ten degrees each against horizontal. In all cases, the slope was kept at eight degrees. Twenty-five female experienced typists were employed, who, using each keyboard, copied a text for 45 minutes. Their body posture was recorded. The subjects adjusted the heights of seat and keyboards according to their preference. The split keyboards with a slant of 25 degrees were clearly preferred. Most subjects also preferred the use of a wrist rest. Under these conditions, operators leaned back in their chairs and elevated their arms; they also assumed a "more natural" hand posture, particularly shown in a reduction of lateral deviation of the wrist.

Fraser, T.M. (1981). Tenosynovitis of Occupational Origin - A Review with Recommendations. (Report dated 15 July 1981 for NCR Canada Ltd.) Waterloo, Ontario: Safetywork Inc. Centre for Occupational Health and Safety, University of Waterloo.

Tenosynovitis (or shoulder-arm syndrome) results from excessive and repeated motions of the tendons in conditions in which the joints involved are excessively angled, for example when the hand is deviated to the right or left, or angled backwards or forwards at the wrist. Specifically, among office workers, the problem occurs in typists, VDT operators, accounting machine operators, and other keyboard users. Common in the days of mechanical typewriters, when the need for key force was greater, this disease can occur readily also with electrical keyboards, in which the required force is less. This fact suggests that joint angulation and posture is at least as important as the force required (p. 4). Excessive angulation occurs, at least in part, because of a design and layout of keyboard, equipment, desk or table, and seating that fails to take into account the limitations of human anthropometry (p. 5). Excessive backward angulation of the wrist arises because of an incompatibility between the height of the top of the keyboard and the operator's seat. For an ideal working position, the forearms should be horizontal or with a slight downward slope. A smooth rest area should be available for the butt of the hand, so that support is provided with minimal interference in access to the keyboard. The rest area should be large enough so that the hand does not rest on an edge (p. 6). The other primary factor involved in the causation of tenosynovitis is "overusage", which is mostly due to keying force and travel, and keying frequency (p. 7).

IBM (1981). Ergonomics and Visual Display Terminals. A Guide for Managers. No location given. IBM Corporate Health & Safety Dept.

This guide discusses workplace design, lighting, seating, visual fatigue, eyeglasses and contact lenses, radiation, work/rest cycles, work design/organization, and training.

IBM (1981?). Ergonomics Handbook. Armonk, NY: International Business Machines Corporation. (The publication date is not given in this brochure, but the most recent literature cited therein is of 1981.)

This handbook discusses the environment, work design and organization, as well as the back, the hand, anthropometrics, and posture of the operator. The resulting recommendations are specifically applied to a VDT workstation (pages 62ff).

Zipp, P., Haider, E., Halpern, N., Mainzer, J. and Rohmert, W. (1981). *Experiments on the Ergonomic Design of Keyboards* (in German). *Zbl. Arbeitsmedizin*, 31, 326–330.

This is a report on experiments performed on various keyboards. In particular, EMGs (electromyograms) are said to provide the criteria for judging the muscular effort involved. EMGs provide information about minimal muscular efforts to achieve optimal joint angles, in which the static muscular effort is minimized. Based on EMG considerations, it is suggested that the keyboard be split into two symmetrical halves, which are tilted sideways and slanted in the top view. These recommendations are argued to be to be similar to those by Klockenberg (1924) and Kroemer (1964).

Litterick, J. (1981). *QWERTYUIOP – Dinosaur in a Computer Age*. *New Scientist*, 8 January Issue, 66–68.

This article starts with the statement that “today’s keyboard is ideal for people with arms coming out of their chest, and fingers all the same length”. It then relays the known shortcomings of the keyboards, mostly with respect to the layout of the keys, and their large number, which causes stressful activities and postures. The author explains the original layout, based on the 1887 design by Sholes, and then indicates a number of proposed improvements, such as by Dvorak, Kroemer, Maltron, the British Post Office, as well as the Antel (Stewart) technique, which is similar to the one developed by IBM (Rochester and Bequaert, 1978a,b – listed in Table 3).

Smith, M.J., Cohen, B.G.F. and Stammerjohn, L.W. (1981). *An Investigation of Health Complaints and Job Stress in Video Display Operations*. *Human Factors*, 23, 387–400.

A questionnaire survey dealing with working conditions, job stress factors, health complaints, and psychological mood state was filled out by about 250 VDT operators and 150 non-operator control subjects. While little difference in psychological mood states were found between the two groups of subjects, VDT operators reported higher levels of job stress and health complaints. The health complaints dealt mostly with visual, musculoskeletal, and emotional problems. Of the musculoskeletal problems, most were muscular in nature.

Stammerjohn, L.W., Smith, M.J. and Cohen, B.G.F. (1981). *Evaluation of Workstation Design Factors in VDT Operations*. *Human Factors*, 23, 401–412.

An ergonomic evaluation was conducted at VDT workstations at five work sites in the San Francisco area. Three issues were addressed, namely illumination and luminance, dimensions of the workstation, and workstation features of special interests. Also, a questionnaire was distributed to a sample of the employees at each work site. In addition to recommendations regarding illumination, glare, and other visual aspects, it is recommended, as a result of the investigation, that workstation dimensions including keyboard and screen height, knee angle, and chair dimensions and characteristics, be specifically assessed in the future.

Miller, I. and Suther, T.W. (1981). *Preferred Height and Angle Settings of CRT and Keyboard for a Display Station Input Task*. In *Proceedings of the 25th Annual Meeting of the Human Factors Society* (pp. 492–496). Santa Monica, CA: Human Factors Society.

Thirty-seven operators were taken individually through a step-by-step sequence to determine their preferred adjustments of seat height, keyboard height, keyboard slope angle, CRT height, and CRT tilt angle. The subjects adjusted seat height and backrest according to their individual preferences, and then sat at a comfortable distance from a VDT workstation. The height for the keyboard support was adjusted according to the individuals’ instructions. Likewise, keyboard slope angle, CRT height, and CRT angle were all adjusted in the same manner. Then the subject was given a page of text to transcribe, and urged to make additional adjustments in the workstation arrangements. After the trial was completed, measurements of the preferred furniture settings were recorded. In analyzing the recorded settings, a negative but high correlation between keyboard angle and seat height was found. The preferred angles range from 14 to 25 degrees in slope. The correlation between slope and operator stature was also negative. The researchers conclude that adjustability of keyboard slope is essential to accommodate individual preferences. Furthermore, the keyboard should be of minimal thickness.

Tynan, P.D. (1981). *A Method of Evaluating the Ergonomic Qualities of Computer Workstations*. In *Proceedings of the 25th Annual Meeting of the Human Factors Society* (pp. 497–499). Santa Monica, CA: Human Factors Society.

The author describes a procedure to evaluate ergonomic aspects of computer workstations that consist of five steps:

1. Build a full-scale mock-up of the workstation and models of the equipment.
2. Arrange the workstation in a manner that seems ergonomically sound.
3. Evaluate the ergonomic quality of the workstation with a paper-and-pencil test.
4. Repeat steps 2 and 3 modifying the configuration to better test results.
5. Based on the final evaluation, decide whether new models should be built and the process repeated.

The paper-and-pencil test, mentioned above, consisted of rules for good ergonomic design collected from human factors books, past research at NCR, and from proposed European guidelines and regulations, specifically as published by Cakir, Hart, and Stewart (1979, listed in Table 3, and 1980, listed above). Each such rule was transformed into an item on the test. The author states that the procedure was found to be simple to use and take relatively little time. It provided a good evaluation of the proposed design, and allowed easy evaluation of design trade-offs.

Dainoff, M.J. (1981). *Ergonomic and Health Issues Related to Work at VDTs*. In *Proceedings of the 25th Annual Meeting of the Human Factors Society* (p. 316). Santa Monica, CA: Human Factors Society.

This is an overview and introduction to a forum held with six panel members, two from Europe and four from the U.S.A., on major health and ergonomic issues associated with work and video display terminals.

Happ, A.J. and Beaver, C.W. (1981). *Effects of Work at a VDT-Intensive Laboratory Tasks on Performance, Mood, and Fatigue Symptoms*. In *Proceedings of the 25th Annual Meeting of the Human Factors Society* (pp. 142–144). Santa Monica, CA: Human Factors Society.

In an experiment with ten subjects unpaced video-coding at low load was performed for two hours. Subsequently, the subjects completed a checklist regarding fatigue, vigor, and tension. The preferred angles range from 14 to 25 degrees in slope. The correlation between slope and operator stature was negative, but at -.43 instead of -.71, as found between keyboard angle and seat height. The researchers conclude that adjustability of keyboard slope is essential to accommodate individual preferences.

Benz, C., Grob, R. and Haubner, P. (1981). Designing VDT Workplaces. (In German: Gestaltung von Bildschirm-Arbeitsplaetzen.) Cologne, Germany: TUV Rheinland. This is the original version upon which the 1983 English text is based (see listing below.)

Eikelberger, R.J. (1981). Selecting the Right Keyboard. Machine Design, November Issue, 1-8.

The author discusses keyboard features such as faceplate, keycap, key switches, printed circuit board, and related electronics. Key layouts can either be standard with offsets between key rows in regular or irregular fashion or may be custom-designed with fairly little cost. The author describes several switching techniques, such as capacitive, membrane, or mechanical contacts, Hall effect, conductive elastomeric, and saturable core. These are described in detail, including the specific statement that key switches with "snap action" for tactile feedback are available.

Gentner, D.R. (1981). Skilled Finger Movements in Typing. (Technical Report N00014-79-C-0323). La Jolla, CA: Center of Human Information Processing, University of California, San Diego, CA.

Ten skilled typists were studied while they transcribed English text. While all typists showed stable patterns of performance, there were significant individual differences among the typists. Inter-key press times for two-finger digraphs were particularly variable. The observed differences corresponded to differences in the independence of within-hand finger movements. The starting times of movements were six times as variable as the ending times, suggesting that it is the completion rather than initiation of the movement that is controlled in skilled typing.

Hirsch, R.S. (1981). Procedures of the Human Factors Center at San Jose. IBM Systems Journal, 20, 123-171.

This is a description of projects performed at the IBM Human Factors Center in San Jose, CA. Many of these studies concerned keyboard operation. In the summary, the author states that the examples were to show how the IBM Human Factors Center goes about to evaluate products to determine that the man-machine interfaces are optimized. For this, the task must be evaluated, the application environment simulated, the data collected and analyzed. As a subsequent step, the gathered information has to be incorporated into design recommendations.

Malt, L.G. (1981). Keyboard Arrangement. Patent No. 4244659. United States Patent Office.

This patent concerns a keyboard in which the keys are arranged to take into account the frequency of occurrence of letters so that the keys that occur most frequently are positioned for easiest operation. Keys are grouped for the left and right hand, forming two separate keyboards. The keys are arranged on a concave surface.

Simonelli, N.M. (1981). Interdigit Keying Intervals on a Membrane Keyboard. In Proceedings of the 25th Annual Meeting of the Human Factors Society (pp. 333-337). Santa Monica, CA: Human Factors Society.

The issue discussed in this paper is how long would so-called "casual users" take to enter data. Fifty-three persons performed a keying task on a flat membrane keyboard. The results showed that there was no effect of gender while there was a small but significant effect of age, and that those people trying to input the fastest might take the longest because they may miss a key. Use of bifocals and "hunting for zero" as well as unfamiliarity with membrane keyboards also tended to increase keying times. Furthermore, the arrangement of the keys on the keypad influenced the keying time.

Grandjean, E. (1982). Ergonomics of VDUs: Review of Present Knowledge. In E. Grandjean and E. Vigliani (Eds.), *Ergonomic Aspects of Visual Display Terminals* (pp. 1-12). London, UK: Taylor & Francis.

VDUs, as a new tool in many work places, show ergonomic problems in terms of work place design, development of technology, work organization, and attitudes of users. This paper addresses in particular the complaints of VDU operators concerning constrained postures. These are said to be due to the fact that the operators must keep their head as well as their hands in practically permanent and fixed positions. The visual distance to the screen or the source document determines the head position, and the position of the hands is determined by the location of the keyboard.

Huening, W., Laeubli, T. and Grandjean, E. (1982). Constrained Postures of VDU Operators. In E. Grandjean and E. Vigliani (Eds.), *Ergonomic Aspects of Visual Display Terminals* (pp. 175-184). London, UK: Taylor & Francis.

In a survey of VDU work places it was found that constrained postures may be associated with physical impairments in hands, arms, shoulders, and neck. The greater the keyboard height, the larger the number of complaints. The incidence rates are reduced if the operators can rest their hands and forearms.

McPhee, B. (1982). Deficiencies in the Ergonomic Design of Keyboard Work and Upper Limb and Neck Disorders in Operators. Journal Human Ergology, 11, 31-36.

The author indicates a quickly rising number of occupational repetition injuries in all states of Australia. The author lists 37 different diagnoses and descriptions. Because of the lack of standardized criteria for defining repetition injuries and the general misinterpretation these conditions and their causes, it appears that many sufferers do not receive appropriate treatment early enough. About ten years ago, most of the overuse syndromes were observed in manufacturing industry, but recently there have been a large number of cases occurring as the result of work on computer terminals with electronic keyboards.

The author lists six factors that, singly or in combination, appear to be the direct causes of repetition injuries:

1. The numbers of movements;
2. The load of force required;
3. The amount of static muscle work used;
4. Stressful postures of the body generally and/or particular joints;
5. The individual's physical capacity;
6. Length of exposure.

Also, there seems to be an increased risk when new demands are placed on the individual, when he or she habitually works beyond his or her capacity, and when personal, social or environmental factors reduce the individual's tolerance. The ergonomic approach to be taken is listed as task design, workplace design, and work organization.

Cohen, K.M. (1982). Membrane Keyboards and Human Performance. In Proceedings of the 26th Annual Meeting of the Human Factors Society (p. 424). Santa Monica, CA: Human Factors Society.

Twenty-one subjects used either a membrane keyboard or a conventional keyboard, each for three consecutive days. Each day, a one-hour session with various typing exercises was completed. Typing performance was found to be not different for non-touch typists, but different for touch typists. Initially, touch typists performed much better with the conventional keyboard. However, they rapidly learned to work with the membrane keyboard, and, within a few sessions, the difference in typing performance was substantially reduced.

Helander, M.G. (1982). Symposium on Human Factors Standards and Guidelines. In Proceedings of the 26th Annual Meeting of the Human Factors Society (p. 788). Santa Monica, CA: Human Factors Society.

This is an overview of the standards development activities in which the Human Factors Society is involved. These include the ISO Committee 159, Ergonomics, the DOD Human Factors Standardization Program, the SAE Human Factors Guidelines, the Nuclear Power Industry Regulatory Guidelines, and the Human Factors Guidelines for Visual Display Terminals.

Norman, D.A. and Fisher, D. (1982). Why Alphabetic Keyboards are Not Easy to Use: Keyboard Layout Doesn't Much Matter. Human Factors, 24, 509–519.

Studies were reviewed that demonstrate the inferiority of alphabetically organized keyboards as compared with either randomly keyboards or the standard QWERTY keyboard. Performance with some alphabetical layouts is quite slow, but with others it is within about two percent of the speed achieved when using the QWERTY keyboard. The Dvorak keyboard shows an improved performance of about five percent over the QWERTY keyboard. It is concluded that it is not worthwhile to use alphabetic keyboards for novice keyboarders, nor to change to the Dvorak layout for experts. The authors say that keyboards can probably be improved, but only by radical redesign of the present key configuration. According to the authors, there is little justification for the present size of the keyboard, for the staggered arrangements of keys, and for other aspects of the keyboard's physical arrangements. The left and right hands should have a mirror-image configuration of keys, and the keyboard halves should be separated. Chords may be used.

Nakaseko, M., Huening, W., Gierer, R. and Grandjean, E. (1982). Ergonomic Development of a New Keyboard Design. In Proceedings, 8th Congress of the International Ergonomics Association (pp. 176–177). Tokyo: Japan Ergonomics Research Society.

The authors report on experiments with models of split keyboards, compared with standard keyboards, in trials with 31 subjects. The subjects preferred the split keyboard ar-

rangement, particularly if tilted laterally ten degrees below horizontal, and slanted at 25 degrees in the top view. Operating such keyboards allowed the subjects to lean back the torso, to hold the arms higher, and to reduce ulnar wrist deviation. Fatigue symptoms were significantly improved.

Emmons, W.H. and Hirsch, R.S. (1982). Thirty Millimeter Keyboards: How Good Are They? In Proceedings of the 26th Annual Meeting of the Human Factors Society (pp. 425–429). Santa Monica, CA: Human Factors Society.

Twelve skilled typists were employed to work one day each on one of four keyboards. Three of the keyboards were similar to the IBM Selectric keyboard, while the fourth one was different. The IBM keyboards were arranged at slope angles of five, 12, and 18 degrees, and their heights at the home row were 30, 38, and 45 mm. The results indicated a higher input rate with higher keyboards. Four of the 12 subjects expressed feelings of discomfort with the less sloped keyboards, while no such feelings were expressed regarding the more sloped keyboards.

Dainoff, M. (1982). Occupational Stress Factors in Visual Display Terminal (VDT) Operation: A Review of Empirical Research. Behaviour and Information Technology, 1, 141–176.

The author reports on a review of the literature involving empirical research (i.e., experimental and field investigations) on stressful aspects of VDT operation. The studies concerned the assessment of visual fatigue and/or performance, musculoskeletal symptoms and operator attitudes towards job demand, and quality of working life. Some investigations also concerned the physical attributes of VDT work places, environmental factors, and psychosocial job factors. The literature reveals that levels of visual and musculoskeletal complaints among VDT operators are high. Field assessments of VDT work places indicate that the majority was sub-optimal with respect to lighting, glare, and dimensions of screen, keyboard, desk, and chair. Yet, evidence of causal linkages between specific ergonomic attributes and specific patterns of symptomatology is found to be lacking.

Suther, T.W. and McTyre, J.H. (1982). Effect on Operator Performance at Thin Profile Keyboard Slopes of 5°, 10°, 15°, and 25°. In Proceedings of the 26th Annual Meeting of the Human Factors Society (pp. 430–434). Santa Monica, CA: Human Factors Society.

A study was performed to determine how keyboard slopes of 5 to 25 degrees affected typing. It was found that the performance was not significantly different at the different slopes, but that the five degree and 25 degree slopes were perceived as uncomfortable, while there was no difference perceived between 10 and 15 degree angles. The authors conclude that a fixed-slope keyboard should be in the range of 10 to 18 degrees, while an adjustable slope range should be between 10 and 25 degrees.

Armstrong, T.J., Foulke, J.A., Joseph, B.S. and Goldstein, S.A. (1982). Investigation of Cumulative Trauma Disorders in a Poultry Processing Plant. American Industrial Hygiene Association Journal, 43, 103–116.

In the introduction it is stated that cumulative trauma disorders in the soft tissues of the upper extremity can be caused, precipitated, or aggravated by repeated exertions with certain hand and wrist posture, particularly in combination with high forces. The authors investigated activities in a poultry processing plant, and recorded the various motions, positions,

and grasps around cutting devices. The investigation indicated methods and procedures for evaluating upper extremity cumulative trauma disorders in the work place. Certain departments were found to account for the largest percentage of CTDs. Stressful exertions and corresponding work elements could be identified. Consequently, improvements in tool and workstation design, and in work practices, were recommended.

Wagner, G.N. (1982). *Synopsis of Ergonomic Standards Information Related to Display Workplaces in the Office Sector*. (Report HFP 82-08.) Dayton, OH: NCR.

This report is a synopsis of ergonomics standards information for display workplaces. It concerns particularly the German DIN 66234 and the ZH1/618 Safety Regulation. Among other topics, the report concerns workstation dimensions (pp. 9–10) and keyboards, including keys (pp. 13–14). Regarding key stroke feedback, the interpretation is that a positive feedback (indication) of switch activation should accompany each key depression (p. 51). The feedback should be audible and/or tactual in addition to visual. Regarding key force, the interpretation is that NCR products should limit key forces to approximately two to three ounces (p. 52). Regarding keyboard thickness, the interpretation is that it should not be more than 30 mm above the work surface at the home row. A palm rest is required if the thickness is more than 30 mm. The palm rest must be equal to the keyboard width, its depth should be between 50 and 100 mm, and its angle should match the angle of the keyboard or be adjustable (p. 58). Regarding keyboard slope, it should be shallow, preferably less than 15 degrees (p. 60).

Haider, E., Rohmert, W. and Halpern, N. (1982). *Ergonomic System for Literature Documentation for the Design of Typewriter Keyboards* (in German). Z. Arbeitswissenschaft, 36 (no pagination).

This is a discussion of the target hierarchies in ergonomic design of keyboards, in terms of energetic and information workload. The energetic work can be subdivided into dynamic and static muscular effort, while the information effort is mostly related towards identification and feedback.

Regarding the energetic workload, subtopics of ergonomic research concern the trunk, upper arm, lower arm, and hand. Regarding the digits of the hand, areas of ergonomic concern are the actuation force, key travel, keyboard layout, arrangement of keys, design of individual keys, distances between keys, key operation frequency, and the association of characters to keys. Regarding the feedback to the operator, this may be provided by visual, acoustic, or tactile means. The article concludes with a list of about 470 publications, ordered according to the following topics:

Dimensions of the Keyboard

Keyboard Arrangement

Keyboard Layout

Support for Arm and Hand

Arrangement of Keys

Design of Keys

Distances Between Keys

Keying Frequency

Keying Force/Force Displacement

Key Travel

Association Between Keys and Characters

Acoustic Feedback

Optical Feedback

Numerical Keyboard.

Helander, M.G. (1982). *Ergonomic Design of Office Environments for Visual Display Terminals*. Report for the National Institute for Occupational Safety and Health. Buffalo, NY: SUNYAB, Department of Industrial Engineering.

The type of task for which a VDT is being used, the task initiation, and the frequency of VDT use are important factors to consider in designing a workplace. For instance, for data entry, the main items of importance are the keyboard and the source document. On the other hand, dialogue tasks and data inquiry rely more heavily on the screen. Trade-offs have to be made in order to choose the best design for multi-purpose workstation. This report addresses the furniture and office illumination issues which must be considered in workplace design. With regard to furniture design, adjustability was found to be important for operator comfort, especially when male and female operators used the same workstation. However, adjustments will not be made unless the operator perceives changes to the workstation as being easy to make. Guidelines are provided for selecting appropriate placement of the screen and of all related working surfaces. Guidelines for the seat, backrest, padding, and arm rests of the operator's chair are also provided. An adjustable document holder is recommended for data entry and dialogue tasks. A summary of office lighting needs and solutions completes the document.

Lundborg, G., Gelberman, R.H., Minter-Convery, M., Lee, Y.F. and Hargens, A.R. (1982). *Median Nerve Compression in the Carpal Tunnel – Functional Response to Experimentally Induced Controlled Pressure*. Journal Hand Surgery, 7, 252–259.

The authors state that the pathophysiology of nerve entrapments, which includes the carpal tunnel syndrome, is an issue of considerable theoretical and clinical significance. A study was undertaken to investigate the biological basis underlying nerve compression by evaluating the functional response of the median nerve at the wrist to various levels of local compression. Sixteen healthy volunteers participated who had no history of carpal tunnel syndrome or other deficiencies. A wick catheter was introduced into the carpal tunnel of the non-dominant hand. Then the hand was placed in a specially designed external compression device and localized pressure was applied to the carpal tunnel. Three different levels of pressure were used, and as soon as the desired pressure level was reached, the catheter was withdrawn. It was found that local pressure did not significantly affect either arterial flow to the hand or venous return. The application of pressure caused little or no pain, and the experiment was easily continued for 30 to 90 minutes.

When tourniquet application around the upper arm was used, the subjects felt paresthesias at a very early stage, usually within 30 seconds, as well as a feeling of pins and needles in the hand, followed by numbness. Total anesthesia occurred within 10 to 15 minutes. Sensory response showed a course similar to the one found with the medium and high levels of wrist compression.

The authors interpret their results as indicative that the acute effects of compression were due to ischemia and not to mechanical deformation. Thus, they conclude that continued ischemia to the median nerve at the wrist level was the true reason for the maintenance of nerve conduction block. Accordingly, ischemia plays a major role in the early stages of nerve compression.

Butterbaugh, L.C. (1982). *Keying Logics for Alphanumeric Keyboards and Human Performance*. In Proceedings of the 26th Annual Meeting of the Human Factors Society (pp. 634–638). Santa Monica, CA: Human Factors Society.

Four different key layouts (logics) for the entry of alphanumeric characteristics were investigated. In experiments with subjects, all layouts were found to be equally accurate, with an error rate of approximately 0.5 percent.

Price, D.L., Fayzmehr, F., Haas, E. and Beaton, R. (1982). *Phase I Executive Summary - Ergonomics of Typewriting*. (Report IBM/SPO-3, for International Business Machines, Lexington, Kentucky.) Blacksburg, VA: Human Factors Laboratory, Virginia Polytechnic Institute and State University.

A literature search showed the factors that affect performance. They are operator characteristics (such as age, training, and cognitive processing), source document characteristics, operator feedback (visual or kinesthetic), equipment design characteristics (including key shape, key force and displacement, keyboard layout, keyboard inclination), and environmental factors. Techniques to measure performance include measures of typing speed and accuracy, muscular fatigue of the operator, efficiency of human movement, measures of operator fatigue, etc. In a field survey, long-term observations were conducted during an eight-hour workday at eight locations. The motion analysis indicated that, as the day progressed, the hand movements during typing were reduced. Also, body posture changed. In general, the hand became more horizontal and more rotated medially, the forearm became more vertical, the elbow was closer to the body, the upper arm was rotated towards the horizontal, the face dropped and the torso rotated toward the copy and leaned farther forward. Resting of the hand increased as the day progressed.

Armstrong, T.J. (1983). *An Ergonomics Guide to Carpal Tunnel Syndrome*. Akron, OH: American Industrial Hygiene Association.

Armstrong states in this Ergonomics Guide of the American Industrial Hygiene Association that carpal tunnel syndrome is one of a family of "occupational illnesses" (p. 1). The purpose of the guide is the prevention of carpal tunnel syndrome through control of occupational factors. "As a general rule, tools and tasks should be designed so that they can be used and performed without causing wrist deviations from side to side or causing the wrist to be flexed or highly extended" (pp. 3, 4). "Carpal tunnel syndrome is a disabling condition of the hand that can be caused, precipitated or aggravated by certain work activities" (p. 6).

Jensen, R.C., Klein, B.P. and Sanderson, L.M. (1983). *Motion-Related Wrist Disorders Traced to Industries, Occupational Groups*. Monthly Labor Review, 106, Issue 9, 13–16. The authors review the 1979 compensation claims of 26 U.S. States. There were 3027 workers' compensation claims for non-impact wrist disorders, which constitute six percent of all compensable cases.

Noyes, J. (1983a). *The QWERTY Keyboard: A Review*. Int. J. Man-Machine Studies, 18, 265–281.

The author provides a detailed discussion of the history of the original "Sholes" keyboard and typewriter. A large number of disadvantages of the QWERTY layout were pointed out in the past. Since 1905, a large number of proposals for keyboard

improvements have been brought forth, both in terms of re-arranging the relative location of keys on the keyboard, and of re-arranging the whole keyboard. In the 1920s, proposals for the re-design of the keyboard itself were made. Hoke proposed to split the keyboard into two and locate the shift keys in the center. In 1930, Gilbert proposed the "Orthographic Keyboard" with keys arranged according to their occurrence, followed by Dvorak's related proposals.

After World War II, a large number of rather sophisticated proposals for re-design were published, such as Griffith's "Minimotion Keyboard" and Maxwell's "Rhythmic Keyboard". Also, various recommendations were made to arrange the keys on the keyboard according to the alphabet. Various kinds of split keyboards were investigated, often in combination with arrangements of the keys according to a "natural" hand configuration. The author concludes with the statement that design and layout of the QWERTY keyboard are not optimal but that it is not easily feasible to modify it because of its common acceptance and use. The article ends with a bibliography of more than 50 publications.

Noyes, J. (1983b). *Chord Keyboards*. Applied Ergonomics, 14, 55–59.

This article presents an overview of the development of chord keying, apparently first proposed by Colombo in 1942, followed by proposals by Dvorak and Levi in 1950 and 1955, respectively. Levi's chord operation, and several of the following ones, were first envisioned for use by the postal services in mail sorting, but others were intended for speed typing (stenography) and for general typing. The author then discusses the advantages and disadvantages of chord keyboards. She argues that a chord keyboard might be of particular interest for application with a personal computer. Specifically, she indicates that the purposeful development of a chord keyboard could prevent a repeat of the "QWERTY episode", where simply the first developed keyboard becomes the standard.

Najjar, L.J. (1983). *Keyboard Human Factors: An Annotated Bibliography*. Report, Systems Research Laboratories, Inc. Eastern Division. Hanover, MD: Systems Research Laboratories, Inc.

A search of the recent professional and technical literature was conducted to assemble information related to human factors in the design of keyboards. This literature search stemmed from the effort to design a computer keyboard in which operator comfort and performance are important considerations. This prompted consulting of the human factors literature early in the design process. Twenty publications, dated from 1967 to 1983, were included in the report. Each of these publications is identified, summarized, and then excerpted with respect to keyboard design features. Findings are cross-referenced and discussed by the author.

AT&T Bell Laboratories (Ed.) (1983). *Video Display Terminals - Preliminary Guidelines for Selection, Installation, & Use*. Copyright 1983, Bell Telephone Laboratories, Inc. Indianapolis, IN: AT&T Technologies, Customer Information Center.

This set of guidelines discusses human factors aspects of VDTs. The handbook is said to have the following purposes:

- * Guiding the selection of new VDTs and VDT-related furniture.
- * Guiding the installation and use of VDTs and VDT-related furniture.

- * Specifying the physical design of offices in which VDTs are used, whether the offices are new or are to be remodeled.
- * Promoting the understanding of problems in the design of office environments in general, and especially of those offices in which VDTs are a significant item of office equipment.
- * The manual discusses proper working postures at VDT workstations, and specific details such as the location of the screen, use of proper furniture to achieve good body postures, different keyboard designs, illumination aspects, etc. In the section on VDT design, different aspects of keyboards are discussed, such as angle, key shape, keyboard design. In the section on "feedback from typing" (p. 40), it is said "feedback from typing is important, especially for less experienced operators. There are two types of feedback: tactile and auditory. Although both can assist unskilled typists, neither is a mandatory design feature." On page 43 it is said that key force should be approximately 0.9 to 5.4 ounce and key travel between 0.03 and 0.2 inch. Keys should have tactile (snap-action) feedback.

Clarke, M.M. and Carroll, L. (1983). *Human Factors Guidelines for Writers of User Manuals*. In Proceedings of the 27th Annual Meeting of the Human Factors Society (pp. 496–500). Santa Monica, CA: Human Factors Society.

There is a fair amount of experimental evidence on how people read and use instructions. Translating this knowledge into useful guidelines on writing instructions is the topic of this paper. It is necessary to tailor the instruction to the reader, using knowledge about human perception, information processing, and output to produce simple, unambiguous instructions that ensure that users quickly, easily, and correctly grasp what they must do (p. 497). A review of existing manuals indicated that they were generally poorly written, their grammar was too complex, and diagrams were unclear.

Murray, J.T., VanPraag, J. and Gilfoil, D. (1983). *Voice Vs. Keyboard Control of Cursor Motion*. In Proceedings of the Human Factors Society 27th Annual Meeting (p. 103). Santa Monica, CA: Human Factors Society.

The authors say that in word processing voice control of cursor position has a potential advantage over keyboard control. This is due to the fact that manual control requires a considerable amount of hand and eye travel, considering particularly the fact that cursor control accounts for 30 percent of all commands used in word processing. In an experimental evaluation of such an assumption it was found that, contrary to expectation, voice control took about twice as much time to position the cursor on a target than keyboard control.

Arndt, R. (1983). *Working Posture and Musculoskeletal Problems of Video Display Terminal Operators – Review and Reappraisal*. American Industrial Hygiene Association Journal, 44(6), 437–446.

A review of the existing literature indicates that postural problems are related to the positioning of the operator in relation to work table, display, and keyboard at VDT workstations. Some of the recommendations in the literature are said to apparently to be based upon extrapolations from earlier office and industrial workstation guidelines. The author discusses in detail keyboard location, chair design, display screen position, document placement, working position and work task, working posture and personal attributes, job design, and administrative procedures.

A special section concerns repetitive-motion injuries. The author states that it has been long recognized that repetitive

motion of the fingers, hands and arms may lead to (repetitive motion) disorders (p. 442). Thus, repetitive motion patterns may place workers at risk, while the position of arms, wrists and hands play important roles.

Miller, W. and Suther, T.W. (1983). *Display Station Anthropometrics: Preferred Height and Angle Settings of CRT and Keyboard*. Human Factors, 25, 401–408.

In a study to determine postural preferences of operators at VDT workstations, participants were selected to assure representation of extremely short and extremely tall persons, as well as of persons of mid-range physical statures. Thirty-seven persons participated.

The procedure was to first adjust seat height and backrest height according to the operators' preferences. The operators then sat on the adjusted chair at a comfortable distance from the workstation. A preferred keyboard height was then determined. Likewise, keyboard slope angle, CRT height, and CRT angle were adjusted to the operators' preference. While the operators then transcribed a page of text, they were urged to make any readjustments deemed necessary. Among the findings is a high and negative correlation between keyboard slope and seat height. Likewise, the correlation between stature and keyboard angle was negative, although not as numerically high. Based on the outcomes, adjustment of the keyboard height between 63 to 78 cm above the floor would include the preferred height settings of 95 percent of all operators.

Grandjean, E., Huening, W. and Pidermann, M. (1983). *VDT Workstation Design: Preferred Settings and Their Effects*. Human Factors, 25, 161–175.

A field study was conducted to assess the preferences of VDT operators regarding their body posture and workstation adjustment settings. Four different companies were involved, and the study was performed during regular work. The operators preferred body postures that are distinctly different from those recommended in textbooks and other publications. For example, neither the preferred screen heights nor the keyboard heights were noticeably influenced by anthropometric data. The body postures actually observed are characterized by marked trunk inclination, i.e., pronounced backward leaning (between about 100 and 120 degrees), accompanied by an increase of upper arm flexion, elbow angle being between 90 and 110 degrees, and elevated hand. Eye levels were lower than expected according to anthropometric data.

Kroemer, K.H.E. (1983a). *Ergonomics of VDT Workplaces*. Ergonomic Guide. Akron, OH: American Industrial Hygiene Association.

Kroemer, K.H.E. (1983b). *Design Parameters for Video Display Terminal Workstations*. Journal of Safety Research, 14, 131–136.

Kroemer, K.H.E. (1983c). *Ergonomics of VDU Workplaces*. Digital Design, 13(2), 25–34.

These three publications discuss the ergonomic design and arrangement of VDT/VDU workstations. Based on estimates of the anthropometry of U.S. civilians, recommendations are presented for design ranges of the furniture (chair, supports for keyboard and display, footrests, armrests) and for the arrangement of the keyboard and the display.

Kaplan, P.E. (1983). *Carpal Tunnel Syndrome in Typists*. JAMA, 250(6), 821-822.

The question is posed: "Is there any evidence that typing as an occupation may contribute to the development of carpal tunnel syndrome (CTS)?" In the answer, the author refers to two studies of patients with CTS. One, published in 1965, concerns more than 1200 patients with CTS which specifically lists clerks and stenographers. The 1975 study specifically lists cashiers/secretaries as occupation of patients with CTS. It is said that tension in flexor tendons of the fingers during wrist flexion generated elevated carpal tunnel pressure. Hence the author concludes: "These studies all point to repetitive wrist flexion as a contributory factor in the etiology of CTS" (p. 822).

Chisvin, S.G. (1983). *Video Display Terminals: An Ergonomic Evaluation of Workstations and Operating Postures*. Master Thesis, Department of Human Kinetics. Windsor, Ontario: The University of Windsor.

A field study was undertaken in a major company to evaluate ergonomic aspects of 21 individual VDT workstations used by female operators. Also, the opinions of more than 1,000 clerical workers were analyzed, of whom nearly 60 percent were VDT operators. The results indicated that VDT operators experience a higher incidence of physical discomfort than other office workers, with the most common complaints being aches and soreness of the neck and shoulder areas. The ergonomic recommendations taken from the literature do not always seem appropriate to VDT operators. Recommendations were made to develop a workstation prototype which could be used either in the sitting or standing posture.

Gopher, D. and Koenig, W. (1983). *Hands Coordination in Data Entry With a Two-Hand Chord Typewriter*. (Technical Report CPL 83-3.) Champaign, IL: Department of Psychology, University of Illinois.

This report describes the results of an experiment performed to investigate acquisition of data entry skill, and operation of a two-hand chord keyboard consisting of two separate five-key panels, one for each hand. Characters are entered by typing chords composed of one to five fingers. Each panel is capable of producing the full set of characters, and hence can be considered as a full and independent keyboard. A study was conducted to determine the best chording principle, the duration of the learning period, the rate of progress, the final level of performance on this system, and the nature of coordination between the hands and simultaneous chord production. The results indicate that representation of letter chords by spatial patterns was superior to the principle of hand symmetry.

Benz, C., Grob, R. and Haubner, P. (1983). *Designing VDU Workplaces*. Cologne, Germany: TUV Rheinland.

This is a comprehensive book of 176 pages, based on the original 1981 German text. It discusses many aspects of VDU workplaces, referring extensively to German DIN standards. In its various chapters the authors first present the workplace as a work system, then discuss basic ergonomic factors, finally provide specific data and advice on workstation design, on the design of the working environment, and on job organization. The book also contains guidance on how to prepare the new VDU user. It presents ergonomic checklists as well as a list of literature and index. The keyboard is discussed in Sect. 4.1.5. An existing keyboard should be as thin as possible, i.e., not more than 30 mm at the home row. An "action point" on the keys is said to be advantageous, since the user receives a tactile acknowledgment of key activation.

Monty, R.W., Snyder, H.L. and Birdwell, G.G. (1983). *Keyboard Design: An Investigation of User Preference and Performance*. In Proceedings of the 27th Annual Meeting of the Human Factors Society (pp. 201-205). Santa Monica, CA: Human Factors Society.

This report describes a study on the effects of several alphanumeric keyboard characteristics, including audible feedback, location, size, and shape of the return key, and the shape and style of keycaps. Eighteen subjects performed a text entry task during which they were not allowed to correct errors. The working time was not significantly different among the keyboards, but with audible feedback, text entry time was significantly faster. The subjects indicated strong preferences for certain sizes, styles, and locations of the return key, and for the styles and shapes of keycaps. Acoustic feedback on keystrokes was subjectively preferred to the absence of feedback.

Francas, M., Brown, S. and Goodman, D. (1983). *Alphabetic Entry Procedure with Small Keypads: Key Layout Does Matter*. In Proceedings of the 27th Annual Meeting of the Human Factors Society (pp. 187-190). Santa Monica, CA: Human Factors Society.

The authors describe two experiments on the entry of alphabetic characters on an enhanced telephone system. This required a small keypad which restricted the users to one or two finger typing activities. Performance was clearly faster on the QWERTY keypad than on an alphabetically organized keypad or a numeric entry pad similar to the telephone pad.

Hochberg, F.H., Leffert, R.D., Heller, M.D. and Merriman, L. (1983). *Hand Difficulties Among Musicians*. JAMA, 249(14), 1869-1872.

The authors describe their experience with 100 musicians who developed occupation-related difficulties. The majority were pianists who experienced hand difficulties in mid-career, with the right upper extremity involved twice as often as the left one. Inflammatory disorders of tendon or synovium were the most common difficulties. The authors refer to the well-known affliction of Robert Schumann, necessitating a variety of orthopaedic devices and then cessation of performance. They say (on page 1872): "An extensive literature details the hand difficulties or cramps that may develop in persons who are required to perform prolonged, highly repetitive, or intricate movements such as writing, telegraph operation, sewing, tobacco picking, and gold beating."

Zipp, P., Haider, E., Halpern, N. and Rohmert, W. (1983). *Keyboard Design Through Physiological Strain Measurements*. Applied Ergonomics, June issue, 117-122.

The physiologically tolerable range of positions of the upper extremity was investigated for typing tasks by recording myoelectric activities of involved muscles. For long-term typing, a split keyboard is recommended, with the halves rotated against each other in the plane view, and declined laterally.

Snyder, H.L. (1983). *Keyboard Design* (pp. 43-58). Stockholm, Sweden: Ericsson Informations Systems AB.

This text describes the "best currently available information" for the arrangement, height and slope of the keyboard, and other important considerations in selecting and using the keyboard of the VDT (p. 45). Electronic keyboards have "made

possible a wide variation in keyboard design from the old mechanical ones. Keys can be better placed in relation to each other resulting in less user hand movement..." (p. 46). It is a "misconception that the QWERTY layout was designed for the convenience of the operator" (p. 48). The "QWERTY layout is not necessarily desirable from a human factors point of view..." (p. 49). "Keyboards that do not fulfil ergonomic recommendations not only result in poor performance but also poor posture. For example the height and angle of the keyboard determine the posture of the operator. If the keyboard is too high, the operator will not be able to adopt a comfortable posture." "...Smooth reflexive keying depends upon a user's ability to generate the pattern of motor signals and to receive information back via the fingers so that perceived and intended activity are matched" (p. 47). A "modular keyboard of several units" that "could be arranged in the desired layout" could cater to variations in user requirements (p. 49). Tactile and auditory feedback provided in the mechanical keyboards must be translated into electronic keyboards (p. 46). Keyboards should be slim, that is 30 mm or less in height (p. 51). The slope angle should be 5 to 11 degrees. Spacing of the keys should be 18 to 20 mm (p. 52). Key force should be 0.25 to 1.5 N and be the same for all keys (p. 53).

NIOSH (Ed.). (1984). *Video Terminals NIOSH Fact Sheet*. External Review Draft, 21 August, 1984. Cincinnati, OH: Author.

This text states that users of VDTs have reported visual problems, musculoskeletal problems, stress, and concerns about radiation emissions and adverse reproductive effects. Visual and musculoskeletal problems are the most frequent complaints. The latter problems are controllable through the use of properly designed furniture and equipment (p. 1). Improving ergonomic conditions reduced visual and musculoskeletal complaints and increased work efficiency (p. 3). Visual and musculoskeletal problems at VDTs can be controlled through ergonomic improvements. This includes also controlling the work regimens, especially interrupting VDT work by rest breaks, at a minimum after two hours of continuous VDT work.

Peters, J.A. (1984). *Fifteen Cardinal Principles to Ensure Effectiveness of Warning System*. Occupational Health and Safety, May Issue, 76-79.

The author states that there are a number of cardinal principles for effective warnings:

- Readability
- Understanding
- Comprehensibility
- Practicality
- Effectiveness
- Behavioral Modification
- Compatibility
- Conspicuity
- Durability
- Reliability
- Reinforcement
- Danger Signal
- Placement
- Novelty
- Type

After discussing these aspects, the author cautions that a warning should be used only when a design remedy is not appropriate, available, or sufficiently effective. A second limitation is that each warning has associated costs, reliability,

service ability, and failure mode consequences that should be carefully considered. Another limitation is the user, i.e., the person who reacts to the warning. The user should be considered in terms of priority, information processing capability, adverse suggestibility, and excessive reliance.

Burke, T.M., Muto, W.H. and Gutmann, J.C. (1984). *Effects of Keyboard Height on Typist Performance and Preference*. In Proceedings of the 28th Annual Meeting of the Human Factors Society (pp. 272-276). Santa Monica, CA: Human Factors Society.

A study was performed to determine if the height of the home row of a keyboard affected performance and preference of typists. Forty-eight skilled typists participated, performing two five-minute typing tests and a 30-minute intervening typing task. They were using eight low-profile keyboards from two manufacturers, with the home row set at 35, 64, 84, and 104 mm above the support surface. The performance measurements included words-per-minute and number of errors. The preference assessments were based on paired comparisons, absolute judgments, and comparative judgments. The results showed no difference in words-per-minute and number of errors during the typing tests.

Joyce, M.S. (1984). *Ergonomic Skills Training: Discussion of Methods Used to Improve Skill and Comfort Levels of Workers Using Video Display Terminals*. In Proceedings of the 28th Annual Meeting of the Human Factors Society (pp. 781-785). Santa Monica, CA: Human Factors Society.

This is a report on the outcomes of "ergonomic skills training" applied in ten hours of classroom instructions to more than 1600 participants. The training consists of physical awareness training, short-term memory training, recommendation to take short breaks several times a day and to perform exercises during these breaks, deep breathing procedures, eye relaxation techniques, and counseling and basic ergonomic principles regarding proper posture at the terminal, and "do it yourself" ergonomic adjustments. Other components include limbering exercises and self-massage of tight muscles. The author claims that on average, performance rate increased by 65 percent, and errors were reduced by 42 percent. Results of pre- and post-training surveys indicated a reduction in headaches from 82 to 72 percent, no changes in fatigue, irritated eyes, blurriness of vision, and eye strain. Back pain was reduced from 81 to 70 percent, painful neck/shoulders from 88 to 80 percent. There were no changes in the occurrence of painful arms/legs, and hand/finger cramps. However, the incidences of stiff/sore wrists increased from 33 to 39 percent, and the loss of feeling in wrists/fingers increased from 16 to 27 percent.

Brunner, H. and Richardson, R.M. (1984). *Effects of Keyboard Design and Typing Skill on User Keyboard Preferences and Throughput Performance*. In Proceedings of the 28th Annual Meeting of the Human Factors Society (pp. 267-270). Santa Monica, CA: Human Factors Society.

A study was conducted with 12 expert and 12 occasional typists to compare user preferences and performance. The keyboards all had the same appearance and layout, but were different in their tactile and auditory operating characteristics. Each subject participated in a full day of testing and evaluation, inputting standard text for one hour on each keyboard. The subjects completed questionnaires and made comparisons at the end of the day.

In conclusion, the authors state that clear and unambiguous tactile and auditory performance feedback were primary determinants of the preference and performance data.

Salthouse, T.A. (1984). The Skill of Typing. Scientific American, 250, 128–135, 148.

The author indicates that on the surface, typing resembles a series of choice-reaction-time tasks. In such tasks, the stimulus-response time, or average latency, or delay, is approximately 200 ms. Yet, this would result in a typing rate of only 48 words per minute, while indeed twice that rate is fairly common. In 1908, Book thought that skilled typing was achieved by expanding the size of units to be typed from single characters to words. This was reinforced by Coover in 1923. Thus, “preview” of text to be typed (“chunking”) is important.

A second major phenomenon is the “overlapping” of the actions of several fingers. While the first finger strikes a key, the second finger is already starting its movement. This was mentioned both by Book (1936) and by Dvorak. Thus, the ability to perform fast keystrokes is both a matter of “perceptual chunking” of the material, and of the skills in performing “overlapping motions”. “The intensive practice necessary to achieve high levels of skill appears, then, to result in the elimination of unnecessary operations, in the ability to execute more than one operation at a time and perhaps in a reduction in the attention demanded of the typists by certain operations” (p. 135).

Gopher, D., Koenig, W., Karis, D. and Donchin, E. (1984). An Information Processing Approach to the Study of Data Entry Skills: The Effects of Representation Rules and Coordination Requirement. (Technical Report, CPL 84-4.) Report for the Personnel and Training Research Programs, Office of Naval Research (Contract N000-14-83-K-0092). Champaign, IL: University of Illinois.

In the course of developing typing skills on a newly developed two-handed chord keyboard (described in 1983 by Gopher and Koenig), the importance of the generation of skilled movements was investigated. These are based on high-level motor programs, and schemes and long-term memory, developed with practice.

Buesen, J. (1984). Product Development of an Ergonomic Keyboard. Behaviour and Information Technology, 3, 387–390.

The development of the STR ergonomic keyboard is described. Its major design objective was to reduce complaints related to keyboard work through substantial improvement of body posture.

A split keyboard represents the logical solution for the achievement of a natural position of the hands and forearms, as proposed earlier, e.g., by Kroemer (1964). For this, an arrangement of two half-keyboards was constructed which allowed continuous adjustments in slant, lateral tilt, and slope. Based on previous experiments, the slant angle between the keyboard halves were set to 25 degrees, the lateral tilt against horizontal to 10 degrees, and the slope against horizontal to 10 degrees. The distance between the keyboard halves was 10 cm. Prototypes were developed and shown to customers, which lead to changes in keyboard appearance. First working prototypes were introduced to the public in 1983.

Rosch, W.L. (1984). Keyboard Ergonomics for IBMs. PC Magazine, October 2 Issue, 112–121.

A brief history of the development of the QWERTY keyboard is presented, including proposals for redesign, particularly by Dvorak. Adding more keys to the original layout increases use problems. No research is known to have been on the subject of adding function keys.

Another problem area is that of the design of keys, and particularly of their force-displacement characteristics. Various graphs are shown that indicate various kinds of tactile feedback, accomplished by changes in resistance during the key travel. A survey of different keyboards available on the market was performed, comparing these to the ones used with the Selectric typewriter, and the IBM PC. It was found, among other things, that on the PC keyboard different keys showed different resistance characteristics, and differences occurred even in the same key when activated in succession. “Interviews with representatives of major keyboard companies... revealed that the companies are ready, willing, and able to build products to nearly any specification” (p. 117).

The comparisons showed only very small differences among the “best” keyboards. Some of the differences are apparently related to the “feel” of the keyboard, i.e., the lack, or presence, of tactile indication of the exact moment the key is activated. “Some kind of tactile feedback is apparently desirable” (p. 119). “Probably the most important conclusion we can draw is that there might not be one single best keyboard. The three fastest keyboards differ substantially in their measured characteristics. Although all three give the typists some degree of tactile feedback, their force versus displacement curves are quite different” (p. 119).

Life, M.A. and Pheasant, S.T. (1984). An Integrated Approach to the Study of Posture and Keyboard Operation. Applied Ergonomics, 15, 83–90.

A survey showed that typists tended to work with their keyboards placed higher than recommended in the ergonomic literature, and to report discomfort in shoulders, neck, and upper back. To investigate these phenomena, an experiment was undertaken with 12 skilled typists. Each performed a typing task on an electronic keyboard placed at elbow level, and 50, 100, and 200 mm above. Three dependent variables were examined, namely performance, comfort, and biomechanical load.

Performance was not affected by keyboard height. With respect to biomechanics, an increase in torque at the shoulder occurred as working height was increased. This means that at higher keyboards more static work must be performed by the musculature of the shoulder to support the weight of the upper limb. This might result in fatigue and discomfort.

Grandjean, E., Huenting, W. and Nishiyama, K. (1984). Preferred VDT Workstation Settings, Body Posture, and Physical Impairments. Applied Ergonomics, 15, 99–104.

In a field study, an adjustable VDT workstation was given to each of 68 operators for one week. The preferred settings, the body postures, and subjective evaluations were assessed. There was practically no correlation between preferred settings and anthropometric data concerning stature or eye height. The majority of the operators tended to lean backward with trunk angles between 97 and 121 degrees. Physical complaints in the neck, shoulder, and back areas diminished in the preferred settings.

Shute, S.J. and Starr, S.J. (1984). *Effects of Adjustable Furniture on VDT Users*. Human Factors, 26, 157–170.

One group of telephone directory operators compared a non-adjustable VDT table with a conveniently adjustable one. The other group compared an easily adjustable chair with a chair that was more difficult to adjust. For the adjustable furniture, selected adjustments were recorded. Also, subjective evaluations were collected by means of questionnaires.

The use of adjustable furniture was endorsed by the participants, and adjustments were done frequently. Yet, two problems with the adjustable tables became apparent. One was insufficient legroom. The other was that tables could not be lowered sufficiently so that the keyboard was at approximately elbow height. This was partly due to the fact that the home row of the keys was 6.7 cm above the support surface.

The authors conclude that a combination of easily adjustable chair and table would have been the most satisfactory solution. Such adjustment would be particularly important if different persons would have to use the same workstation in succession.

Starr, S.J. (1984). *Effects of Video Display Terminals in a Business Office*. Human Factors, 26, 347–356.

A self-administered questionnaire was completed by 211 workers in a business office who used VDTs, and by 148 control subjects doing the same job but using paper documents instead of VDTs. The two groups exhibited few differences in the likelihood and intensity of physical discomfort. Some differences were found in job satisfaction, but most of these favored VDT users. The author concludes, based on this study and on findings of an earlier study, that VDTs appear not to be potent sources of discomfort and dissatisfaction among office workers.

Thomas, M. (1984). *West Germany Beefs Up Its Electronic Office Laws*. Facilities Design & Management, July/August Issue, 82–85.

The author describes the requirements of the West German regulation ZH 1/618 going into effect on January 1, 1985, and its influence on developing U.S. guidelines for equipment and furniture ergonomics.

Cumming, G. (1984). *QWERTY and Keyboard Reform: The Soft Keyboard Option*. Int. J. Man-Machine Studies, 21, 445–450.

The author first quotes Noyes' 1983 statement (see listing above) that the design and layout of the QWERTY keyboard are not optimal for efficient operation, but that it is not feasible to modify the standard keyboard and hence improve it. The main obstacle to change is claimed to be the existing investment in the current QWERTY layout, particularly in terms of the developed skills of its users. The author argues that with current software, assignment of characters to keys can now be easily achieved. "Soft" keyboards include programmable keys. Yet, the author argues that also the physical arrangement of the keys could be changed, such as on the Maltron. Furthermore, chord keyboards could be introduced. In summary, the advent of the soft keyboard provides a new opportunity for keyboard reform.

Helander, M.G., Billingsley, P.A. and Schurick, J.M. (1984). *An Evaluation of Human Factors Research on Visual Display Terminals in the Workplace*. Chapter 3 in F.A. Muckler, A.S. Neal, and L. Strother (Eds.), Human Factors Review: 1984 (pp. 55–129). Santa Monica, CA: Human Factors Society.

This is a critical review of 82 human factors research studies on VDTs. In particular, twenty-eight studies concern visual discomfort, 16 studies concern anthropometry and biomechanics, nine studies concern work organization, 21 studies concern display design, and nine studies concern lighting. Each of these studies is critically analyzed, particularly with respect to study design and subject selection, both of which affect the validity of the results.

In summary, studies on visual and postural fatigue did not provide clear answers to the question whether such fatigue is more frequent for VDT work than for tasks performed without the use of VDT. Work organization and job design factors may be more important than the VDT itself.

Arndt, R. (1984). *Workbook for Ergonomic Considerations in Office Design*. Developed for the National Office Products Association, Office & Contract Furnishings Division. Alexandria, VA: National Office Products Association.

The author discusses in seven chapters ergonomics goals and methods, the environment, video display terminals, workstations, analyses of work tasks, and use of checklists.

For keyboard design, the following variables are especially important:

- * Keyboard angle;
- * Key shape;
- * Feedback from keying;
- * Keyboard layout;
- * Numeric keyboard layout;
- * Keyboard size and thickness;
- * Location of special function keys.

For further information, the reader is referred to a list of suggested readings. Under possible problems the author mentions cumulative trauma disorders "which occur as the result of repetitive motion or chronic irritation of nerves, muscles, tendons, or bones" (page 18). In this context, carpal tunnel syndrome is described. It is recommended that the hands not be rotated outward, but the wrist be kept as straight as possible to avoid carpal tunnel syndrome and tendonitis.

Browne, C.D., Nolan, B.M. and Faithfull, D.K. (1984). *Occupational Repetition Strain Injuries*. Guidelines for Diagnosis and Management. Medical Journal of Australia, 140, 329–332. The authors refer to the rapidly rising rate of compensation claims for repetition strain injuries in Australia. Since the classification of these claims is under "synovitis, bursitis, and tenosynovitis", the reported figures are believed to not even approximate the actual rate of occurrence. RSI is claimed to be not only occupationally induced, but also associated with numerous leisure and sporting activities. The outlook for complete recovery in a non-occupational group is favorable, in contrast to the poor outcome in occupational injuries. The leisure or sport injury can be better avoided by modifying or ceasing the activity when early symptoms appear, while in industry or office, the employee is subject to pressures which may force to delay the reporting of symptoms, with a consequent delay in diagnosis and remedial action.

Among the factors that may combine to generate RSI are:

- * Biomechanical factors, such as constrained posture, frequency of repetitive movement, force used in performing a task, joint position, faults in workstation, equipment, and task, and increased muscle tension associated with stress.

- * Work organization faults, such as the duration of work without rest, bonus and overtime incentives, lack of training, failure of supervision.
- * Personnel (medical and other) incompetence leading to delayed reporting of the injury, incorrect or delayed diagnosis, inappropriate management of the injury.
- * Failure of the employee to report early, due to ignorance of the significance of the symptoms, strong work ethic, fear of losing the job, economic pressure, language barriers, absence of alternative jobs, attitudes of supervisor, health personnel, and fellow workers.

Hadler, N.M. (1984). *Medical Management of the Regional Musculoskeletal Diseases*. Chapter 7, The Carpal Tunnel Syndrome (pp. 143–155). Chapter 8, The Hand and Wrist (pp. 157–201). Orlando, FL: Grune & Stratton.

The author provides an overview of musculoskeletal disorders and the use-relatedness of these conditions. In the conclusion (page 192), the author writes: “The soft-tissue syndromes of the hand are not only numerous, they are common. A characteristic of nearly all these states is that hand usage, or particular hand usages, appear to precipitate symptoms in particular individuals. ...Soft-tissue pain syndromes, however, may plague perhaps one-fifth of manual workers at some point each year, particularly those whose tasks are highly repetitive. ...If particular patterns of usage can be associated with particular illnesses, proscription of these usages can be effected by redesigning tasks. In this way, workers’ discomfort can be avoided and productivity improved.”

Bergqvist, U.O. (1984). *Video Display Terminals and Health*. Scand. J. Work Environ Health, 10, Suppl. 2. Solna, Sweden: Research Department, National Board of Occupational Safety and Health.

This tome deals with all current technical and occupational hygiene issues, including radiation, sound, displays, operator, and environment, and with medical and epidemiological aspects such as eye strain, stress, muscular discomfort, and pregnancy outcomes.

Arndt, R., Sauter, S.L., Dainoff, M., Helander, M.G., Kroemer, K.H.E., Snyder, H.L., Schleifer, L. and Smith, M. (1984). *Health Issues – Video Display Terminals*. (Report PB88 – 126115 /AS). Cincinnati, OH: NIOSH.

This is a manual for use in a course on health issues for VDT supervisors. It primarily concerns the effects of VDT work on the operators’ visual and musculoskeletal systems, but other job stresses are also discussed.

Frey, W.U., Rometsch, J. and Diem, H. (1984). *Keyboard Arrangement*. Patent 4483634. United States Patent Office.

To diminish the physical troubles often occurring with the usual keyboards, the keyboard is split into left and right halves, slanted, and tilted down to the sides. Additional sets of keys are arranged on each side, but not tiled. All key fields are located on planes sloping up away from the operator who uses the slopes to rest the arms.

Louis, W.M. (1984). *Ergonomic Keyboard Apparatus With Left and Right Key Section Separated by an Irregularly Shaped Space*. Patent 5372441. United States Patent Office.

To avoid fatigue and slow performance occurring with the usual keyboards, the keyboard is split into left and right halves, slanted, and sloped. The split is along the right sides

of the keys 5, T, G, and B, resulting in ragged edges on each side of the split. The space bar bridges the gap and can be operated with either thumb. All key fields are located on planes sloping up away from the operator who uses the front parts of the slopes to rest the arms.

Gaydos, J.M. (1985). *A Staff Report on the Oversight of OSHA with Respect to Video Display Terminals in the Workplace*. Report of the Subcommittee on Health and Safety, Committee on Education and Labor, House of Representatives (Serial No. 99-A). Washington, DC: U.S. Government Printing Office.

This report discusses major issues with respect to working at video display terminals, namely radiation and ergonomic issues including the environment, furniture, and use aspects, including musculoskeletal concerns.

Koppa, R.J. (1985). *A Study of Data Entry Keyboards: The 4 × 4 Keypad*. (Technical Note 4-85). Aberdeen Proving Ground, MD: U.S. Army Human Engineering Laboratory.

This is a review of previous keyboards, mostly keypads for data entry. The review begins with keyboard arrangements in the 1920s, but discusses mostly telephone and calculator layouts. A review of previous design solutions is provided, and evaluations of the suitability for efficiency of input is provided. There is also a review of chord versus single-stroke entry. Recommendations for future research are included.

Bartram, D. and Feggou, O. (1985a). *A Pilot Investigation of the Sequential Keyboard* (Report ERG/Z6545/85/4a). Hull, UK: University of Hull.

This is a report on work carried out on behalf of the British Post Office as part of a larger study involving the comparison of four different keyboard designs for destination coding. The current report concerns solely one of these keyboards, namely a ten-key sequential entry device. This is similar to the arrangement of number keys of a standard typewriter keyboard, where each activation of a key produces one discrete character. The study was carried out to examine differences in keying rate and accuracy as a function of the fingers used. It is assumed that such key presses would be most affected by the “strength” and “ease of articulation” of the finger involved. The authors propose a three-digit -number stimulus for the main study to be conducted.

Nakaseko, M., Grandjean, E., Huening, W. and Gierer, R. (1985). *Studies on Ergonomically Designed Alphanumeric Keyboards*. Human Factors, 27, 175–187.

A split keyboard was adjusted in various slant angles, lateral tilts, and distances between the keyboard halves. The proper arrangement of a split keyboard decreased the lateral deviation of the hands. Use of a wrist support allows subjects to lean back. The ergonomically arranged split keyboard was preferred by two-thirds of the subjects. They also stated increased feeling of relaxation when using the split keyboard, as compared to a regular arrangement.

Gopher, D., Karis, D. and Koenig, W. (1985). *The Representation of Movement Schemas in Long-Term Memory: Lessons from the Acquisition of a Transcription Skill*. Acta Psychologica, 60, 105–134.

This paper reports on the results of three experiments with a newly designed two-hand chord typewriter keyboard. The experiments concerned coding principles and the spatial arrangements of the keyboard panels.

Lynch, G. (1985). *American National Standard for Human Factors Engineering of Visual Display Terminal Workstations: A Report to the Society*. In Proceedings of the Human Factors Society 29th Annual Meeting (p. 987). Santa Monica, CA: Human Factors Society.

This is an introduction to the presentations of several members of the HFES/ANSI committee that is developing an American National Standard for Human Factors Engineering of Visual Display Terminal Workstations. This standardization work, started in 1983, is said to be in the final stages of acceptance as a national standard.

Thomas, J.C. (1985). *Human Factors in IBM*. In Proceedings of the Human Factors Society 29th Annual Meeting (pp. 611–614). Santa Monica, CA: Human Factors Society.

This is a "...report on some of the ways that IBM attempts to promote good human factors in their products" (p. 611). IBM is claimed to be guided by certain fundamental beliefs, which are respect for the individual, pursuit of excellence, and best customer service. Producing products with good human factors is argued to benefit IBM concretely. Products which help make users more productive and happier help keep IBM itself highly productive and keep low rates of turn-over, absenteeism and accidents. While IBM sponsors inter-divisional and university research on human factors and related work and cognitive psychology, one cannot wait until "all" the evidence is in before trying to ensure that best guesses are based on current science (p. 612).

Stewart, T. (1985). *Ergonomics of the Office*. Ergonomics, 28, 1165–1177.

This is a description of the history of office work, from the monks in their cells copying religious texts to the paperwork factory of today. While there is much to keep the traditional ergonomist occupied in today's office in trying to improve the design of equipment, work places and working environments, new models of the office, and related work, need to be developed. This would avoid many of the unproductive aspects, and negative impacts, of today's technology and procedures.

Oxenburgh, M.S., Rowe, S.A. and Douglas, D.B. (1985). *Repetition Strain Injury in Keyboard Operators*. Journal of Occupational Health and Safety – Australia and New Zealand, 1(2), 106–112.

A comparison was made between VDT operators who had reported symptoms of repetition strain injuries and operators who had no symptoms. No significant differences were found between these groups in terms of workstation design, or adjustment. However, there was a difference between the groups in terms of the time spent at the keyboard, with the likelihood of injury arising rapidly after more than five hours of work per day. This finding has led to the formulation of policies for prevention. "VDU Guidelines" were widely disseminated in 1983 and a training program was developed to include workstation adjustment and work practices. The results were a marked reduction in the days of absence from work. The authors conclude that the interaction between physical and psycho-social factors needs to be addressed to make prevention of repetition strain injuries successful.

Bartram, D. and Feggou, O. (1985b). *An Evaluation of Four Different Keyboard Designs for Foreign Destination Coding Desks* (Report ERG/Z6545/85/4c). Hull, UK: University of Hull.

This is a report on work to examine the use of four keyboards: the current two-finger chord keyboard, a new two-finger chord keyboard (with two key pads each of four rows of four keys), a ten-key sequential entry keyboard and a ten-key chord keyboard. Matched groups of subjects, four in each group, practiced for six weeks (approximately one hour per day) on the four keyboards. The results showed clear differences between the keyboards. The sequential keyboard produced the slowest and most inaccurate responses. Performance on the currently used chord keyboard was worse than that on the new two-finger chord design.

Westgaard, R.H. and Aaras, A.A. (1985). *The Effect of Improved Workplace Design on the Development of Work-Related Musculo-skeletal Illnesses*. Applied Ergonomics, 16, 91–97.

Following the implementation of ergonomics interventions at the workplaces in a Norwegian factory, sick leaves taken by the employees were reduced, especially for musculo-skeletal illnesses. This was accompanied by an increase in productivity.

McKenzie, F., Stormont, J., Van Hook, P. and Armstrong, T.J. (1985). *A Program for Control of Repetitive Trauma Disorders Associated with Hand Tool Operations in a Telecommunications Manufacturing Facility*. American Industrial Hygiene Association Journal, 46, 674–678.

Repetitive trauma disorders were the leading cause of lost time and compensation expenses at a telecommunications manufacturing plant with 6600 employees. An ergonomic task force, a training program, improvements in the design of workstations and tooling, and schemes of restricted work were introduced. This substantially decreased the number of lost and restricted workdays.

Knave, B.G., Wibom, R.I., Voss, M., Hedstroem, L.D. and Ulf, O.V. (1985). *Work with Video Display Terminals Among Office Employees*. Scand J. Work Environ Health, 11, 457–466.

This is a report on an epidemiological health investigation of about 400 VDT operators and 150 selected referents (office employees not using VDT) in Sweden. Occupational exposure factors were measured by the investigators, and subjective symptoms and discomfort in relation to the eyes, arms, back, skin and headache were derived from questionnaires. Information about previous and current diseases was also obtained. The aim of the investigation was to compare the occurrence of symptoms and discomfort in VDT operators and referents and to study the relationships among symptoms and discomforts.

There was no significant difference between the VDT group and the referents with respect to the occurrence of current and previous diseases. However, the VDT operators reported significantly more eye discomfort symptoms than the referents. In the study, musculoskeletal discomfort was found to be higher for women, but no difference was observed between VDT workers and referents. However, there was a tendency towards higher discomfort for shoulder, neck, and back in VDT operators. Significant correlation coefficients were found between the scores of eye discomfort, headache, and musculoskeletal discomfort. Both the total number of work hours per day, and the time spent at the VDT, seemed to be related to disorders.

Burry, H.C. and Stoke, J.C.J. (1985). *Repetitive Strain Injury*. New Zealand Medical Journal, 98, 601–602.

The term repetitive strain injury (RSI) refers to groups of muscle and tendon injuries caused by continuous use of hand, leg or body movements, in repetitive or continuous work. It also describes pain in neck, shoulder and arm due to postural strain. This is currently a vast problem in Australia but not in the USA. Similar occupational disorders have been identified since 1818. RSI includes postural neck and shoulder pain, lateral epicondylitis and muscle attachment injuries, nerve entrapments (such as carpal tunnel syndrome), tenosynovitis, synovitis and traumatic arthritis, bursitis, reflex vasomotor dystrophy, and diffuse aching of the forearm muscles. The three most common causes of RSI are poorly designed work stations, long periods of work involving rapid repetitions of finger movements, and the arms held virtually rigid with no relief from the strain.

Brown, D.A., Coyle, I.R. and Beaumont, P.E. (1985). *The Automated Hettinger Test in the Diagnosis and Prevention of Repetition Strain Injuries*. Applied Ergonomics, 16, 113–118. The Hettinger Test (listed in Table 2) measures changes in hand skin temperature following vibration. It is able to separate people who have confirmed RSI from those who have not been so injured. Therefore, the test should be able to predict whether persons are susceptible to such injury.

Kirschenbaum, A., Friedman, Z. and Melnik, A. (1986). *Performance of Disabled Persons on a Chordic Keyboard*. Human Factors, 28, 187–194.

A chord keyboard for use with one hand was designed to minimize physical exertion and to inhibit unwanted psychomotor reaction, so as to facilitate its use by disabled persons. After relatively short training, the disabled persons were able to successfully operate the keyboard and to interact with computers.

Gilad, I. and Pollatschek, M.A. (1986). *Layout Simulation for Keyboards*. Behaviour and Information Technology, 5, 273–281.

This article provides a brief overview of previous developments of typewriter keyboards, and subsequently presents a simulation tool for evaluating such layouts, which is claimed to provide a “cheap and effective way to convert any given layout to any desired one” (p. 273).

Gopher, D. (1986). *Experiments with a Two Hand Chord Keyboard – The Structure and Acquisition Process of a Complex Transcription Skill* (Technical Report HEIS-86-5, Grant N00014-84-0012, NR667-510, of U.S. Office of Naval Research). Haifa, Israel: Technion – Israel Institute of Technology.

This report summarizes experimental work with a new two-hand chord keyboard that was developed to provide an alternative to the existing standard keyboard. The report consists of four parts. The first three parts describe the results of experiments in which the new system was employed for the study of the properties of motor memory and the process of learning a complex transcription skill. The last part reviews the results of the effort to apply the principles that have emerged in this research to the development of a new typewriter. In conclusion, the author says that chord keyboards appear to be better data entry devices in the age of computers than previous standard keyboards.

Silverstein, B.A., Fine, L.J. and Armstrong, T.J. (1986). *Carpal Tunnel Syndrome: Causes and a Preventive Strategy*. Seminars in Occupational Medicine, 1(3), 213–221.

Within the last decade there has been an increased interest in carpal tunnel syndrome as an occupational disease. The interest involves concerns about the occupational causes of CTS, losses in productivity over time, rising worker's compensation costs, and ineffective or unnecessary treatment. While valid scientific information is not available to adequately address many of these concerns, sufficient information is available to discuss the known causes of CTS, and the components of a preventive strategy. Six major occupational factors have been proposed as causal factors in CTS: repetitive movements of the wrist or fingers loading the tendons in the carpal tunnel, forceful contraction using these tendons, extreme flexion or extension of the wrist including pinching or other awkward postures, mechanical stresses over the course of the median nerve in the hand, exposure to vibration, and use of poorly fitting gloves.

While it is not clear which of these factors are more important, the following factors are reviewed: force and repetitiveness, posture, vibration, mechanical stress, gloves and cold. Also, non-occupational factors are reviewed in detail.

A prevention strategy is discussed, divided into three sections. “Primary” prevention is meant to avoid the development of mild symptoms. For this, repetitiveness, posture, vibration, mechanical stresses, and gloves and cold are to be controlled. “Secondary” prevention is aimed to avoid the development of impairments. “Tertiary” prevention aims to avert recurrence.

Rosch, W.L. (1986). *New Looks for Replacement Keyboards*. PC Magazine, January Issue, 203–220.

This article describes three new keyboards that aim to solve the problem of how to key-in data faster and with less fatigue. The keyboards examined are the WICO SmartBoard, the KB5153, and the Maltron KIBM.

Standard Telephon und Radio AG, ITT (Ed.). (1986). *The Ergonomic STR-Keyboard*. Zuerich, Switzerland: Author.

This brochure describes the reasons for the development of the Ergonomic STR-Keyboard, namely to avoid unnatural positions of hand, arm, and torso, and to avoid static muscle work, tendon/muscle stress, reduction of blood flow, fatigue, pain, irritation, and rheumatic illnesses. The brochure describes how to arrange the workstation for use with the STR keyboards, including adjustment of the chair, positioning of the keyboard, and adjustment of the display.

Hagberg, M. and Sundelin, G. (1986). *Discomfort and Load on the Upper Trapezius Muscle When Operating a Wordprocessor*. Ergonomics, 29, 1637–1645.

EMG measurements of the trapezius muscle, and discomfort ratings, were obtained on six VDT word processor operators during their work. Each operator was studied during three work periods of three to five hours on different days. During one of the work periods, pauses of 15 seconds were introduced every six minutes. It was found that musculoskeletal discomfort during VDT work increased with the number of work hours. The introduction of short pauses on the muscular activities did not show a demonstrable effect on the EMG measurements, although the discomfort ratings were improved.

Armstrong, T.J. (1986). *Ergonomics and Cumulative Trauma Disorders*. Hand Clinics, 2, 553–565.

The author provides an overview of the development of the understanding of cumulative trauma disorders that were first described in 1717 by Ramazzini. In the 1960s, cumulative trauma disorders in “epidemic” proportions were reported. The author discusses repetitiveness, force, and mechanical stresses as factors of tendon disorders, as well as the effects of posture, vibration, and temperature.

Armstrong, T.J., Radwin, R.G., Hansen, D.J. and Kennedy, K.W. (1986). *Repetitive Trauma Disorders: Job Evaluation and Design*. Human Factors, 28, 325–336.

This is a discussion of the underlying reasons for repetitive trauma disorders, similar to the 1986 article by Armstrong.

Fine, L.J., Silverstein, B.A., Armstrong, T.J., Anderson, C.A. and Sugano, D.S. (1986). *Detection of Cumulative Trauma Disorders of Upper Extremities in the Workplace*. Journal of Occupational Medicine, 28, 674–678.

Three issues are addressed: (1) the objectives and specific features of surveillance of upper extremity cumulative trauma disorders in the workplace, (2) the use of existing data sources for surveillance, and (3) the use of questionnaire and physical examination data for surveillance. The efforts are to elucidate the causal factors of cumulative trauma disorders. The analysis of existing records is generally less costly, but the reliability of the data is difficult to assess. Standardized questionnaire and physical examinations, on the other hand, may increase the reliability of the results, but are associated with additional costs.

Sauter, S.L., Chapman, L.J. and Knutson, S.J. (1986). *Improving VDT Work. Causes and Control of Health Concerns in VDT Use*. University of Wisconsin: Department of Preventive Medicine.

This book reviews characteristics of ergonomic furniture and displays. It challenges the notion that VDT work must be necessarily associated with discomfort. There are many practical, low cost measures that can reduce everyday physical stress of VDT work. These measures can eliminate many of the unsatisfactory physical conditions experienced by workers. Individual characteristics, the nature of the job, and the design of the equipment at work place all play a role in affecting worker comfort. Problems are likely to arise from long periods of highly routine duties (p. iii).

The book is divided into 12 chapters, including chapters addressing musculoskeletal strain in general, back strain, neck and shoulder strain, problems in arms and hands, leg discomfort, eye strain and its prevention. The book concludes with several appendices, including checklists and exercises, and relaxation technique. With respect to arm and hand problems in VDT work, particular attention is paid to the irritation of the median nerve (carpal tunnel syndrome). Bending of the wrist is said to constrict the opening of the carpal tunnel. Compression of its content may damage the median nerve, a problem that is said to be one of the most serious problems that might occur in VDT use. Sustained and extreme bending of the wrist can bring about a carpal tunnel problem. Among the causes, even the “normal” typing posture involves an outward deviation (ulnar deviation), and thus increases pressure in the carpal tunnel. Another problem is the extension of the hand. Such hand posture is “common with a tall keyboard and no way to support the palms or wrists other than on a table top” (p. 29). Solutions are the use of a thin keyboard, and possibly

of wrist rests. Pressure at the front edge of a desk or VDT chassis may be substantial because a small area absorbs the full weight of the limb (p. 30).

Marsh, D.R. (1986). *Use of a Wheel Aesthesiometer for Testing Sensibility in the Hand-Results in Patients with Carpal Tunnel Syndrome*. Journal of Hand Surgery, 11B(2), 182–186.

The author describes a new device that can test the sensibility in the hand of patients with CTS before and after surgical decompression.

Statshaelsen (Ed.) (1986). *Computerization, Man and the Work Environment*. Karlskrona, Sweden: Swedish Foundation for Occupational Health and Safety for State Employees. This report of 136 pages discusses, in considerable detail, information on the impact of the computerization of work, and provides guidance on practical measures of industrial health and ergonomics. The report consists of several sections, including sections that address issues of occupational health, psychosociology, and technology/ergonomics. Two chapters concentrate on work with VDTs. The report also includes questionnaires and checklists.

Regarding the keyboard, it is said that the height of the keyboard should be 30 millimeters or less (p. 97). The key actuation force should be between 0.25 and 1.5 Newtons. Feedback must be provided in terms of a “snap action”, meaning a sudden release of key resistance at the actuation point of the key travel (p. 98). If a keyboard is too high to permit wrist resting on the table, a wrist support should be provided (p. 106).

Fry, H.J.H. (1986a). *Overuse Syndrome in Musicians - 100 Years Ago*. A Historical Review. The Medical Journal of Australia, 145, 620–625.

Overuse syndromes in musicians were extensively reported a century ago. Clinical features and results of treatment, which were recorded in considerable detail, match well the conditions occurring today. The medical literature that is reviewed here extends from 1830 to 1911 and includes 21 books and 54 articles. “The disorder was said to result from the overuse of the affected parts. ...No serious case was put forward for a psychogenic origin, though emotional factors were believed to aggravate the condition... Total rest from the mechanical use of the hand was the only effective treatment recorded” (p. 620)

Fry, H.J.H. (1986b). *Physical Signs in the Hands and Wrists Seen in the Overuse Injury Syndrome of the Upper Limb*. Australian and New Zealand Journal of Surgery, 56, 47–49.

Overuse injury is produced by many hand-use intensive activities performed by musicians, keyboard operators, process workers, writers, and others. The physical symptoms are common to all groups. There are two major symptoms: pain and mild fluctuating depression. Pain must come from some identifiable structure, and in the case of overuse injury syndrome from those structures that are over-worked.

Fry, H.J.H. (1986c). *Overuse Syndrome in Musicians: Prevention and Management*. The Lancet, September 27, 723–731.

The author provides a historical perspective, indicating that overuse syndromes in musicians were well known in the 19th century. Then, clinical features are discussed and related to playing techniques and practices. Prevention and management are discussed.

Fry, H.J.H. (1986d). *What's in a Name? The Musician's Anthology of Misuse*. In *Medical Problems of Performing Artists* (pp. 36–38). And: *Incidence of Overuse Syndrome in the Symphony Orchestra* (pp. 51–55). Philadelphia, PA: Hanley & Belfus.

The author discusses overuse injuries experienced by musicians, including those related to tendons and tendon sheaths, and the carpal tunnel. In terms of overuse syndromes, the grading of the severity of the injuries explained, followed by incidence statistics. The activities in playing certain instruments with respect to overuse illnesses are enumerated.

Armstrong, T.J., Fine, L.J. and Silverstein, B.A. (1986). *Cumulative Trauma Disorders and Keyboard Workstations – A Literature Review*. (Report for IBM Purchase, New York.) Ann Arbor, MI: The University of Michigan.

The authors first describe the various types of cumulative trauma disorders, then refer to epidemiological studies of which they say, on page 6: “They have yielded useful information primarily because the very strong relationship between occupational factors and the prevalence of these disorders.” Among the reported risk factors for CTDs, the authors list (in Table 2.6, page 14): repetitive and forceful exertions, posture (shoulder, forearm, wrist), pressure on the wrist and base of the palm, and constrained working postures. They also recognize the presence of “occupational neurosis” or “mass psychogenic illness”, and the relationships between subjective complaints and job stress (page 15). “An excess morbidity of chronic muscle, tendon and nerve disorders has been reported in nearly all hand intensive activities subjected to medical scrutiny including use of keyboard data entry devices for word processing and key punching. Use of keyboard data entry devices involves exposure to reported factors of cumulative trauma disorders including repetitive and sustained exertions and certain postures” (p. 17). The report ends with a listing of approximately 60 publications.

Hodges, A.N. (1986). *Adjustable Keyboard*. Patent No. 4597681. United States Patent Office.

This patent concerns an adjustable keyboard which is divided into sets of keys that can be individually adjusted in slope, tilt, and height.

Pollatschek, M.A. and Gilad, I. (1986). *The Customizable Keyboard*. *Behaviour and Information Technology*, 5, 283–287.

Based on the need to modify the QWERTY keyboard (discussed by Gilad and Pollatschek, 1986), the authors present means to customize any keyboard cheaply and effectively.

Seligman, P.J., Boiano, J. and Anderson, C. (1986). *Minneapolis Police Department* (HETA 84-417-1745). Cincinnati, OH: NIOSH.

This is the report on a NIOSH health hazard evaluation performed at the Minneapolis Police Department, based on occurrences of carpal tunnel syndrome and tendonitis of the wrist. CTD and tendonitis of the wrist were found to occur more frequently among police transcribers than in clerical staff. Factors that contributed were related to the positions of the hands and wrists during the typing of documents. Recommendations for improving the situation include the provision of wrist rests, chairs that can be quickly adjusted, typing tables that can be adjusted, and education of the employees about the proper use and adjustment of chairs and work surfaces to achieve the best ergonomic position.

Smith, M.J. (1986). *Mental and Physical Strain at VDT Workstations*. Transcript of a Presentation at Ergodesign: An International Conference on Design and Ergonomics. Montreux, Switzerland. October 21–24, 1986.

In this 27-page document, the author provides a state-of-the-art report regarding potential health effects of VDT use. The author discusses the following issues: reproduction and radiation, vision, musculoskeletal problems and stress. Regarding musculoskeletal problems, he states that they “... have become the foremost health concern associated with VDT use” (p. 9). Improper wrist bending and repetitive keying could produce tenosynovitis and/or carpal tunnel syndrome, as stated by Arndt in 1983 and 1986 and supported by research findings (p. 10). “There is an abundance of anecdotal evidence that VDT use can lead to serious wrist problems such as carpal tunnel syndrome given high keying demands...” (p. 10). While initially there was a lack of research evidence to support this contention, recent findings indicate that disability from wrist pain and tenosynovitis is prevalent among VDT users with high keying requirements. These disorders are the precursors of more serious wrist disorders, which take more time to develop. Rapid repetitive movements and static loading or sustained isometric contraction of muscles are postulated causes of repetitive strain injuries of the shoulder-neck area and the wrists-hands (p. 11). In conclusion, the author states that the research evidence for disabling health problems of the musculoskeletal system associated with VDT work is increasing. Field studies have demonstrated that large numbers of VDT users report a variety of muscular complaints, most notably of the back, neck/shoulders, and hand/wrists (pp. 15, 16).

Chatterjee, D.S. (1987). *Repetition Strain Injury – A Recent Review*. *Journal of Society of Occupational Medicine*, 37, 100–105.

The author reports, in detail, the knowledge about repetition strain injury, starting with Ramazzini's 1713 report on this condition among clerks and scribes. He states that there is no general agreement regarding the cause of RSI. The risk factors may be non-occupational as well as occupational. To avoid RSI, the author recommends proper work design, proper structure, and “competent personnel for proper case management.”

Hocking, B. (1987). *Epidemiological Aspects of “Repetition Strain Injury” in Telecom Australia*. *The Medical Journal of Australia*, 147, September Issue, 218–222.

This report provides details on the occurrences of RSIs in Australia, which showed a peak in 1984. Out of about 5000 clerical positions, there were 1421 reports of RSI. The relationship of RSI to keystroke rate, age, job duration, and posture is questioned.

Ferguson, D. (1987). *“RSI”: Putting the Epidemic to Rest*. *The Medical Journal of Australia*, 147, September Issue, 213–214.

This article critically evaluates the use of the term “repetition strain injury” (RSI) which may imply injury and cause when, in fact, neither may exist. It also describes the “Australian epidemic of RSI” in the late 1970s and early 1980s. In terms of the attributions of cause and attitudes to this syndrome, a change from blaming the epidemic on fixed postures and repeated actions at the VDT to an emphasis on mental and social factors is mentioned.

Richardson, R.M.M., Telson, R.U., Koch, C.G. and Chrysler, S.T. (1987). *Evaluation of Conventional, Serial, and Chord Keyboard Options for Mail Encoding*. In Proceedings of the 31st Annual Meeting of the Human Factors Society (pp. 911–915). Santa Monica, CA: Human Factors Society.

A study was performed to integrate a suitable data entry device in a workstation display unit. Keying performances were evaluated with a one-hand calculator keyboard, a two-hand serial keyboard, and a two-hand chord keyboard. Twelve subjects were employed, obtained from an agency for office temporaries. At reporting time, not all results were available, but the task was claimed to involve “maximum” speed, accuracy, and versatility of encoding destination addresses with “minimum” stress, fatigue, and training for the operators.

Silverstein, B.A., Fine, L.J. and Armstrong, T.J. (1987). *Occupational Factors and Carpal Tunnel Syndrome*. American Journal of Industrial Medicine, 11, 343–358.

This is a report of a study of 652 active workers in 29 jobs from seven industrial work sites. It was found that high force and highly repetitive work were strongly associated with carpal tunnel syndrome. Force and repetitiveness appeared to have a multiplicative effect, increasing the risk more than five times than either factor alone.

Wiklund, M.E., Dumas, J.S. and Hoffman, L.R. (1987). *Optimizing a Portable Terminal Keyboard for Combined One-Handed and Two-Handed Use*. In Proceedings of the 31st Annual Meeting of the Human Factors Society (pp. 585–589). Santa Monica, CA: The Human Factors Society.

An experiment was conducted to determine the effect of employing either one or both hands on typing speed and accuracy, depending on key spacing. Twelve subjects participated, selected to represent the range of typing skills expected from potential users of a down-sized keyboard. Each subject typed 12 paragraphs, six with two hands on all six keyboards investigated, and six with one hand on all six keyboards. Each trial was timed with a stopwatch. At the end of the test, each subject was asked about preferences for keyboards, and the level of fatigue they felt.

The subjects preferred the feel of the keyboard that had the most travel and kinesthetic feedback, which was a standard size keyboard. The next most preferred was a slightly down-sized keyboard. They judged a downsized calculator keyboard to be the most fatiguing.

Grandjean, E. (1987a). *Design of VDT Workstations*. Chapter 11.1 in G. Salvendy (Ed.), Handbook of Human Factors (pp. 1359–1397). New York, NY: Wiley.

This chapter describes, in considerable detail, design recommendations for VDT workstations. One section concerns the ergonomic design of keyboards (starting on page 1391). It contains a listing of dimensions of keyboards for which, it is said, “general agreement among experts (exists)”. It mentions a keyboard height above the desk of 30 mm at the middle row, and resistance of the keys of 0.4 to 0.8 N, with key displacements between 3 and 5 mm.

Raij, D., Gopher, D. and Kimchi, R. (1987). *Perceptual and Motor Determinants of Efficient Data Entry*. In Proceedings of the Human Factors Society 31st Annual Meeting (pp. 820–824). Santa Monica, CA: Human Factors Society.

Two separate scales were developed to assess the influence of motor and perceptual factors in chord typing. Two experi-

ments were performed, one to investigate the motor difficulty of executing chord combinations of five fingers, the other to investigate the perceptual difficulty associated with the spatial pattern created by each chord. The results of the two experiments were combined into regression equations that indicated about 60 percent of the reaction time variability.

Armstrong, T.J., Fine, L.J., Goldstein, S.A., Lifshitz, Y.R. and Silverstein, B.A. (1987). *Ergonomics Considerations in Hand and Wrist Tendinitis*. Journal of Hand Surgery, 12, No. 5, Part 2, 830–837.

The authors provide a historical perspective of hand and wrist tendinitis in workers, starting with the remarks of Ramazzini in 1717, continuing with Gray’s 1893 “Anatomy”, then referring to de Quervain’s 1895 description of the disease later named after him. In the 20th century, a fairly large number of reports refer to repetitive trauma injuries, starting in 1927 with Zollinger. The authors conclude their historical review with stating that there are non-occupational factors, such as congenital defects, acute injuries, chronic diseases, aging, gender, and recreational factors. Among the occupational factors are (in addition to acute injuries) repetitive exertions, forceful exertions, mechanical insult, awkward postures, and vibrations (p. 831).

The authors then describe a cross-sectional study undertaken by them. Out of 652 workers, 45 (6.9 percent) showed tendinitis. An analysis of the data indicated that the odds ratio of disease for high-force, high-repetitive workers was 29 times higher than of low-force, low-repetitive workers. Posture and vibration were not significant factors. Based on these findings, combined with a discussion of the strains in tendons, the authors recommend that ergonomic measures be taken to reduce the repetitiveness and forcefulness of industrial work.

Greenstein, J.S. and Arnaut, L.Y. (1987). *Human Factors Aspects of Manual Computer Input Devices*. Chapter 11.4 in G. Salvendy (Ed.), Handbook of Human Factors (pp. 1450–1489). New York, NY: Wiley.

This is a comprehensive chapter about human factor aspects of computer input devices, including keyboards, touch screen devices, graphic tablets, mice, trackballs, joysticks, and light pens.

Regarding keyboards, the advantages and disadvantages of the standard QWERTY layout are discussed. Among the disadvantages, the following are noted: overloading of the left hand, overloading of certain fingers, too little typing done on the home row, excessive row hopping, many common words typed by the left hand alone. The authors state that size and spacing of keys on general-purpose alphanumeric keyboards are largely based on design conventions rather than on empirical data. Most keyboards appear to exhibit a rapid buildup of force as the key is depressed, with a reduction in the required force in the region of activation, followed by a second increase in force thereafter. The general recommendations for activation force are 0.9 to 5.3 ounces of force (0.25 to 1.47 N), and the total displacement 0.05 to 0.25 inches (1 to 6 mm) (p. 1461). Regarding feedback with full-travel keyboards, kinesthetic-proprioceptive-tactual feedback is claimed to be the major source of feedback, especially for skilled users. Visual feedback is important during training, and for error correction and editing tasks. It does not affect input accuracy and speed. Supplementary auditory feedback may provide modest enhancement, but inherent feedback is typically sufficient (pp. 1463–4).

Blair, S.J. and Bear-Lehman, J. (1987). *Editorial Comment: Prevention of Upper Extremity Occupational Disorders*. Journal of Hand Surgery, 12A(5) Part 2, 821–822.

Cumulative trauma disorders may develop in the soft tissues, most frequently in the tendons, muscles, and nerves or possibly the vascular system in response to repeated overexertions and excessive movements of the human body. The various nerve compression syndromes, tenosynovitis, tendonitis, and muscle strains may be included in this classification if their cause can be traced to a mechanical origin. The U.S. Bureau of Labor Statistics lists occupational illnesses associated with repeated trauma as follows: in 1980, 23 300; in 1983, 26 000; in 1984, 34 700 (p. 821). The number and cost of non-traumatic cumulative trauma disorders have risen. Many countries have experienced and reported similar problems. Australia had an epidemic of such disorders in the past several years (p. 822).

Louis, D.S. (1987). *Cumulative Trauma Disorders*. Journal of Hand Surgery, 12A(5), Part 2, 823–825.

The author describes the “predictable and inexorable course of the worker” ...developing a...“variety of classic symptoms, such as numbness and tingling in the median nerve distribution...” (p. 824). This is caused by on-the-job exposure that usually involves repetitive or hand-intensive activities, although recreational activities may also play a part in the genesis of the symptoms. If symptoms do not respond to conservative treatment, then hand surgery may be indicated.

Silverstein, B., Fine, L. and Stetson, D. (1987). *Hand-Wrist Disorders Among Investment Casting Plant Workers*. Journal of Hand Surgery, 12A(5), Part 2, 838–844.

A re-survey of 136 workers in a casting plant was performed three years after the initial survey. Nearly 25 percent of the originally surveyed workers had moved to different jobs. Workers with hand-wrist disorders in jobs that required high force and high repetition rates tended to transfer out of these jobs more frequently than those that had low-force and low-repetitive jobs. Ergonomic changes did not seem to change the prevalence of chronic hand-wrist disorders, which is thought to be possibly due to the fact that the ergonomic changes did not substantially alter the forces and repetitiveness of the jobs.

Responding to criticisms in previous studies, psychosocial factors were addressed in this investigation. However, no significant differences were observed and the reports of job satisfaction or job interest between groups with and without reported CTDs were not substantially different. Thus, at least in this plant, psychosocial factors did not appear to be important predictors.

Bleeker, M.L. (1987). *Medical Surveillance for Carpal Tunnel Syndrome in Workers*. Journal of Hand Surgery, 12A(5), Part 2, 845–848.

Carpal tunnel size was examined as a possible wrist factor for carpal tunnel syndrome. In a study, seven of 14 electricians had symptoms of CTS. Measuring the cross-section of the wrist by computerized tomography indicated that affected workers had, statistically, a smaller cross-sectional area than non-affected workers. Wrist circumference was not a predictor of the smallest carpal canal area.

Feldman, R.G., Travers, P.H., Chirico-Post, J. and Keyserling, W.M. (1987). *Risk Assessment in Electronic Assembly Workers: Carpal Tunnel Syndrome*. Journal of Hand Surgery, 12A(5), Part 2, 849–855.

The authors describe the results of an on-site survey in an electronic assembly plant, a biomechanical analysis of work aspects, and a special neurologic assessment. On the basis of the findings, specific recommendations to improve the conditions and to prevent the development of CTS were made. These recommendations include the introduction of mechanical assistance devices, reduction in the weight of tool and equipment, and reduction in the reach distances.

Gomer, F.E., Silverstein, L.D., Berg, W.K. and Lassiter, D.L. (1987). *Changes in Electromyographic Activity Associated with Occupational Stress and Poor Performance in the Workplace*. Human Factors, 29, 131–143.

Surface EMGs were recorded during machine-paced keyboard operation. They were found to be indicative of localized muscle fatigue in the forearms, even although the work did not require very forceful muscular exertions.

Rossignol, A.M., Morse, E.P., Summers, V.M. and Pagnotto, L.D. (1987). *Video Display Terminal Use and Reported Health Symptoms Among Massachusetts Clerical Workers*. Journal of Occupational Medicine, 29, 112–118.

This is a report of a cross-sectional epidemiologic study on 1,545 Massachusetts clerical workers. The study indicated increased prevalence of adverse conditions pertaining to vision, musculoskeletal discomfort, and headaches among clerical workers who used VDTs. The magnitude of the increased prevalence was dose-dependent.

Seror, P. (1987). *Tinel's Sign in the Diagnosis of Carpal Tunnel Syndrome*. Journal of Hand Surgery, 12-B(3), 364–365.

The author states that Tinel's sign (“tingling sensation”) has been used as one of the indicators of CTS. The author describes that a positive Tinel's sign was found in the 63 percent of patients with electrophysiology proved carpal tunnel syndrome, but it was also positive in 45 percent of controls. Thus, the diagnostic value of Tinel's sign for CTS diagnosis is questioned.

Ilg, R. (1987). *Ergonomic Keyboard Design*. Behaviour and Information Technology, 6(3), 303–309.

A systematic breakdown of design alternatives concerning keyboards is presented. This includes the following features:

- * Distance between hands;
- * Geometric dimension spacing;
- * Horizontal key spacing;
- * Keyboard tilt (lateral inclination);
- * Slope (frontal inclination);
- * Curvature longitudinal to the fingers;
- * Curvature transverse to the fingers;
- * Depth of depression (coarse test);
- * Depth of depression (precise test);
- * Force (coarse test);
- * Force (precise test);
- * Key design (in side view);
- * Key design (in top view);
- * Key size.

These parameters were systematically investigated over a period of five years under “real life conditions”. Forty subjects, at three levels of user experience, participated. Performance data were reported, as well as acceptance data.

A marketable ergonomic keyboard was developed that utilizes, among other findings, the separation of left and right hand keyboards, which are laterally tilted. A key travel of four millimeter was chosen, with a key force between 0.2 and 0.7

Newtons. A hand rest was incorporated because all subjects desired it.

Sauter, S.L., Chapman, L.J., Knutson, S.J. and Anderson, H.A. (1987). Case Example of Wrist Trauma in Keyboard Use. Ergonomics, 18, 183–186.

This report describes injury to the pisiform bone in the wrist, and to the dorsal sensory branch of the ulnar nerve of a VDT operator. The person had performed data entry work by keyboard for 20 years and used to rest the wrist against the leading edge of the keyboard. The author suggests the self-evident design principle that there should not be a sharp pressure point toward the wrist. It appears advisable for the edge of the keyboard or wrist rest to be gently contoured, and/or for the surface to be lightly padded.

Grandjean, E. (1987b). Ergonomics in Computerized Offices. London, UK: Taylor & Francis.

This book of 226 pages describes main ergonomic aspects in the design of computerized offices, especially computer workstations. The main topics are vision, lighting, photometry, design of VDT workstations, noise, stress, and radiation. The book concludes with a list of ergonomic recommendations for VDT workstations. Grandjean explicitly recommends wrist supports while keyboarding (pp. 138–141), keyboards with a home row no higher than three cm above the support surface, with the keyboard freely movable according to the operator's personal needs (p. 151), and split keyboards (pp. 151–153).

Herzog, S. and Herzog, B. (1987). Biomechanical Keyboard Structure and Method. Patent 4 669 903. United States Patent Office.

This patent concerns the design of a keyboard in which keys are selectively positioned and aligned to allow proper posture of the left forearm and hand.

Kiser, D.M (1987). Physiologic and Biomechanical Factors for Understanding Repetitive Motion Injuries. Seminars in Occupational Medicine 2(1), 11–17.

This text explains, in physiologic and biomechanic terms, how repetitive motion injuries develop. Body systems adapt to the demands of the task up to their maximum capacity. Good tool and task design decreases work demands, poor design increases them.

After a discussion of aerobic and anaerobic muscle work, of muscle mechanics including fatigue, and of biomechanic aspects, it is explained how seemingly "...low-load hand activities can result in significant stress on muscle and tendons. ... Rapid changes in direction, as occur during cyclic motions at high frequencies, result in high acceleration of body segments. The rapid and cyclic wrist extension-flexion described earlier is one example of a situation where the muscle forces could become excessive because of the rapid changes in direction (p. 15). ... it can be seen that when the wrist is in a neutral position there tends to be little interference between the tendons, nerves, and other supporting tissues, even during forceful contraction of the finger flexor muscles. However, tasks that force the wrist into extension, flexion, radial, or ulnar deviation, may lead to less efficient transmission of muscle contractile force to the fingers or result in direct compression or irritation of the median nerve or synovial tissue (p. 16)."

Goldstein, S.A., Armstrong, T.J., Chaffin, D.B. and Matthews, L.S. (1987). Analysis of Cumulative Strain in Tendons and Tendon Sheaths. Journal Biomechanics, 20, 1–6.

In 12 human cadaver hands, 25 digitorum profundus flexors were put under step stress and cyclic loadings. Using special strain transducers it was found that an accumulation of strain occurs in the tendinous tissues.

Patkin, M. (1988). Hand and Arm Pain in Office Workers. Modern Medicine of Australia, October 1988 Issue, 66–76.

This paper discusses the various types of cumulative trauma injuries to the hand-arm region. For prevention, it is suggested that ergonomic job design and cooperation between employees, management, and the physician are important. This paper discusses the necessity for proper medical diagnosis and the consideration of "work-attributed symptoms...are often a way for expressing personal problems, sometimes ones arising from work" (pp. 67, 68).

Kroemer, K.H.E. (1988, reprinted 1991). VDT Workstation Design. In M. Helander (Ed.), Handbook of Human-Computer Interaction (pp. 521–539). Amsterdam, NL: Elsevier Science Publishers (North Holland).

This is a survey of ergonomic measures for the design of workstations. Emphasis is put on achieving a suitable posture which is largely the result of the interactions among the placement of visual targets (such as the computer screen), of input devices (such as the keyboard) and of the seat. It is postulated that ill-designed and ill-arranged computer workstations may lead to health complaints and attitude problems, while ergonomic conditions further well-being, both physically and psychologically.

Pfeffer, G.B., Gelberman, R.H., Boyes, J.H. and Rydevik, B. (1988). The History of Carpal Tunnel Syndrome. Journal of Hand Surgery, 13-B(1), 28–34.

In 1880, Putnam published probably the first report on a previously not described condition of pain: paresthesiae in the median nerve distribution of the hand. He saw a large number of cases in his out-patient department of the Massachusetts General Hospital. Disturbance of the sensibility of the skin was common, giving rise to numbness, especially at night, affecting one or both hands. In 1893, Schultz, and in 1909, Hunt described similar events. Hunt produced a set of publications on the topic in 1911, 1914, and 1950. Yet, these authors, and especially Hunt, focused on the motor branch of the median nerve, and de-emphasized sensory symptoms, as did Marie and Foix in 1913.

The authors describe how, from the 1900s on into the 1940s, the emphasis continued to be on the motor branch of the median nerve, and on blood supply to the hand, in spite of the fact that Learmonth in 1933 published a paper on the use of decompression of the carpal tunnel by cutting a ligament. In 1941, Woltman followed that example at the Mayo Clinic by discussing how compression of the median nerve may be due to an increase of the soft tissue content of the carpal tunnel. Again at the Mayo Clinic, Cannon and Love in 1946 reported on carpal tunnel release on nine surgically treated patients. In 1947, neurologists and clinicians, followed by surgeons, diagnosed compression of the carpal tunnel content, and its release by surgical procedures on the ligaments became commonly accepted. Brains, Wilkinson, and Wright, in 1947, published an influential paper that inspired Phalen, in 1950, to publish the first of a series of articles on carpal tunnel syndrome which did much to popularize the diagnosis and treatment of CTS in the medical community. His definition of CTS was soon expanded to include all causes of median nerve compression in the carpal tunnel.

"By 1960, median nerve compression in the carpal tunnel had become the most frequently diagnosed, best understood and most easily treated entrapment neuropathy in medical practice" (p. 34).

Kiesler, S. and Finholt, T. (1988). The Mystery of RSI. American Psychologists, 43, 1004–1015.

The authors discuss the appearance and nature of the RSI problem in Australia. The computer-related health epidemic known as repetitive strain injury RSI was rampant in Australia. The authors speculate that the spread of RSI in Australia is indicative of a larger social problem, especially in typing pools. There exists a negative psychosocial condition, which is overlaid with repetitive keyboarding. A second reason is, in the opinion of the authors, that RSI has become a labor-relations issue, and specifically a concern of feminist groups in Australia. Another reason, according to the authors, is that RSI has received official validation by the Australian medical establishment. Finally, the authors claim that the Australian press has played a large role in popularizing RSI. The authors close by stating that they do not intend to suggest that RSI is a scam. Yet, they believe that if the work environment was better and the jobs more satisfying, RSI complaints would not be as important. "The epidemic in Australia ostensibly involves RSI, but we hypothesize that it is really related to bad work conditions and an unfulfilling work life" (p. 1012). "RSI stands at the intersection of three traditional categories of illness: physical overuse injury, psychophysiological syndrome, and hysterical or conversion illness...RSI, falling somewhere in these categories, is a classic demonstration of some abnormalities and inadequacies of the existing paradigm of clinical knowledge. It seems clear that the professional and public definition of an illness/injury such as RSI does not occur in a vacuum and that it is important to try to understand its social, organizational, economic, and political context" (p. 1013).

Foster, L. and Fry, H. (1988). Conflicts in Medical Knowledge – The Case of "RSI". Science and Technology in Australia and New Zealand, 19(4), (pagination not legible).

The authors discuss, in great detail and clarity, the terminology applied to the "RSI epidemic" in the early 1980s in Australia, and the underlying causes. A very large number of cases were suddenly reported in Australia, which did not fit any of the commonly taught and written recognized specific clinical entities. Medical practitioners who had little experience in the management of the cases, were asked to see scores of patients. The nature of the disorder itself caused difficulty because one of the symptoms was pain, but there were often no conventional physical signs. None of the conventional treatments seemed to work and the patients did not get better. Even if they returned to work, it was only a matter of time before they had to go off work again. There appeared to be no way of dealing with the situation, and there were no modern textbooks or journal articles to help the medical practitioner. Thus, the term "repetition strain injury" (RSI) was introduced which, however, included a large number of previously properly defined and termed symptoms and also non-traditional aspects that referred to psychological stress at the workplace, ease of obtaining welfare payments, etc.

In 1985, a strongly worded letter appeared in the Medical Journal of Australia in which RSI was called a disease unique to Australia that had no physical signs. The letter writer expressed in 1986 the opinion that "conversion hysteria" was the most likely explanation of the symptoms. The authors state

that the literature available at that time would have shown that these statements were not correct, but they nevertheless made an impact. Around 1986, another letter appeared in the Medical Journal of Australia that stated RSI to be an occupational phenomenon, which had no physical basis and was uniquely Australian. Thus, the impression was gained that RSI was essentially a disorder of employed people with low job satisfaction. The use of the term RSI was officially buried at the end of 1986 by the Occupational Health and Safety Commission of Australia.

The authors discuss how a number of conventions in medicine were broken, especially that terminology should remain as specific as possible. Eleven specific clinical entities were known and well described in the literature, but later included in an unspecific manner in the term RSI. The eleven "non-controversial" clinical entities include, among others, carpal tunnel syndrome, nerve entrapment, osteoarthritis, and tenosynovitis. Furthermore, the authors list three "controversial" clinical entities, namely conversion hysteria, malingering, and overuse syndrome, which were also commonly included in the RSI set.

Gopher, D. and Raij, D. (1988). Typing with a Two-Hand Chord Keyboard: Will the QWERTY Become Obsolete? IEEE Transactions on Systems, Man, and Cybernetics, 18, 601–609. The chord keyboard investigated consists of two separate panels, one for the left and one for the right hand. On each panel, five regular binary keys are located in a hand-configured fashion. For the experiments, the chord keyboards were arranged vertically, and the regular QWERTY keyboard horizontally. Subjects used either both parts of the chord keyboard, or only the right panel for right-hand inputs only. The results indicated that the learning curves on the single- and two-hand chord keyboards were faster than on the standard QWERTY keyboard. Learning on the one-hand or two-hand chord keyboards was essentially the same. The authors conclude that substantial gains can be made by employing chord keyboards of the kind used in this experiment, particularly since the chord operation has cognitive advantages over the traditional QWERTY layout.

Carr, R.L. (1988). Determination of Forces Applied on a Calculator Keypad Through a Computer Interface. Unpublished MS Thesis, Texas Tech University, Lubbock, TX.

Thirty subjects, 15 male and 15 female, pushed prescribed keys on a Texas Instruments (TI-60) calculator. The results showed that males, on the average, exerted about 12 percent of their maximal finger forces, while females, on average, exerted about 16 percent of their maximum capability. Different keys were pressed with different forces.

Human Factors Society, Inc. (Ed.). (1988). American National Standard for Human Factors Engineering of Visual Display Terminal Workstations (ANSI/HFS 100-1988). Santa Monica, CA: Human Factors Society.

This standard is divided into several sections. In Sect. 1 (p. 3) it is stated that this technical standard specifies conditions that have been established as representing acceptable implementation of human factors engineering principles and practices. Section 7 (pp. 35–40) is devoted to keyboards, claimed to be simple adaptations of the basic typewriter keyboard (p. 35). In the discussion of keyboard height and slope (p. 37), it is said that keyboards, in combination with their supporting surface, chair, and other furniture elements, shall permit users to adopt and maintain arm positions stated in Sect. 8.4.1. In

that section it is said that the user shall be permitted to adopt a posture in which the elbow angle is between 70 and 135 degrees (with the upper arm essentially vertical). The keyboard slope shall be between 0 and 25 degrees. Some typists prefer to use palm rests with keyboards. However, such palm rests should be an optional feature (p. 37).

In the section on key travel (7.9, p. 39), it is said that keys shall have a maximum vertical displacement between 1.5 and 6.0 mm. The preferred displacement is between 2 and 4 mm.

Regarding key force (7.10, p. 39), it is said that the maximum force to depress keys shall range between 0.25 and 1.5 N. The preferred key force is between 0.5 and 0.6 N. The reference for key force measurement is section 9.19, p. 77. There, the following is stated: "The maximum force to the" make "point shall be measured in the geometric center of the key top strike area along the same axis as the key travel".

Regarding keying feedback (7.11, p. 39): "Actuation of a key shall be accompanied by tactile or auditory feedback or both. If there is only one, tactile feedback is preferred". Further, in the discussion: "It is recommended that key forces should increase with vertical displacement to approximately 40 percent of the total displacement, followed by a sharp and substantial reduction in force over approximately 20 percent of the total displacement, followed by an increasing force over the remaining displacement. As the key is displaced, there should be a definite 'breakaway' force to acknowledge to the operator that the key stroke has in fact been registered. Subsequent to this reduction in required force, the force should then increase again to the bottom of the key travel. Elimination of either the breakaway force reduction or the secondary force increase for cushioning may result in a slower keying activity, higher error rates, and increased operator fatigue" (p. 39).

Rose, M.J. (1988). Toward an Ergonomic Keyboard Design for Input of Chinese Text. Pre-Prints, International Conference on Ergonomics, Occupational Safety and Health and the Environment, 24-28 October 1988 (pp. 153-163). Beijing, PRC: The Chinese Society of Metals/Australian Darling Downs Institute of Advanced Education.

Based on the experiences made in the "western world", recommendations are made concerning keyboards for the input of Chinese text. The maximal number of keys should be approximately 44. Posture of the hands, to optimize strength and reduce static muscle contraction, should be such that the hands are not fully pronated. The wrist should be kept approximately straight, and ulnar deviation should be avoided. To split the keyboard, and to tilt each side downward may be a suitable solution. Also, the two sides should be slanted with respect to each other to allow a straight wrist position. Keys may be arranged to provide a convex shape to the fingers.

Baidya, K.N. and Stevenson, M.G. (1988). Effects of Rest Breaks on Local Muscle Fatigue During Repetitive Work. Ergonomics International, 88, 421-423.

The authors explain that, according to previous research, EMG measurements, particularly changes in the center frequencies, are sensitive to fatigue. An experiment was conducted in which seven subjects performed wrist extensions, at a rate of 45 per minute, for a period of 30 minutes. The torque against which the wrist had to work was 0.25 Nm, which was, on average, only three percent of the maximal voluntary contraction of the subjects. The results showed that the center frequency change was quite sensitive to fatigue, in spite of the relatively light work.

Hobday, S.W. (1988). A Keyboard to Increase Productivity and Reduce Postural Stress. In F. Aghazadeh (Ed.), Trends in Ergonomics/Human Factors V (pp. 321-330). Amsterdam, NL: Elsevier.

The author explains that current keyboards often give rise to cumulative trauma disorders. Such events could be avoided by using a better keyboard. A particular problem is the "outward turning" (ulnar deviation) of the wrists. Splitting the keyboard and separating the two parts could reduce ulnar deviation. Furthermore, arranging the key tops so that they form a convex dish-shape would allow easier finger movements. The resulting keyboard, called Maltron, has been refined since 1977.

Green, R.A., Briggs, C.A. and Wrigley, T.V. (1988). Preferred Posture of Keyboard Operators as Determined by Using Long-Term Computerized Video Based Analysis. Ergonomics International, 88, 439-441.

The authors observed the locations of joint centers of keyboard operators using a video camera. Each recording ran for four hours during actual working periods. Eight hours of video recordings of 15 keyboard operators were obtained. The comparison of the results obtained indicated large variability at wrist, elbow, shoulder, trunk and hip, similar to the ranges previously recorded by Grandjean et al. (1983) and Starr et al. (1985). The authors conclude that it is important that furniture in the working environment should cater to a full range of preferred postures.

Bjoerksten, M.G. (1988). Neck and Shoulder Disorders Among Medical Secretaries in Relation to Muscular Activity During Typewriting. Ergonomics International, 88, 381-383.

Three hundred eighty-one female medical secretaries from two hospitals, who had been on the same task for the last 12 months, answered specific questionnaires regarding the locomotor system. In a second study, 12 healthy female medical secretaries performed three different typing tasks. EMG measurements were taken from the right trapezius muscle. The results of the first study showed a high prevalence of neck problems (53 percent), and shoulder problems (47 percent), together with low back problems (42 percent). The EMG data indicated different levels of activities depending on the typing task.

Molan, R. and Sikorski, J.M. (1988). Do Keyboard Operators with R.S.I. Work Differently to Their Pain-Free Workmates? Ergonomics International, 88, 390-392.

Thirty-one data entry operators were studied, particularly with respect to right forearm and shoulder. Subjects were videotaped while they keyed. Great variation was observed in work style. No significant differences were found in body movements, including shoulder, wrist, thumb, and finger. Production rates, pause rates, and key stroke rates were similar. The only difference was found in the use of the right ring finger, which was significantly more frequent in the persons free of pain.

Erdelyi, A., Sihvonen, T., Helin, P. and Haenninen, O. (1988). Shoulder Strain in Keyboard Workers and Its Alleviation by Arm Supports. Int. Arch. Occupational and Environmental Health, 60, 119-124.

Twenty experienced word processors and typists participated in an experiment. Twelve were suffering from shoulder and neck pains, while eight had no problems. Subjects were seated in various postures, particularly elbow angles, and provided

with arm supports. EMG signals at the trapezius muscles were recorded while the subjects performed keyboard work. The EMG activities were reduced with larger elbow angle, and pain sufferers showed less EMG activity when using arm supports. The authors conclude that the elbow angle should be larger than 100 degrees, and that supports for the arms are reasonable means of reducing muscular activities in the shoulders. The authors conclude that use of computerized surface EMG as a suitable means to find individually suitable conditions.

Itani, T., Otani, T., Babazono, T., Ogawa, T. and Aoyama, H. (1988). Electromyographic Evaluation of the Muscular Load in Shoulder and Forearm Muscles During Different Types of VDT Work Among Trained and Untrained Operators. Ergonomics International, 88, 433–435.

Four subjects participated in this study. Two of them could perform touch typing, while the other two could not. While they were typing manuscripts, surface EMGs were obtained from the trapezius and extensor carpi radialis brevis of both sides. The activity levels of the extensor carpi muscles were lower in the untrained operators than in the trained operators. However, the activity levels of the trapezius muscles were higher in the untrained operators.

Krueger, H., Hinnen, U., Guggenbuehl, U. and Laeubli, T. (1988). Musculoskeletal Disorders at Different Work Places. 88 Ergonomics International, 445–447.

Questionnaires were distributed at office work places with and without VDUs, work places with cash desks, and work places with microscopes and loupes, and the workers underwent a medical checkup. This report concerns only subjective complaints. Their number rises with daily working time. They also increase along with the repetitiveness and monotony of the work. The authors conclude that their results support the hypothesis that repetitive and monotonous work is one reason for musculoskeletal complaints.

Nathan, P.A., Meadows, K.D. and Doyle, L.S. (1988). Occupation as a Risk Factor for Impaired Sensory Conduction of the Median Nerve at the Carpal Tunnel. Journal of Hand Surgery, 13-B(2), 167–170.

A sample population of employees randomly selected from four industries was surveyed. A medical history questionnaire was administered, and nerve conduction speeds of both hands were measured in all subjects. In the results obtained on 471 employees, no consistent association was found between the type and level of occupational hand activity and the prevalence or the severity of slowing of sensory conduction of the median nerve at the carpal tunnel. Slowing was not associated with bi-manual occupational hand activity, and the length of employment did not influence the occurrence of impaired sensory conduction.

Langley, L.W. (1988). Ternary Chord-Type Keyboard. Patent No. 4775 255. United States Patent Office.

The patent describes a keyboard with keys that have three possible positions. The preferred design has two sets of four ternary keys, one set for each hand. This keyboard of eight keys does not require movement of the fingers from one key to another.

Kroemer, K.H.E., Fathallah, F.A. and Langley, L.W. (1988). A New Keyboard with Chorded Keys. In Proceedings of the 32nd Annual Meeting of the Human Factors Society (pp. 724–726). Santa Monica, CA: Human Factors Society.

Experiments on a new ternary chord keyboard (US Patent 4775 255, Langley 1988) indicated fast keying performance.

Putz-Anderson, V. (Ed.) (1988). Cumulative Trauma Disorders: A Manual for Musculoskeletal Diseases of the Upper Limbs. London, UK: Taylor & Francis.

This manual describes and defines cumulative trauma disorders (CTDs), especially of the upper extremities. The stated purpose is of enabling non-medical personnel to recognize them and to present strategies to prevent their occurrence.

The manual is divided into three parts. Part 1 defines CTD categories and describes details of CTDs and the afflicted body organs, especially the main groups of tendon, nerve and neurovascular disorders. It describes structures of the hand and arm to help locate and identify CTD symptoms. Among the examples of activities with work-related CTDs are typing, keypunching and cashiering with these “occupational factors” listed: “static, restricted posture, arms abducted/flexed, high speed finger movement, palmar base pressure, ulnar deviation” (p. 23).

Part 2 presents survey and recording methods for determining the number of work site occurrences of CTDs or early symptoms thereof. Job analysis is said to be particularly useful for identifying work conditions and tools that may cause or contribute to CTDs.

Part 3 presents two major strategies for controlling or preventing CTDs: management and engineering, often used in combination. Guidelines for ergonomic re-design rely on the principle “to make the job fit the person, not to make the person fit the job” by redesigning the job or the tool to reduce the job demands of force, repetition and posture (p. 85).

Helander, M. (Ed.) (1988). Handbook of Human-Computer Interaction. Amsterdam, NL: Elsevier Science Publishers (North Holland).

A comprehensive treatment of practically all topics of concern regarding the design of computers, and their use, with respect to the human operator, as known to engineers and behavioral scientists in the middle 1980s.

Burnette, J.T. and Ayoub, M.A. (1989a). Cumulative Trauma Disorders. Part 1. The Problem. Pain Management, July-Aug. Issue, 196–209.

This is a review of 96 CTD-related publications, apparently meant to become a program describing a computer-based integrative model to facilitate collection and interpretation of the necessary information (p. 196). The review concerns the definition of CTD, their prevalence and cost. Ensuing, the pathology and etiology of CTDs is discussed. Though it is said that the pathology is not known, it is usually attributed to inflammation, impaired blood flow, or peripheral neuromuscular fatigue. Mechanical failure, depletion of synovial fluid around the tendons, and disruption of metabolism are also mentioned. Both upper and lower extremities can exhibit symptoms of CTDs. Reduction of full function in median, ulnar, and radial nerves is said to bring about serious impairment.

The final sections of the review concern treatment and rehabilitation, and prevention. Re-designing of job and workplace are claimed to be of paramount importance.

Burnette, J.T. and Ayoub, M.A. (1989b). Cumulative Trauma Disorders: Part II. Assessment of Risk. Pain Management, Sept.-Oct. Issue, 256-264.

This review builds on the previous Part I. It constitutes the basis for a program that can assimilate the knowledge and skills of several disciplines in addressing all aspects of work. Such an ideal CTD model would be reliable and consistent, valid in construct and content, easy to use, cost effective, relevant to existing records, and would have a modular nature to allow incorporation of new data. Finally, it should supply the ability to measure work in all dimensions, and static loadings as well as dynamic and intermittent activities. The proposed model, called CTD-123, is presented in this article. In its simplest form, the model indicates risk as job stress divided by moderating factors.

The model input factors include information about the persons involved, their postures, strengths and torque requirements and cycle time characteristics. The risk-moderating factors concern the working environment, workplace design, treatment and rehabilitation of CTD sufferers, and employee screening and conditioning. The model outputs are in terms of force, frequency, and joint strain. The overall CTD risk is calculated, together with the total job risk.

Field tests were performed using a series of videotapes of eight jobs. The results indicated that the CTD-123 program accurately predicted the order of risk for each joint of all eight jobs.

Ayoub, M.A. and Wittels, N.E. (1989). Cumulative Trauma Disorders. International Reviews of Ergonomics, 2, 217-272. This is a review of nearly 2000 publications related to CTDs. In the "overview" section, CTDs are discussed and defined. They are said to share the following common characteristics:

1. They are not a result of sudden or spontaneous injury, i.e., they are not accidental.
2. They result from sustained or repetitive application of low stress (trauma) over time.
3. They may also result from stress applied to previously injured or already diseased structures (p. 219).

In the section on "who is at risk?", twenty kinds of jobs with high risks are listed. These include typing, key punching, and cashing. As occupational factors, the following are listed: static, restricted posture, arms abducted/flexed, high speed finger movement, palmar base pressure, ulnar deviation (p. 219).

The authors say that "CTDs are the final outcome of a long series of pain and discomfort. ...As pain is allowed to progress unchecked, the development of CTDs is almost certain. ...Most estimates of CTD prevalence are considered to be underestimates because they are based on the late stage or compensation cases" (p. 221). "Most of the reported disorders occurred in the upper extremity, with the hand/wrist area accounting for about 60% of all the cases." ...The involvement of the shoulder paralleled that of neck and upper back. "...The disorders were attributed to a number of causes; chief among them was repetitive and forceful motions." All the causes cited reflect various types of ergonomic deficiencies in job content, in workplace design and/or matching individuals to their respective jobs (p. 222).

Concerning the development of the problem, CTDs are said to be not new, but to have been around for centuries. The first case of carpal tunnel syndrome dates back to 1853. De Quervain's tendonitis was reported in 1895. Yet, in spite of the prevalence of CTDs in many industries, and its association with significant personnel and monetary loss, there are no en-

forced regulations or requirements to control them. In the absence of CTD-specific standards, OSHA relies on its "general duty clause" to ensure that employers provide a workplace free from ergonomic deficiencies. But in Sweden, a specific standard directly related to CTD became effective in 1984. Other countries, notably Japan, designated CTDs as a significant problem that warranted an industry-wide program of research and intervention. In the USA, NIOSH named CTDs in 1983 as one of the ten major hazards affecting people at work (pp. 226-228). The review then continues to discuss the pathomechanics of CTDs in terms of a stress/strain relationship of viscoelastic responses of soft body tissues, rate of stress application, and static and dynamic situations.

The authors continue to describe the aetiology. CTDs are said to be the direct consequence of soft tissue failure, either due to physical overload or as a by-product of impairment of periphery circulation, or as a result of infection and systemic disease. Four major factors are said to be the causative bases for CTDs: heavy load, mental stress, disturbance of periphery circulation, and changes in sensory and motor nervous function. The authors state that, at this time, there is not sufficient knowledge to attach a threshold level to the stress/strain values that may be tolerated by various body tissues.

Hadler, N.M. (1989). The Roles of Work and of Working in Disorders of the Upper Extremity. Bailliere's Clinical Rheumatology, 3, 121-141.

The author states two extreme positions. One is that any specific regional arm discomfort in a worker is to be ascribed to the workers task. The opposite position is that such discomforts are disorders of ordinary life, which are manifest as a degree of work incapacity because usage exacerbates the discomfort. These two points of view lead to very different consequences and ramifications.

The author cites a number of previous researches, and focuses in particular on arm pain in the workplace, as investigated by the Occupational Health and Safety Engineering department at the University of Michigan (specifically, the research performed by B.A. Silverstein for her dissertation). According to Hadler, the basic tenet of this investigation is that forceful and/or repetitive exertions of the upper extremity can, over the course of time, cause or precipitate clinically important disorders. "There are no impressive, let alone incontrovertible, data to support the contention that any upper extremity usage, within reason, is damaging.... In fact there is precious little evidence to suggest that such usage increases the likelihood of symptoms beyond that experienced in ordinary living" (p. 129). The author continues to discuss other incidences, such as shoulder/neck pain in the workplace, and carpal tunnel syndrome. Hadler states that, until recently, CTS was a rarity. "With this background, the current brouhaha regarding CTS comes as a surprise. ...I am convinced that this dramatic change in the implications of the concept of CTS can be ascribed to the efforts of two men: T.J. Armstrong and his mentor at the University of Michigan, D.B. Chaffin" (p. 135).

In conclusion, the author says: "It is clear that many people suffer frequent episodes of discomfort in discrete musculoskeletal regions when they are otherwise well - episodes I have termed 'regional musculoskeletal illnesses'. ...Most are fleeting predicaments and, while they might give us pause, seldom dramatically impact on the quality of our lives. ... But you must have the patience and self-assurance to await remission. That is why we are accustomed to coping with many of

these episodes without even complaining to a spouse, let alone choosing to be a patient" (pp. 137, 138).

Center for Office Technology (Ed.). (1989). Latest Studies on VDTs (2nd ed.). Washington, DC: Author.

This volume is a collection of abstracts of studies by scientists, academicians, and health experts around the world. The abstracts were divided into six main sections. The first five focus on visual research and display quality, musculoskeletal research and workstation design, stress and job design, radiation, and pregnancy outcome. The sixth part contains general reviews of the research on VDTs, recommendations and guidelines. Altogether, about more than 130 papers are compiled in this volume.

Morita, M. (1989). Development of New Keyboard Optimized from Standpoint of Ergonomics. In M.J. Smith and G. Salvendy (Eds.), *Work With Computers: Organizational, Management, Stress and Health Aspects* (pp. 595–602). Amsterdam, NL: Elsevier.

The author first indicates the well-known ergonomic shortcomings of the standard keyboard. He then establishes a number of requirements which include ease of learning, fast operation, and lengthy operation not resulting in excess fatigue. With these requirements in mind, the author describes a new layout in which the keyboard is divided into two sections, one for the left and one for the right hand. The keys are arranged in configurations that are similar to the natural position of the fingertips and the thumb tips. The keys are distributed within these areas. The new keyboard called "Bungo 3M II" is meant to be used both in the English and the Japanese mode.

Knight, L.W. and Retter, D. (1989). Datahand: Design, Potential Performance, and Improvements in the Computer Keyboard and Mouse. In *Proceedings of the Human Factors Society 33rd Annual Meeting* (pp. 450–454). Santa Monica, CA: Human Factors Society.

This paper presents the design and preliminary evaluation of a new computer key and spatial entry system, called Datahand. It is intended to improve data entry and control. Also, it is intended to provide possible reduction of CTDs problems with conventional keyboards. This design also splits the keyboard in halves for the left and right hand, but the keys are "wrapped around the user's fingers" as opposed to a conventional two-dimensional keyboard. Each unit has a contoured palm and hand support which makes it more comfortable, and also graspable, so that it can be moved like a mouse.

Green, R.A. and Briggs, C.A. (1989a). Effect of Overuse Injury and the Importance of Training on the Use of Adjustable Workstations by Keyboard Operators. *Journal of Occupational Medicine*, 31(6), 557–562.

The authors used a questionnaire to evaluate the use of adjustable workstations by 514 keyboard operators. Actual anthropometric dimensions and workstations settings were measured for 15 operators. No difference was found between sufferers and non-sufferers of overuse injury in the adjustability of their equipment. However, sufferers were less satisfied with the equipment. The authors conclude that the design of equipment must be based on suitable anthropometric data, but the operators must also be trained in the correct use of the equipment. This information should be provided not only in writing, but also verbally, for example by their direct office supervisors.

Green, R.A. and Briggs, C.A. (1989b). Anthropometric Dimensions and Overuse Injury Among Australian Keyboard Operators. *Journal of Occupational Medicine*, 31(6), 747–750.

Seventy subjects were selected at random from a larger population of keyboard operators. Thirty-five of these were classified as sufferers, 35 as non-sufferers of overuse injury. On each, anthropometric dimensions were measured, and the test-retest reliability of the measurements was established. The anthropometrics of male sufferers (8) and non-sufferers (13) were not different from each other. However, among women, there were significant differences between non-sufferers (22) and sufferers (27). The sufferers had greater hip widths and seat breath, and more were overweight. There was also a tendency for shorter limb segment lengths.

Lockwood, A.H. (1989). Medical Problems of Musicians. *The New England Journal of Medicine*, 320(4), 221–227.

There were references to medical problems among musicians more than 100 years ago. Instrumentalists are susceptible to a variety of disabling problems. About 130 000 Americans are instrumentalists. Of these, more than 4000 are members of the International Conference of Symphony and Opera Musicians (ICSOM) in 48 orchestras. In a questionnaire to these members, with 2122 responding, 76 percent reported having had at least one medical problem severe enough to affect performance and 36 percent reported four severe problems (the maximum foreseen in the questionnaire). The prevalence was highest among string players (nearly 75 percent) and lowest among percussionists, about 32 percent. The majority of musician patients suffer from cumulative trauma effects. Women are more commonly affected than men. Most problems are with the muscle-tendon unit, but other overuse syndromes involve bones, joints and their bursae, and muscles. Musculotendinous overuse is most common in the hand, wrist, and forearm, but also occur in the shoulder.

The following are typical complaints:

- * Keyboarders – extensors of the wrist and fingers.
- * String players – flexors and extensors of the left and right hand. Shoulder and neck problems.
- * Clarinet, or bow, and English horn players – Webb-space muscles between the thumb and index finger.
- * Flute players – shoulder, radial side of the left index finger.
- * Trumpet and or bow players – venous return to the heart, larynx and pharynx pouches.

Related to neural problems, pain, loss of strength, and sensory abnormalities are common among musicians. Carpal tunnel syndrome is the most common entrapment neuropathy observed in many musicians. Other sites of nerve entrapment are found in the forearm. Medical injuries related to music playing are an important problem that is likely to have implications even during the early phases of musical training. Almost half of a group of high school musicians reported symptoms of overuse injuries (p. 226).

Kroemer, K.H.E. (1989). Cumulative Trauma Disorders: Their Recognition and Ergonomics Measures to Avoid Them. *Applied Ergonomics*, 20(4), 274–280.

This paper reiterates and summarizes findings about CTD phenomena and their associated activities, both on and off the job. Ergonomic measures to avoid possible causes for CTD are listed.

Williams, T.L., Smith, L.A. and Herrick, R.T. (1989). Exercise as a Prophylactic Device Against Carpal Tunnel Syndrome. In *Proceedings of the Human Factors Society 33rd*

Annual Meeting (pp. 723–727). Santa Monica, CA: Human Factors Society.

Thirty female employees took an exercise program. Measures of grip strength, Phalen's tests, and thermograms did not show statistically significant improvements after the exercise treatment.

Sind, P.M. (1989). The Effects of Structural and Overlay Design Parameters of Membrane Switches on the Force Exerted by Users. Unpublished Dissertation, Virginia Polytechnic Institute and State University, Blacksburg, Virginia.

Two experiments were conducted to evaluate the effects on applied force of structural design parameters and feedback conditions in membrane switch touch pads. Twelve male and 12 female subjects keyed one hundred four-digit sequences using different touch pads. The same task was completed by nine males and nine females in a second experiment to evaluate feedback conditions. The results indicate that the forces applied by the subjects were correlated with the required "actuation force of the switch inherent in the touch pad". However, the specific membrane switch structures had a significant affect on the applied forces, even when the required actuation forces were controlled. Feedback conditions inherent in the touch pads also had an effect on the forces applied. The following human factors design recommendations for touch pads are provided:

- * The lowest required activation force appropriate for the application should be used.
- * When auditory feedback is provided, supplemental kinesthetic feedback is redundant.
- * Even when auditory feedback is provided, supplemental tactile feedback is desirable.

Rempel, D., Lopes, J., Davila, R. and Davis, B. (1989). Cumulative Trauma Disorders Among Visual Display Terminal Users at a Newspaper Company (Report dated September 13, 1989). Berkeley, CA: Hazard Evaluation System and Information Service; and Sacramento, CA: California Occupational Safety and Health Administration.

A symptom survey and ergonomic evaluation were conducted on VDT users at a large newspaper company. With a 63 percent participation rate, 136 out of 217 VDT users completed the symptom survey. Of these, approximately 61 percent reported work related pain or discomfort during the previous 12 months. The primary sites of pain or discomfort were the hands, forearms, upper arms, and upper back. Common complaints included symptoms consistent with forearm or wrist tendonitis or carpal tunnel syndrome. The prevalence rate of lower arm tendonitis or carpal tunnel syndrome was calculated to be at least 26 percent. This was significantly higher than the prevalence rate of similar cases in an industrial work force.

A detailed ergonomic evaluation was conducted. Significant findings were that most keyboard support surfaces were not adjustable in height, hard copy support surfaces were inadequate in size and location, wrist rests or forearm supports were either not present or not properly adjusted, leg room was inadequate, and chairs were not easily adjustable. Employees had not had adequate training on methods for preventing musculoskeletal injuries.

National Occupational Health and Safety Commission (Ed.) (1989). Guidance Note for the Prevention of Occupational Overuse Syndrome in Keyboard Employment. Canberra, ACT: Australian Government Publishing Service.

This 50-page brochure provides comprehensive advice to prevent overuse syndrome associated with keyboarding in terms of the organization and design of job tasks and practices as well as of workstation and equipment design and arrangement. Training and education as well as case management are also discussed.

The 1980s saw a wave of computers entering modern offices, rapidly sweeping out the remaining mechanical and even the newer electric typewriters. IBM introduced its first PC in 1981. Only ten years later, it stopped producing typewriters (Forbes Magazine Vol. 166, No. 13 of 13 November 2000, page 234). The literature demonstrates a series of similar events.

The 1980s decade is characterized by several main developments:

1. The occurrence of "CTD epidemics" in certain office occupations in North America and Europe, and especially in Australia, where computer keyboards were suddenly and massively employed. See, for example, the publications by Maeda, Huenting and Grandjean 1980; Laeubli, Nakaseko and Huenting 1980; Price et al. 1982, McPhee 1982; Browne, Nolan, and Faithfull 1984; Oxenburgh, Rowe, and Douglas 1985; Hocking 1987; Ferguson 1987; Rossignol et al. 1987; Nishiyama 1987, Kiesler and Finholt 1988; Green and Briggs 1989.
2. The investigation of the pathological nature of CTDs, of their causes, and hence of ways to prevent them from occurring in certain professions, especially keyboarding. The following publications are particularly indicative of the state of knowledge: Fraser (1981), Lundborg et al. (1982), Armstrong (1983), Arndt (1983), Kaplan (1983), Hadler (1984), NIOSH (1984), Brown, Coyle, and Beaumont (1985), Hagberg and Sundelin (1986), Armstrong, Fine, and Silverstein (1986), Smith (1986), Chatterjee (1987), Gomer et al. (1987), Seror (1987), Kiser (1987), Pfeffer et al. (1988), Foster and Fry (1988), Bjoerksten (1988), Molan and Sikorski (1988), Putz-Anderson (1988), Helander (1988), Burnette and Ayoub (1989), Ayoub and Wittels (1989), and Rempel, Lopes et al. (1989). However, Hadler expressed, in 1989, disbelief that keyboard operation causes repetitive disorders.
3. The design of keyboards was reviewed by Litterick in 1981, and in great detail by Noyes in 1983. Explicit human engineering guidelines were published for operator-centered design of keyboards and keyboard workstations, and for the proper ergonomic ways to perform keying work. See, for example, Smith, Cohen, and Stammerjohn (1981), Miller and Suther (1981), Benz, Grob, and Haubner (1981), Wagner (1982), Haider, Rohmert, and Halpern (1982), Huenting, Laeubli, and Grandjean (1982), Dainoff (1982), Snyder (1983), Kroemer (1983), Arndt (1984), Stewart (1985), Sauter, Chapman, and Knutson (1986),

Seligman, Boiano, and Anderson (1986), Statshael-san (1986), Grandjean (1987b), Ilg (1987), Greenstein and Arnaut (1987), Rose (1988), Helander (1988), and (Australian) National Occupational Health and Safety Commission (1989).

4. Many new patents were issued for keyboards that either maintained the basic features of the conventional flat QWERTY keyboard but claimed ergonomic improvements, or attempted to achieve substantial changes in order to facilitate operation and improve performance. However, the 1988 American National Standard for Human Factors Engineering of Visual Display Terminal Workstations (ANSI/HFS 100-1988) explicitly required (on page 35) that a conventional QWERTY keyboard layout be used, with commonly straight rows and columns of keys added on its sides, mostly to the right.

To summarize the development during the 1980s decade, local "epidemics" of musculoskeletal diseases occur to groups of keyboarders. The health complaints are, at least partially, attributed to certain design and use features of keyboards. Human engineering guidelines address operator-centered design of keyboards and workstations, and ergonomic ways to perform keying work.

6 From 1990 to 1999

Annotated list of publications on keyboards and keyboarding from 1990 to 1999

Zapp, W. (1990). *Keyboard Actuable with the Aid of the Fingers of at Least One Hand*. Patent 4,201,489. United States Patent Office.

The patent describes a key that can be "rocked", i.e., tilted in four directions by force applied with the fingertip. A rest for the ball of the thumb or the wrist is provided. References are made to similar patents and publications, such as by IBM (Conway) of 1976, and to other patents of 1950, 1961, 1971, 1972, and 1973.

Faubert, F.M. and Pritchard, J.H. (1990). *Upper Limb Disorders Among Keyboard Users – A Manufacturer's Prospective*. In Proceedings of the 1990 HUSAT Training Conference "Upper Limb Disorders in Keyboard Users" (pp. 1–8). London: HUSAT Training.

In early 1981, the authors' company started the design of a new keyboard for the Coyote Workstation. They attempted to comply with the recommendations put forth by Cakir et al. (1980) and the German standard ZH 1/1618, (1980). The keyboard was not in compliance in all respects. The authors state that Repetitive Strain Injury (RSI) is a well-known condition related to repetitive muscle strain which, although found related to many activities, is particularly evident in the use of computer terminals. "The bottom line - more time doing the same thing, using the same muscles, maintaining the same position all day long. RSI, anyone?" (p. 2).

Since 1980, the company was aware of three RSI episodes of significant magnitude. The first occurred in Australia, in late 1984. The second occurred in Southern California, in early 1985. More than 200 people reported RSI problems there. The third RSI occurrence was in London, starting in late 1988.

In conclusion, the authors acknowledge that the keyboard must be closely examined in its relation to RSI problems. Improvements such as repositioning keys, using different keyboard shapes, and placing the keyboard differently in space might be beneficial.

Hughes, J. (1990). *Current Approaches to the Development of Standards for Input Devices*. In Proceedings of the 1990 HUSAT Training Conference "Upper Limb Disorders in Keyboard Users" (pp. 69–75). London: HUSAT Training.

The author provides information about the evolution of keyboards from the Sholes design to modern keyboards in which nearly all layout aspects are subject to change, thanks to improved technology. The author then indicates current attempts to develop standards for keyboards, particularly of the International Standards Organization. This includes in particular work on IS 9241 "Ergonomic Requirements on VDTs".

Stewart, T. (1990). *The European Directive Concerning the Minimum Safety and Health Requirements for Work with Visual Display Units*. In Proceedings of the 1990 HUSAT Training Conference "Upper Limb Disorders in Keyboard Users" (pp. 77–89). London: HUSAT Training.

The author reviews the draft of the European Directive that was published in 1988 and has since then undergone a number of revisions. The directive applies to the entire workstation, including display, keyboard, document holder, chair, desk, the environment, etc. The operator may be anyone who uses a VDT, even if on a part-time basis. The ANNEX contains a variety of requirements for display, keyboard, desk, chair, lighting, reflections and glare, noise, heat, radiation, and operator/computer interface.

Lachnit, H. and Pieper, W. (1990). *Speed and Accuracy Effects of Fingers and Dexterity in 5-Choice Reaction Tasks*. *Ergonomics*, 33, 1443–1454.

Two experiments were conducted (with 168 and 40 students, respectively) to determine the speed and accuracy in 5-choice reaction tasks of the five digits of the dominant hand. The subjects were divided into typists, pianists, and controls. Five colored squares were used as stimuli. The results showed that in all three groups, thumb and little finger showed significantly shorter reaction times than the other fingers. There was no interaction with dexterity. Pianists showed the shortest reaction times of all groups.

Ayoub, M.A. (1990). *Ergonomic Deficiencies: I. Pain at Work*. *Journal Occupational Medicine*, 32, 52–57.

The author starts out by indicating that pain at work, and discomfort, are related to a poorly designed workplace or poorly structured job. An ergonomically deficient workplace may not cause immediate pain, because the human body has a great capacity for adapting to a poorly designed workplace or structured job. However, in time, the compounding effect of job and/or workplace deficiencies will surpass the body's coping mechanism causing the inevitable: physical symptoms, emotional stress, low productivity, and poor quality of work (p. 52). The ergonomic deficiencies are shown in terms of extreme posture, excessive force, concentration of stress, and static loading. Cumulative trauma disorders may occur. Manifestations may take these forms: sudden and unexpected increase in the number of reported cases, increase in the number of employees out of work because of disability, high number of employees seeking medical care, and increasing rate of turnover or transfers from some jobs (p. 53).

Hadler, N.M. (1990). *Cumulative Trauma Disorders. An Iatrogenic Concept*. Journal of Occupational Medicine, 32, 38–41.

In this editorial, the author considers the issue of “whether any particular musculoskeletal usage which in and of itself is reasonable, comfortable, and customary, and which can be repeated without undue distress, can lead to specific musculoskeletal damage over time” (italics in the original, p. 38). The author questions that such tasks as meat packing or data entry can cause CTDs. “This is not a clinical event; it is a sociopolitical phenomenon. It is flying, even gaining speed, in the face of considerable and reasonable information that it is more than flawed - it is wrong” (p. 38).

Council of the European Communities (1990). *Council Directive of 29 May 1990 on the Minimum Safety and Health Requirements for Work with Display Screen Equipment (Fifth Individual Directive Within the Meaning of Article 16 (1) of Directive 87/391/EEC)*. Official Journal of the European Communities, 33, (L156, 21 June 1990), L156/13–18.

This Directive lays down minimum safety and health requirements for display screen equipment, meant to be a workstation that comprises a display, keyboard or other input device, and software, and optional accessories, peripherals, work chair, work desk, work surface, and the immediate work environment (Article 2).

Article 3 requires that employers perform an analysis of workstations in order to evaluate safety and health condition. Any risks found must be remedied by the employers. This requirement applies to all workstations put into service after 31 December 1992, and all workstations put into service before that date must be adapted to comply with the minimum requirements within four years (Articles 4 and 5).

Within the minimum requirements, spelled out in the Annex, the following apply to the keyboard:

The keyboard shall be tiltable and separate from the screen so as to allow the worker to find a comfortable working position avoiding fatigue in the arms or hands.

The space in front of the keyboard shall be sufficient to provide support for the hands and arms of the operator.

The arrangement of the keyboard and the characteristics of the key shall be such as to facilitate the use of the keyboard.

Helme, R.D., Gibson, S.J. and LeVasseur, S.A. (1990?). *The Role of the Primary Afferent Nerve Pathway in Repetitive Strain Injury: A Report to the National Occupational Health and Safety Commission*. National Research Institute of Gerontology and Geriatric Medicine, Mount Royal Hospital, Department of Medicine, University of Melbourne. (No date given, but the newest publication cited in the references is 1990.)

This study was conducted to determine, by actual measurement, whether persons with chronic upper limb and neck pain related to occupation (repetitive strain injury, cervicobrachial syndrome) showed different physiological data, in terms of peripheral pain receptor activity, than persons without pain, in response to a noxious thermal stimulation.

Eighty-three volunteer subjects with chronic upper limb and neck pain related to RSI participated. The control group was composed of 75 age-matched pain-free persons. The results indicated that the subjects with pain had higher levels of anxiety, depression, anger, psychological confusion, and feelings of fatigue in comparison to the control subjects. Yet, there was little evidence of hypochondriasis or dependence on others for control of health.

Low, I. (1990). *Musculoskeletal Complaints in Keyboard Operators*. Journal of Occupational Health and Safety – Australia and New Zealand, 6(3), 205–211.

A survey, performed by means of a self-administered questionnaire, of 210 keyboard operators in Brisbane, Australia, showed that an epidemic of keyboard-related conditions has peaked in 1985/1986. Females were more frequently affected than males. The difference in prevalence of symptoms may be due to job factors rather than anatomical differences. The variety of treatments may reflect the diversity of conditions attributed to keyboard use and the relative ineffectiveness of the treatments in at least some of the cases. Symptoms commonly settled with a change in the workstation, work practices, or the type of work. Multiple regression analysis showed that the time per day that had been spent doing keyboard work, and the number of years of keyboard use, were significantly associated with the development of symptoms (p. 205).

Thompson, D.A. (1990). *Analysis of the TONY! Variable Geometry VDT Keyboard*. In Proceedings of the Human Factors Society 34th Annual Meeting. (pp.365–369). Santa Monica, CA: Human Factors Society.

A study to determine the optimal configuration of a split QWERTY keyboard was done using the TONY! Variable Geometry keyboard. EMG signals and operator discomfort were reported. The keyboard geometry was manipulated to represent a conventional, flat, linear keyboard, a neutral, undeviated rest position, and two forearm positions that essentially eliminate pronation. Eight experienced subjects with no history of wrist injury or other pathology were involved. The outcome of the study suggested that an 18 degree opening angle (slant) and a 30 to 60 degree lateral angle (tilt) minimized EMG activity and subjective discomfort.

Heyer, N., Checkoway, H., Daniell, W., Horstman, S. and Camp, J. (1990). *Self Reported Musculoskeletal Symptoms Among Office Video Display Terminal Operators*. In H. Sakurai et al. (Eds.), *Occupational Epidemiology* (pp. 255–258). Amsterdam, NL: Elsevier.

This study was designed to estimate the prevalence of vision and musculoskeletal complaints among VDT operators in the Seattle/Washington area, and to evaluate the association between VDT workstation design and/or job task, and the prevalence of reported symptoms. The information was obtained through visits to the work units, and a self-administered questionnaire. Data concerning 702 female VDT operators (defined as working four or more hours per day with a VDT) were compared with data from 52 female non-VDT operators (defined as persons who work two hours or less per day at a VDT). Employees satisfaction was generally high, and the equipment was adequate, work schedules and break policies were similar.

It was found that VDT operators had a generally higher prevalence of symptoms than non-VDT employees, with differences most pronounced for symptoms in eyes, neck, arm, and hand. However, there was no association observed with the number of hours worked at a VDT. VDT operators doing entry tasks had increased symptoms of the back, neck, arm, and hand compared to operators doing interactive tasks.

Burt, S., Hornung, R. and Fine, L. (1990). *Health Hazard Evaluation Report HETA-89-250-2046, Newsday, Inc., Melville, NY*. Cincinnati, OH: National Institute for Occupational Safety and Health.

A study was undertaken by NIOSH in Newsday offices to estimate the prevalence of CTD among employees. A questionnaire survey was conducted at four office buildings in the fall of 1989. Forty percent (331 employees of 834 people participating) reported symptoms consistent with upper extremity cumulative trauma disorders during the past year. Hand/wrist symptoms were the most prevalent, followed by symptoms in the neck, elbow/forearm, and shoulder. Percent of time spent keyboarding was significantly associated with symptoms. Working as a reporter was significantly associated with symptoms. Women were at higher risk than men for neck symptoms, but not for symptoms in other locations. Symptoms of the hand/wrists, elbow/forearm, shoulder, and neck were all associated with typing on computer keyboards. As percent of time typing or typing speed increased, the likelihood of symptoms meeting the case definition increased.

IBM (Ed.) (1990). Health and Safety Aspects of Visual Displays. Somers, NY: IBM Corporation, VDT Ergonomics Project Office.

This brief guide is intended to answer questions about VDTs with respect to radiation, vision, cumulative strain disorders, stress, and ergonomics.

Guggenbuehl, U. and Krueger, H. (1990). Musculoskeletal Strain Resulting from Keyboard Use. In L. Berlinguet and D. Berthelette (Eds.), *Work With Display Units* 89 (pp. 124–126). Amsterdam, NL: Elsevier.

Experiment showed that the keying rhythm seems to follow set motor programs in the range of 3 to 5 Hz. If the rhythm cannot be maintained, the muscular load of the flexors and extensors rises. The physical characteristics of the keys (e.g., its force requirements) must suit the motor programs to avoid high musculoskeletal loads. A high typing frequency increases the musculoskeletal load.

McPhee, B. (1990). Musculoskeletal Complaints in Workers Engaged in Repetitive Work in Fixed Postures. Chapter 5 in M.I. Bullock (Ed.), *Ergonomics: The Physiotherapist in the Workplace* (pp. 101–132). Edinburgh, UK: Churchill Livingstone.

The author traces the history of musculoskeletal injuries related to repetitive work, leading up to the so-called Occupational Cervicobrachial Disorders in Japan during the 1960s, and the so-called Repetitive Strain Disorders in Australia in the 1980s, later usually called Occupational Overuse Syndrome. Subsequently, causes and risk factors are discussed, as are their analysis, management and means of prevention. A two-year epidemiological study of neck and upper extremity disorders in keyboarders is described.

Center for Office Technology (Ed.) (1991). Latest Studies on VDTs, 3rd edition. Rosslyn, VA: Author.

Among about 150 reviews, this issue contains an abstract of Arndt, R. (1983). *Working Posture and Musculoskeletal Problems of Video Display Terminal Operators - Review and Reappraisal.* American Industrial Hygiene Association Journal, 44, 437–446. Three critical interfaces between the VDT and the operator are noted in this article: the hand and keyboard, the body and chair, and the eyes and visual targets, especially the monitor. Early keyboard height recommendations seem to have been based on anthropometric data and biomechanical principles. Workers prefer postures which elevate the forearms and permit a slightly back-leaning trunk. High back chairs allow these postures to be maintained, if

adjustable. Adjustable screens are also important considerations for meeting postural and visual requirements of VDT work. "Ultimately, the reduction of musculoskeletal problems among VDT operators, as well as other workers, will depend not only upon considerations of workstation designs, but also on the physical demands (eg repetitiveness, postural constraints), and psychological demands (eg productivity pressures) of the job. Jobs must be designed to fit the physical, behavioral, and psychological limitations, as well as the anthropometric and biomechanical characteristics of workers" (p. 39).

Cushman, W.H. and Rosenberg, D.J. (1991). Human Factors and Product Design. Amsterdam, NL: Elsevier.

This book is written to show how human factors technology can be applied during product design and development, to improve product usability, user-product performance, user satisfaction, and product safety - as stated in the preface. Chapter 9 (pages 179–204) is devoted to keyboards and input devices for computers. Keyboard technologies, key characteristics, key arrangements, ergonomic keyboards, and the design of special purpose keyboards are discussed in detail. The second half of the chapter compares the conventional computer keyboard, chord keyboard, mouse, trackball, graphic tablet, joystick, touch screen, light pen, and voice as input devices.

The keys on contemporary microcomputer and VDT keyboards have displacements of about 3.5 mm, with some zero-travel keyboards also available, where the actual travel is about 0.2 mm. Tactile feedback is important for some tasks such as touch typing and high volume data entry.

The recommended key operating force is between 0.25 and 1.5 N. The key resistance should initially increase as the key is depressed but then drop shortly once the switch has been closed. "This sharp transition permits the user to 'feel' that the key has been pressed sufficiently...while the benefits of this type of force/travel relationship are generally acknowledged, some additional improvement appears to be possible by using keys with a double-peak force/travel function and more gradual transitions..." (p. 182).

Regarding "special purpose keyboards", the authors state that such custom keyboard design should begin with a consideration of the user and environment. Subsequently, a task analysis should be performed to pinpoint the specific functions that the keyboards must provide. The analysis may indicate that certain keyboard technologies should be eliminated from further consideration. Then attention may be focused on the issues concerning the assignment of functions to keys, keyboard size, and key arrangements (p. 190).

Rempel, D. and Gerson, J. (1991). Fingertip Forces While Using Three Different Keyboards. In *Proceedings of the Human Factors Society 35th Annual Meeting* (pp. 253–255). Santa Monica, CA: Human Factors Society.

The authors state that while the relationship between chronic musculoskeletal disorders with prolonged keyboard use is controversial, known factors are repetition, forceful exertions, awkward joint postures, direct mechanical pressure, and prolonged constrained posture, which may be applicable to keyboard use.

Nine touch typists and one hunt-and-peck typist typed alphabetic sentences for approximately ten minutes each on three keyboards. Vertical forces applied to the keyboards were recorded. Three QWERTY keyboards were tested, which differed only in their key force and displacement characteristics. The total displacements of the keys were about 3.5, 4.5, and

6.5 mm, respectively; while the forces exerted at the "make" point of the keys, before the break-away in flexion, were approximately 40, 45, and 90 g, respectively.

The forces actually applied by the subjects were between 2.2 and 4.7 times larger than required for key activation. The keyboard with the lowest required actuation force also had the lowest actual applied forces. Conversely, the keyboard with the highest required actuation force was activated with most force.

The authors conclude that the force-displacement characteristics can affect the fingertip forces applied during typing. Reducing the force required for activation and the force for total key travel can reduce the peak fingertip force, in this study as much as 18 percent. A relatively minor reduction in applied fingertip force will significantly reduce the daily cumulative force applied by the fingers.

Kroemer, K.H.E. (1991a). Experiments with the TCK – A Keyboard with Built-in Wrist Rest and Only Eight Keys. In W. Karwowski and J. W. Yates (Eds.), *Advances in Industrial Ergonomics and Safety III* (pp. 537–542). London, UK: Taylor & Francis.

This is a report on experiments with a ternary chord keyboard. Ten subjects learned to use this keyboard and worked on it for approximately 20 hours each. The highest input speeds observed exceeded 180 characters per minute. The TCK is said to possibly be a true competitor for the traditional QWERTY keyboard.

Draganova, D. (1991). Musculo-skeletal Complaints in Some Professions. Abstracts, International Conference on Occupational Musculo-Skeletal Disorders and Prevention of Low Back Pain (p. 21). New York, NY: New York College of Osteopathic Medicine.

A questionnaire study was performed to determine the occurrence of occupational musculo-skeletal complaints in 288 women. Musculo-skeletal complaints were experienced by 85 percent of the dress-makers, 78 percent of the VDT data entry operators, 63 percent of the assemblers, 44 percent of the draft women, and 32 percent of the monitoring panel operators. The most frequent complaints were 63 percent in neck and shoulder, 60 percent in low back, and 35 percent in fingers and wrists.

Caple, D.C. and Betts, N.J. (1991). RSI-It's Rise and Fall in Telecom Australia 1981–1990. In Proceedings, 11th Congress of the International Ergonomics Association (pp. 1037–1039). London, UK: Taylor & Francis.

Telecom Australia experienced a major epidemic of repetition strain injury (RSI) during the period from 1983 to 1988. Almost 4800 cases were reported during these five years out of a work force of about 80 000 employees. The peak occurred during 1985 with 2149 cases reported. While this epidemic affected all occupational groups, telephonists reported the highest incidence, with 48 percent of all cases while constituting only eight percent of the work force. The occurrence of RSI at Telecom Australia followed, apparently, the introduction of a keyboard based system to replace the "Chord and Plug" system previously used.

The authors interviewed more than 100 persons from the telephone work centers. They found no single causal factor for the RSI epidemic, but state a number of factors. These include the community concern about RSI, inconsistent medical management, problems with the management of technology change, the occurrence and length of work periods/rest breaks, the workload, specialization of task performance, some of which

were highly repetitious (with cycle times of 15 seconds) or boring. Furthermore, training was inconsistent, and visits by outside experts resulted in increased number of cases.

The same duties were performed in a large number of locations throughout Australia, with some places reporting no cases, while others had 50 percent of the operators receiving treatment for RSI. In terms of furniture, "ergonomic" chairs and adjustable workstations were in wide use prior to the RSI epidemic, and the same furniture remained in use during the epidemic and since its decline.

No correlations were found between keystroke rate and the number of RSI cases. The rate for telephonists was typically less than 1000 per hour.

The results of the interviews identified that the prevention strategies that were most effective toward the decline of cases were relaxation of performance measures and of management control over workload, multifunctional job design, and introduction of rest breaks.

Hahn, K., Chin, D., Ma, P. and Rebello, S. (1991). Cumulative Trauma Disorder Reductions in the Industrial Workplace. A Systems Approach. In W. Karwowski and J.W. Yates (Eds.), Advances in Industrial Ergonomics and Safety III (pp. 147–154). London, UK: Taylor & Francis.

The authors state that CTDs are a large and growing hazard in the workplace. They affect soft tissue mostly in three regions: neck and shoulder, hand and wrist, and lower back. CTDs are said to be the result of sustained or repetitive applications of low stress over time on fatigued or previously damaged tissues. In order to combat the occurrence of CTDs in Canada, the authors represent a scheme in which analyze, evaluate, and measure CTDs and, consequently, design equipment for CTD reduction.

Major problems related to workstation design are stressful postures, vibrations, repetitive job exertions, and the loading in terms of force (type, magnitude, direction). The approaches to reduce CTDs are divided into four ways: training and education, medical treatment, job design and placement, and workstation design. Workstation design is said to be the most effective way to reduce CTDs.

Armstrong, T.J., Werner, R.A., Waring, W.P. and Foulke, J.A. (1991). Intra-Carpal Canal Pressure in Selected Hand Tasks. A Pilot Study. In Proceedings of the 11th Congress of the International Ergonomics Association (pp. 156–158). London, UK: Taylor & Francis.

This is a report on measurement of intra-carpal canal pressure in a healthy male subject. The subject performed three kinds of tasks: selected static postures of the hand, gripping an object, and moving an object.

In the first experiment, the wrist was fully extended, and the fingers were then fully extended. Then, the wrist was fully flexed, and the fingers were fully flexed, then extended, and then flexed again. Extension of wrist and fingers together resulted in much increased carpal canal pressure, while flexion of wrist and fingers together brought about a much smaller increase. Flexion of the wrist together with extension of the fingers had an intermediate pressure increase as a consequence. Thus, it appears that extension of wrist and fingers is most likely to increase intra-carpal canal pressure. In the next set of experiments, increasingly strong grip exertions were performed, such as when grasping hand tools and handles. There were small increases in carpal pressure with increasing grip force, but altogether the pressure appeared to be less strongly enhanced by gripping than by certain postures. In the last ex-

periment, a 3.6 kg one-liter plastic bottle was grasped, transported over 36 cm and then put down on a table. Grasping of the bottle increased the pressure strongly (similarly to wrist and finger extension).

The authors conclude that it is feasible to measure intra-canal pressure. Secondly, during routine work tasks, pressures may be developed within the canal that are strong enough to adversely affect the median nerve.

Haegg, G.M. (1991). Lack of Relation Between Maximal Force Capacity and Muscle Disorders Caused by Low-Level Static Loads-A New Explanation Model. In Proceedings of the 11th Congress of the International Ergonomics Association (pp. 9–11). London, UK: Taylor & Francis.

The author presents a new model for muscular disorders at low levels of exertion. First, when motor units are recruited to build up muscle force, this is done according to a fixed order in which low threshold motor units (Type 1) are initially recruited and remain active until total relaxation of the muscle. Second, certain Type 1 fibers seem to be affected selectively, not in a traumatic way but apparently by metabolic disturbance. Both findings together explain that those fibers that are contracted over long periods of time, that is those at the bottom of the recruitment pyramid, are affected even at low levels of exertion. This model would also explain why a person's maximal force capacity plays no role in the development of muscular disorders. While a stronger muscle uses a smaller fraction of the total muscle capacity, it still calls up the same "first-called" muscle fibers as in a weaker muscle. Thus, it is not so important to reduce the magnitude of static load levels in muscles as it is to limit the duration of the loading and increase the number of pauses.

Moore, A. Wells, R. and Ranney, D. (1991). Quantifying Exposure in Occupational Manual Tasks With Cumulative Trauma Disorder Potential. Ergonomics, 34, 1433–1453.

While work-related chronic musculoskeletal disorders (CTDs) are recognized as a major occupational health problem, the injurious dose is difficult to determine, mostly because job demands are not quantified in terms of forces, postures, and the like. For example, while it has been shown that both force and repetition individually increase the prevalence of CTD, their combined effect is multiplicative. However, it is not clear whether the situation becomes better or worse if one halves the force but doubles the repetition, or vice versa. The authors propose that there may be two distinct injury mechanisms at work: one is the injury to muscle under static (low repetition/low level) loads, the other is injury to structures, particularly tendon sheaths, under high repetition conditions. According to these considerations, the authors list different injury mechanisms, and, accordingly, different external risk factors for carpal tunnel syndrome, tenosynovitis, tendonitis, and chronic muscle strain. These injury mechanisms, and their associated external risk factors, are different combinations of force, movement, internal friction, axial force in tendons, fatigue, and overuse in terms of constant or dynamic use.

Armstrong, T. (1991). Upper Limb Cumulative Trauma Disorders, Localized Fatigue, & Ergonomic Control Programs. In Proceedings, Occupational Ergonomics: Work Related Upper Limb and Back Disorders (pp. 1–1 to 1–25). San Diego, CA: AIHA, San Diego Local Section.

The author discusses terminology, common traits, and morbidity patterns of Work Related Upper Limb and Back Disorders (WRULBDs) and provides a listing of historical sources.

There are many possible causes of WRULBDs, and diverse individual aspects of the afflicted persons can contribute to the occurrence of the disorders. Factors at work include repeated or sustained exertions, forceful exertions, contact stresses, certain postures, low temperatures, and vibrations. All patients used computer keyboards. Pathomechanics and physiology provide indications of a cause-effect relationship.

Rose, M.J. (1991). Keyboard Operating Posture and Actuation Force, Implications for Muscle Over-Use. Applied Ergonomics, 22, 198–203.

The author determined, in ten subjects, relaxed posture of fingers in which the fingers are bent in all their joints. The author presumes that keyboard users will seek to rest their hand/finger muscles, with the resting done either on the proximal edge of the keyboard housing or on the supporting work surface. If the fingers are kept on the key, the key resistance should be large enough to support the weight force of the fingers. The author also measured the forces transmitted to the keys by 60 subjects when either the hand/fingers were not supported at all, supported at the wrist at the level of the key tops, or supported on a surface 30 mm below key top level. No support resulted in highest fingertip forces, support below the key top height brought about the second highest key top forces, while wrist support at the level of the keys caused the lowest key forces.

The author concludes that keys should have a resistance of at least 1.1 N if no wrist or arm support is available to avoid accidental key actuation. If arm or wrist support is provided at about key top level, a resistance of 0.7 N is sufficient to provide resting support for the fingers.

Kroemer, K.H.E. (1991). Cumulative Trauma Disorders in Computer Operators. In Proceedings of the 11th Congress of the International Ergonomics Association (pp. 727–729). London, UK: Taylor & Francis.

The author presents six cases in which operators of keyboards developed CTDs. While each case is different, all patients used computer keyboards. There are clear indications of a cause-effect relationship, even if the exact relation to repetitiveness, forcefulness and posture is not yet clear.

Apple Computer, Inc. (Ed.) (1991). Creating a Healthy Work Environment. Cupertino, CA: Author.

Under the heading of "Health Concerns Association with Computer Use", it is said that using a computer can make muscles sore and stiff, and that there is the musculoskeletal concern of repetitive stress injuries (RSIs), also known as cumulative trauma disorders (CTDs). "These problems can occur when a certain muscle or tendon is repeatedly overused and forced into an unnatural position. The exact causes of RSIs are not totally understood, but an addition to an awkward posture, such factors as the amount of repetition, the force used in the activity, and the individual's physiology and lifestyle may affect the likelihood of experiencing an RSI." "...RSI...is a wrist problem called Carpal Tunnel Syndrome, which may be aggravated by improper use of computer keyboards" (p. 2).

The keyboard should be at the height of the elbows, and the forearms may be parallel to the floor or the hands raised so that the forearms are inclined up to 20 degrees above horizontal if one sits upright, or even held more steeply if one leans back. The wrists should be held "floating" above the keyboard, as in playing a piano. A wrist pad may be used for rest periods. One should use a "light touch" (pp. 6, 7).

Stock, S.R. (1991). *Workplace Ergonomic Factors and the Development of Musculoskeletal Disorders of the Neck and Upper Limbs: A Meta-Analysis*. American Journal of Industrial Medicine, 19, 87–107.

This is a review of 54 studies relevant to the question of whether epidemiologic evidence exists of the relationship between workplace ergonomic factors such as repetition, force, static muscle loading, and extreme joint position and the development of muscle, tendon, and nerve entrapment disorders. Of the reviewed studies, three were found to be of specific significance. The author concludes that the studies “provide strong evidence of a causal relationship between repetitive, forceful work, and the development of musculoskeletal disorders of the tendons and tendon sheaths in the hands and wrists, and nerve entrapment of the median nerve at the carpal tunnel” (p. 817).

Sauter, S.L., Schleifer, L.M. and Knutson, S.J. (1991). *Work Posture, Workstation design, and Musculoskeletal Discomfort in a VDT Data Entry Task*. Human Factors, 33, 151–167.

Self-reported data on musculoskeletal discomfort were collected from more than 900 VDT users, and worker posture and workstation were measured for 40 of these users. Effects of ergonomics features on musculoskeletal discomfort were clearly evident from the evaluations. Arm discomfort was associated with high keyboards.

Akagi, K. (1992). *A Computer Keyboard Key Feel Study in Performance and Preference*. In Proceedings of the Human Factors Society 36th Annual Meeting (pp. 523–527). Santa Monica, CA: Human Factors Society.

The author conducted a study with 24 touch-typists who used four keyboards that were exactly the same except in key force-travel characteristics. Two keyboards had keys with linear springs, with resistances of 43 and 71 g. The other two keyboards had keys with a “snap” resistance, one at 36 and the other at 71 g.

The results indicated that the keyboards were not associated with any significant typing speed differences. Yet, the typing errors were least with the high resistance keyboards, 23 percent higher with the low resistance keyboards, with the snap-action showing three percent more errors than the high resistance linear keyboard. The highly resistant linear keyboard was the least liked, and the other three keyboards were about equally accepted. However, the keyboard with the highest resistance was the one on which more participants typed fastest than on any other keyboard.

Hadler, N.M. (1992). *Arm Pain in the Workplace. A Small Area Analysis*. JOM (February issue), 113–119.

The author indicates that similar keyboard usage has resulted only in a few, but not in all U. S. states to increased complaints about arm pains. He argues that the computer keyboard was designed with the comfort and efficiency of the operator in mind. He does not agree with the statements of ergonomic experts that the lack of tactile, visual, or auditory feedback of key activation was associated with CTDs.

The author says that one can do violence to the musculoskeletal structure, and that such exposure in industry would be unconscionable abuse. He believes that the CTD hypothesis appears sensible, but runs counter to established biology. While repetitive use of the upper extremity can alter hand structure, this does not lead to increased prevalence of any musculoskeletal disease, and certainly not to carpal tunnel syndrome. Yet, repetitive use can lead to soreness at the end

of the day. Furthermore, since we all suffer from regional musculoskeletal illness as the years go by, usage, whether repetitive or not, can lead to symptoms. He states that the issue is not the damage, but how to minimize mobility particularly at the workplace. Clearly, task design cannot be ignored. Arm discomfort is real, but the quality of experience is perturbed. Psychosocial aspects overwhelm workstation design in regard to safety of video display terminal workers (p. 118).

VDT News (Ed.). (1992). *Various Editorial Discussions of VDT/CTD Issues*. VDT News, 9, November/December issue, 1–15.

This issue is almost exclusively dedicated to the issue of overuse syndromes in working with computer workstations. In particular, much discussion is directed at contributions presented at the Work With Display Units '92 (WWDU '92) Congress in which links were implicated, by numerous presenters, between CTDs and physical and/or psychosocial conditions.

Lee, K., Swanson, N., Sauter, S., Wickstrom, R., Waikar, A. and Mangum, M. (1992). *A Review of Physical Exercises Recommended for VDT Operators*. Applied Ergonomics, 23, 387–408.

This article presents an evaluation of 127 exercises that have been recommended for prevention of musculoskeletal discomfort among VDT/office workers. It was found that more than one-third of the exercises was conspicuous and potentially embarrassing to perform. More importantly, a number of the exercises posed potential safety hazards, exacerbated biomechanical stresses common to VDT work, or were contraindicated for persons with health problems.

There are “generic” stresses common to most VDT work, particularly constrained postures that impart static loads to neck, back, shoulders, and upper extremities. To counteract these, an exercise program for VDT users should include the components of stretching shortened or tensed muscles, mobilization of the spine, strengthening or contraction of stretched and weakened muscles, and improvement of venous blood return from lower extremities. Of the exercise programs evaluated, most focused on the stretching and relaxation aspects, but failed to address the remainder of the needs, particularly the strengthening aspect to prevent musculoskeletal discomfort. However, such exercises are more intrusive and demanding than flexibility and relaxation exercises, and require breaks and exercise areas.

Bernard, B., Sauter, S., Petersen, M., Fine, L., and Hales, T. (1992). *Health Hazard Evaluation Report* (HETA 90-013-2277). Los Angeles Times, Los Angeles, California. Cincinnati, OH: National Institutes of Occupational Safety and Health.

This is a report on an evaluation of upper extremity work-related musculoskeletal disorders (WRMDs) among employees at the Los Angeles Times using video display terminals. Upper extremity symptoms were evaluated through the use of a self administered questionnaire. Forty one percent of the participants reported symptoms of WRMDs. Neck symptoms were most frequently reported, followed by hand/wrist, shoulder, and elbow symptoms. While the odds of having neck and shoulder WRMDs were mostly related to stress factors, the odds of having hand/wrist WRMDs were primarily related to time spent typing on computer keyboards.

130 employees were selected out of the 199 persons with hand/wrist cases, and their outcomes were compared with

those of 99 control employees. It was found that the risk factors associated to WRMDs among these persons were primarily the female gender, and the percentage of time spent typing on the computer keyboard. The investigation provides evidence that increasing time spent on computer keyboards is related to the occurrence of WRMDs, particularly for symptoms and physical findings in the hand/wrist area. For the hand and wrist, psychosocial variables were not as strong predictors as the job task variables (p. 4).

Grant, A.H. (1992). Computer Work Station With Mnemonic Keyboard. Patent 5119078. United States Patent Office.

The invention includes a keyboard that can be moved along the face of a display. The keyboard is split, the halves slanted and tilted down to the sides. The keys at the side ends of the keyboard and the cursor control keys are designed as mnemonic icons.

Flanders, M. and Soechting, J.F. (1992). Kinematics of Typing: Parallel Control of the Two Hands. Journal of Neurophysiology, 67, 1264–1274.

Translational and rotational motions of fingers and wrist while typing were recorded. When typing a single letter, all fingers of the hand were in motion, as was the wrist, and the overall pattern was highly repeatable, generating a “signature” for a given letter and subject. The motion of the other hand, typing preceding or succeeding letters, was not related to the motion of the first hand.

Soechting, J.F. and Flanders, M. (1992). Organisation of Sequential Typing Movements. Journal of Neurophysiology, 67, 1275–1290.

Typing motions are executed letter by letter, in a serial fashion. When consecutive motions are done with different fingers, the motions can overlap. Movement planning encompasses strings of letters.

Rempel, D.M., Harrison, R.J. and Barnhart, S. (1992). Work-Related Cumulative Trauma of the Upper Extremity. JAMA, 267, 838–842.

Work-related cumulative trauma disorders account for more than 50% of all occupational illnesses in the U. S. A. The authors describe their pathophysiology, epidemiology, clinical evaluation, medical management, and prevention.

Rempel, D., Dennerlein, J.T., Mote, C.D., and Armstrong, T. (1992). Fingertip Impact Loading During Keyboard Use. In Proceedings of NACOB II, The Second North American Congress on Biomechanics (pp.425-426). No location given: The American Society of Biomechanics.

A new technique is available to measure the fingertip force during the keystroke. Preliminary data show three distinct phases during the stroke: switch compression, finger impact, and pulp compression.

VDT News (Ed.). (1993). Ergonomic Keyboards. VDT News, January/February issue, 1–15.

This issue contains several articles that describe eight existing “ergonomic keyboards” and three others to be marketed soon. Furthermore, it discusses various aspects of CTDs, which are related to physical and psychophysical conditions.

McAtamney, L. and Corlett, E.N. (1993). RULA: A Survey Method for the Investigation of Work-Related Upper Limb Disorders. Applied Ergonomics, 24, 91–99.

RULA (Rapid Upper Limb Assessment) is a survey method for ergonomic investigations of workplaces where work-related upper limb disorders are reported. RULA uses diagrams in which body postures are indicated to fall into certain areas, which are numbered. One observes the existing postures, and records the associated number. All of the scoring numbers (Group A: upper arm, lower arm, wrist, wrist twist; Group B: neck, trunk, legs) are entered into a diagram and result in a combined grand score. A similar procedure is performed to end up with an overall score for muscle use and force. These data are then combined in a final grand score. The higher the resulting final grand score, the higher (more urgent) is the resulting action level.

Schoenmarklin, R.W. and Marras, W.S. (1993). Wrist Motions in Industry: Variance Between Jobs and Subjects. In Proceedings of the Human Factors and Ergonomics Society 37th Annual Meeting (pp. 649–653). Santa Monica, CA: Human Factors and Ergonomics Society.

The wrist motions of 40 industrial workers, who performed highly repetitive, hand-intensive jobs of high and low risks for CTDs, were recorded. Position, angular velocity and angular acceleration of the wrists were measured in three plains. For all kinematic variables, the variance between subjects within jobs accounted for a substantial amount, and often for the major part, of variance. Yet, acceleration was the one variable that best predicted membership to either the high or low risk group.

Ranney, D. (1993). Work-Related Chronic Injuries of the Forearm and Hand: Their Specific Diagnosis and Management. Ergonomics, 36, 871–880.

The author indicates that in Canada, during 1989, there were more than 20 000 work-related injuries to the elbow, wrist, hand, and finger. This includes both acute and chronic injuries. Yet, the author cautions against the use of such terms as RSI, CTD, etc. because they imply statements of causation necessary but not sufficient for diagnosis. It is important to first identify the tissue injured, then the nature of pathology, and finally the cause. For example, muscles may show tears or fatigue, tendons microtears and synovial thickening, and nerves hypoxia.

For such diagnosis, the author describes specific steps and procedures to be undertaken. This leads to a rational approach to treatment, which is recommended to be early and aggressive.

Wright, W.C. (1993). Diseases of Workers. Translation of B. Ramazzini's 1713 “De Morbis Articum”. Thunder Bay, Ontario: OH & S Press (ISBN 0-9696816-0-7).

Preceded by a thorough and useful introduction, this is an easily readable, interesting modern translation of the classic Latin text. Ramazzini describes occupational diseases and injuries that, around 1713, were known to appear in certain professions and work situations. Each profession or typical activity is discussed in a separate chapter. Of particular interest here is the chapter on “Diseases of scribes and notaries”, pages 253–256.

Angelaki, D.E. and Soechting, J.F. (1993). Non-Uniform Temporal Scaling of Hand and Finger Kinematics During Typing. Experimental Brain Research 95, 319–329.

Keystroke kinematics of the hand and fingers usually, but not always, varied with the typing rate. The variations in motions occurred mostly during the period before the key was hit.

Pascarelli, E.F. and Kella, J.J. (1993). *Soft-Tissue Injuries Related to Use of the Computer Keyboard*. JOM J. Occupational Medicine, 35(5), 522–532.

Physical examinations, history, and keyboarding habits are reported for 53 disabled heavy computer users (35 subjects were female).. The conventional flat keyboard encourages placing the wrists on desk or table, often in dorsiflexion and/or ulnar deviation. Strain on forearm and hand muscles, especially when splaying the fingers to reach far keys, starts a cascade of events leading to muscle damage and tendinitis (p. 525). The following typical postures were identified: dorsiflexion of the wrist while resting it on the table with the fingers arched (the “leaner”), use of index and middle fingers to strike keys (“hunt and peck”, the “pointer”), hitting the keys hard (the “clacker”), each associated with specific health deficiencies.

Keyserling, W.M., Stetson, D.S., Silverstein, B.A. and Brouwer, M.L. (1993). *A Checklist for Evaluating Ergonomics Risk Factors Associated With Upper Extremity Cumulative Trauma Disorders*. Ergonomics, 36, 807–831.

A 2-page checklist was developed and tested for determining the presence of ergonomic risk factors in the following categories: repetitiveness, local mechanical contact stress, forceful manual exertions, awkward upper extremity posture, and hand tool use.

Quill, L.L. and Biers, D.W. (1993). *On-Screen Keyboards: Which Arrangements Should Be Used?* In Proceedings of the Human Factors and Ergonomics Society 37th Annual Meeting (pp. 1142–1146). Santa Monica, CA: Human Factors and Ergonomics Society.

A one-row alphabetic arrangement of keys was superior to the 3-row QWERTY layout. Use of a mouse was faster than using arrow keys.

Smutz, P., Serina, E. and Rempel, D. (1994). *A System for Evaluating the Effect of Keyboard Design on Force, Posture, Comfort, and Productivity*. Ergonomics, 37, 1649–1660.

A measuring system was developed to determine the effectiveness of alternative keyboards on finger force, wrist posture, operator comfort and keying performance.

Carter, J.B. and Banister, E.W. (1994). *Musculoskeletal Problems in VDT Work: A Review*. Ergonomics, 37, 1623–1648.

This paper offers a discussion of the possible causes of musculoskeletal pain related to the design of workstation, chair and keyboard, operator selection, training, conditioning, posture, and rest breaks.

Wells, R., Moore, A., Potvin, J. and Norman, R. (1994). *Assessment of Risk Factors for Development of Work-Related Musculoskeletal Disorders (RSI)*. Applied Ergonomics, 25, 157–164.

An instrumentation system was developed to measure the risk factors for the development of work-related musculoskeletal disorders, by combining a video image of the worker with superimposed quantitative information on risk factors.

Fernstroem, E., Ericson, M.O. and Maler, H. (1994). *Electromyographic Activity During Typewriter and Keyboard Use*. Ergonomics, 37, 477–484.

With reference to Lundervold's 1951 measurements, EMG activities in six muscles of the forearm and shoulder were measured in 8 typists while keyboarding on mechanical, electromechanical, and several electronic keyboards. The mechanical device caused higher strain than the other keyboards. Using a palm rest did not decrease the strain, nor did the use of a split and tilted (at 20 deg) keyboard.

Hales, T.R., Sauter, S.L., Peterson, M.R., Fine, L.J., Putz-Anderson, V., Schleifer, L.R., Ochs, T.T. and Bernard, B.P. (1994). *Musculoskeletal Disorders Among Visual Display Terminal Users in a Telecommunications Company*. Ergonomics, 37, 1603–1621. (Edited version of NIOSH (1992) Health Hazard Evaluation Report HETA 89-299-2230, U.S. West Communications.)

In a cross-sectional study of 533 users of VDTs, associations between workplace factors and musculoskeletal disorders were investigated. The hand-wrist area was most often affected. Tendon-related disorders were most frequent, followed by nerve entrapment syndromes. Psychosocial factors seemed to play a role in the occurrence.

Gerard, M.J., Jones, S.K., Smith, L.A., Thomas, R.E. and Wang, T. (1994). *An Ergonomic Evaluation of the Kinesis Ergonomic Keyboard*. Ergonomics, 37, 1661–1668.

Muscular activities in keyboarders using a Kinesis keyboard are reduced compared to users of an IBM PS/2 keyboard. Learning to achieve high performance is fast on the Kinesis model. Armstrong, T.J., Foulke, J.A., Martin, B.J., Gerson, J. and Rempel, D.M. (1994). Investigation of Applied Forces in Alphanumeric Keyboard Work. American Industrial Hygiene Association Journal, 55, 30–35.

Keys on three computer keyboards were instrumented to measure the forces that were theoretically needed to activate them, and the forces that were actually applied to them while keying a text. The peak forces actually exerted to the keys were found to be 2.5 to 3.9 times the forces required to activate the keys.

Martin, B.J., Armstrong, T.J., Foulke, J.A., Natarajan, S., Klinenberg, E., Serina, E. and Rempel, D. (1994). *Finger Force During Computer Keyboard Work. Part I: Relation of Keyboard Reaction Force to Finger Flexor Muscles Surface EMG*. In Proceedings of the 12th Triennial Congress of the International Ergonomics Association, Vol. 2 (pp. 198–200). Mississauga, ON, Canada: Human Factors Association of Canada.

EMG data were found to be poor in predicting individual dynamic finger forces. The estimated peak typing forces averaged about 9% of those exorable in a maximum voluntary contraction, even although the actual typing forces were about 4.6 times the needed key activation force.

Rempel, D., Klinenberg, E., Serina, E., Martin, B., Armstrong, T., Foulke, J. and Natarajan, S. (1994). *Finger Force During Computer Keyboard Work. Part II: Relation of Keyswitch Make Force to Applied Force and Surface EMG*. In Proceedings of the 12th Triennial Congress of the International Ergonomics Association, Vol. 2 (pp. 201–203). Mississauga, ON, Canada: Human Factors Association of Canada. Typing with keys that had a “make” force of 0.28 and 0.56 N did not result in different applied finger tip forces or finger flexor EMGs, but both increased significantly when the make force was 0.83 N.

Kroemer, K.H.E. (1994). "Alternative Keyboards" and "Alternatives to Keyboards". In Proceedings of the 4th International Scientific Conference Work with Display Units, Vol. 3 (pp. C1-C7). Milano, Italy: University of Milan.

This paper provides a discussion of the shortcomings of traditional keyboards for current computer use and of possible better ways to transfer information from the human to the computer.

McMulkin, M.L. and Kroemer, K.H.E. (1994). *Usability of a One-hand Ternary Chord Keyboard*. Applied Ergonomics, 25, 177-181.

Five subjects learned to operate a one-hand ternary chord keyboard with 18 characters in about 30 minutes. After about 60 hours of practice, their average keying was 170 characters per minute.

Rudakewych, M.Z., Valent, L. and Hedges, A. (1994). *Field Evaluation of a Negative Slope Keyboard System Designed to Minimise Postural Risks to Computer Workers*. In Proceedings of the 4th International Scientific Conference Work with Display Units, Vol. 3 (pp. C17-C19). Milano, Italy: University of Milan.

A negative keyboard slope significantly improved the postures of hand, wrist, forearm, upper arm and the sitting posture in all 19 subjects.

Rempel, D. and Horie, S. (1994a). *Effect of Wrist Posture During Typing on Carpal Tunnel Pressure*. In Proceedings of the 4th International Scientific Conference Work with Display Units, Vol. 3 (pp. C27-C28). Milano, Italy: University of Milan.

Pressure inside the carpal tunnel during typing was measured in six healthy typists while they typed on a standard keyboard arranged in different slopes to produce wrist flexion of 15 degrees and extension of 0, 15, 30, and 45 degrees. Increasing deviations from zero showed increasing pressure.

Rempel, D. and Horie, S. (1994b). *Effect of Keyboard Wrist Rest on Carpal Tunnel Pressure*. In Proceedings of the 4th International Scientific Conference Work with Display Units, Vol. 3 (pp. C29-C30). Milano, Italy: University of Milan.

Pressure inside the carpal tunnel during typing was measured in seven healthy typists while they typed on an Apple extended keyboard. Resting the wrist either on the wrist rest or on the table while typing produced significantly increased pressure over keeping the hand "floating".

Owen, R.D. (1994). *Carpal Tunnel Syndrome: A Products Liability Prospective*. Ergonomics, 37, 449-476.

The author focuses on Carpal Tunnel Syndrome (CTS) from an U. S. legal perspective. He concludes that CTS is preventable, both on and off the job. Administrative and engineering controls can reduce hazardous exposure. People should be kept productive and healthful by fitting their activities to them.

Burastero, S., Tittiranonda, P., Chen, C., Shih, M. and Rempel, D. (1994). *Ergonomic Evaluation of Alternative Keyboards*. In Proceedings of the 4th International Scientific Conference Work with Display Units, Vol. 3 (pp. C15-C16). Milano, Italy: University of Milan.

A methodology was developed to compare the results of biomechanical and performance data and subjective assessments during the use of various keyboards.

Cakir, A. (1994). *Do Split Keyboards Help to Reduce Strain?* In Proceedings of the 4th International Scientific Conference Work with Display Units, Vol. 3 (p. C24). Milano, Italy: University of Milan.

An Apple adjustable keyboard was compared to a standard keyboard both in a laboratory study and in a 6-month field study. Ergonomic design can improve postural comfort and well-being, and reduce fatigue.

Faucett, J. and Rempel, D. (1994). *VDT-Related Musculoskeletal Symptoms: Interactions Between Work Posture and Psychophysical Work Factors*. Am. J. of Industrial Medicine, 26, 597-612.

VDT operators of a newspaper experienced musculoskeletal problems the severity of which was associated with work posture, job characteristics, and length of exposure. Psychosocial factors at work affected the symptoms.

Marras, W.S. (Ed.) (1994). *VDT-Related CTD Research Symposium. (A report of the presented papers.)* Columbus, OH: The Biodynamics Laboratory, The Ohio State University.

This report provides a presentation and discussion of VDT-related CTD concerns and of their prevention by ergonomics measures.

Rempel, D., Johnson, P., Smutz, P. and Tal, R. (1994). *Finger Forces Measured During Computer Mouse Use*. In Proceedings of the 4th International Scientific Conference Work with Display Units, Vol. 3 (pp. C28-C29). Milano, Italy: University of Milan.

An Apple mouse was instrumented to measure the forces applied by the digits of the hand. During moving and pointing, mean pinch forces were about 0.5 N, but 1.4 N while dragging and almost 4 N when lifting and dragging.

Armstrong, T.J. and Lackey, E.A. (1994). *An Ergonomic Guide To Cumulative Trauma Disorders of the Hand and Wrist*. Fairfax, VA: American Industrial Hygiene Association. This booklet "is intended to make readers familiar with cumulative trauma disorders (CTDs) and review workplace risk factors that might lead to CTDs of the hand and wrist. This guide is concerned primarily with prevention of CTDs through control of occupational factors" (page 1). Clerical and data and word processing occupations are mentioned (page 4) among those in which CTDs are prevalent. A common trait of all these is the reliance on hand work. Of the six factors that are consistently implicated as causes of CTDs, "repeated and sustained exertions" are preeminent.

Grant, A.H. (1995). *Prehensile Positioning Computer Keyboard*. Patent 5416498. United States Patent Office.

The patent describes a keyboard that is split, and the halves are slanted and tilted down to the sides. The center-mounted space bars can be activated in several directions. The keyboard sections can be set to different slope angles.

Moore, J.S. and Garg, A. (1995). *The Strain Index: A proposed Method to Analyze Jobs for Risks of Distal Upper Extremity Disorders*. American Industrial Hygiene Association Journal, 56, 443-458.

Six task descriptors (intensity, duration per cycle, effort per minute, wrist posture, speed of exertion, duration per day) are rated, and multipliers are assigned to each. The product of the multipliers is the Strain Index.

Hedge, A., McCrobie, D., Land, B., Morimoto, S. and Rodriguez, S. (1995). *Effects of Wrist Rests, Keyboard Trays, and a Preset Tilt/Down System on Wrist Posture, Seated Posture, and Musculoskeletal Discomfort*. In Proceedings of the Human Factors and Ergonomics Society 39th Annual Meeting (pp. 630–634). Santa Monica, CA: Human Factors and Ergonomics Society.

Using a keyboard with a negative slope improved the postures of wrist and body and reduced musculoskeletal discomfort, and was strongly preferred, in comparison to a conventional arrangement with positive slope.

Masali, M. (1995). *The Degenerate Son of the Typewriter*. *Ergonomia* 3/4, 14–15.

Anthropologists have long known and written about the body's natural tendency to look downward to close targets, with accompanying bends of neck and trunk, rather than look forward or even upward to such target as a computer screen.

Fogleman, M. and Brogmus, G. (1995). *Computer Mouse Use and Cumulative Trauma Disorders of the Upper Extremities*. *Ergonomics* 38, 2465–2475.

Insurance claim statistics of cumulative trauma disorders from 1986 to 1993 show a growing problem associated with the use of the computer mouse. No claims were recorded until 1989, but 219 claims appeared in 1993.

Cakir, A. (1995). *Acceptance of the Adjustable Keyboard*. *Ergonomics* 38, 1728–1744.

Split keyboards with adjustable tilt angles can improve postural and general comfort and reduce fatigue, as shown in short- and long-term studies.

Yoshitake, R. (1995). *Relationship Between Key Space and User Performance on Reduced Keyboards*. *Applied Human Science*, 14, 287–292.

Experiments showed that the center-to-center distance between keys can be reduced from 19 to 16.7 mm for fast typists with large fingers, and to 15 mm for typists with small fingers, without reducing their performance.

Hedge, A. and Powers, J.R. (1995). *Wrist Postures While Keyboarding: Effects of a Negative Slope Keyboard System and Full Motion Forearm Supports*. *Ergonomics*, 38, 508–517.

Subjects typed on a traditional 101-key keyboard placed on a 68-cm high horizontal surface in the following conditions: (a) with arm support, (b) without arm support, and (c) with the keyboard placed at a negative slope of 12 degrees. Ulnar deviation did not change, but dorsal wrist extension was 13 degrees in conditions (a) and (b) and reduced to minus 1 degree with (c).

Hoffmann E.R., Tsang, K.K. and Mu, A. (1995). *Data-entry Keyboard Geometry and Keying Movement Times*. *Ergonomics*, 38, 940–950.

On simulated keyboards with different key sizes and key spacings, the subjects' movement times were shortest when the spacing approximated the size of the finger pad.

Bergqvist, U., Wolgast, E., Nilsson, B. and Voss, M. (1995a). *The Influence of VDT Work on Musculoskeletal Disorders*. *Ergonomics*, 38, 754–762.

In an examination of 353 office workers, the occurrence of muscle problems was not different in VDT-workers and non-VDT workers. However, the combination of VDT work with

more than 20 hours per week and some other factors was associated with risk. Among others, these factors included limited rest breaks and repetitive movements.

Bergqvist, U., Wolgast, E., Nilsson, B. and Voss, M. (1995b). *Musculoskeletal Disorders among Visual Display Terminal Workers: Individual, Ergonomic, and Work Organizational Factors*. *Ergonomics*, 38, 763–776.

A cross-sectional study of 260 VDT office workers identified factors associated with the occurrence of musculoskeletal problems. Among the organizational factors were flexible rest breaks, task flexibility and overtime. Among the ergonomic factors were static work posture, hand position, repetitive movements, and keyboard and VDT vertical position.

Grieco, A., Molteni, G., Occhipinti, E. and Piccoli, B. (Eds.) (1995). *Work with Display Units 94*. Amsterdam, NL: Elsevier.

This book contains papers presented at the 1994 WWDU conference. Among these are discussions of “folk hazards and health panics,” musculoskeletal disorders including psychosocial and psycho-physiological work factors, and keyboard and workplace ergonomics.

Kuorinka, I. and Forcier, L. (Eds.) (1995). *Work Related Musculoskeletal Disorders: A Reference Book for Prevention*. London, UK: Taylor & Francis.

This book, written by experts in the field, provides an overview of the state of knowledge regarding work-related musculoskeletal disorders with the purpose to determine engineering, managerial, and behavioral means for their prevention. The discussion provides “...ample and consistent evidence that a variety of localized musculoskeletal symptoms are associated with work related risk factors of repetition, physical load, certain prolonged postures and vibration” (page 131). “...some personal attributes may contribute to the development of work related musculoskeletal disorders; however, this does not diminish the importance of work exposure contributions” (page 136).

Honan, M., Jacobson, M., Tal, R., and Rempel, D. (1996). *Changes in Wrist Postures During a Prolonged Typing Task*. In Proceedings of the Human Factors and Ergonomics Society 40th Annual Meeting (pp. 629–631). Santa Monica, CA: Human Factors and Ergonomics Society.

Over a 4-hr period of intensive typing, wrist posture did not change in users of a split and tilted keyboard, but right wrist extension and left forearm pronation did change when using a conventional keyboard.

Habes, D.J. (1996). *Upper Extremity Cumulative Trauma Disorders. Current Trends*. Chapter 29 in A. Bhattacharya and J.D. McGlothlin (Eds.), *Occupational Ergonomics* (pp. 581–603). New York, NY: Dekker.

The author defines and describes CTDs, their development due to activities on and off the job, their incidence rates and costs, and ergonomic means to control them.

Sommerich, C.M., Marras, W.S. and Parnianpour, M. (1996). *Observations on the Relationships Between Key Strike Force and Typing Speed*. *American Industrial Hygiene Association Journal*, 57, 1109–1114.

Experiments showed that, when typing at preferred speed, there is no relationship between speed and key strike force.

But if the individual's typing speed is affected by external factors, strike forces tend to increase or decrease, with similar changes in typing speed.

Martin, B.J., Armstrong, T.J., Foulke, J.A., Natarajan, S., Klinenberg, E., Serina, E. and Rempel, D. (1996). *Keyboard Reaction Force and Finger Flexor Electromyograms During Computer Keyboard Work*. Human Factors, 38, 654–664.

Average peak typing forces were about 10% of MVC and 5.4 times the needed key activation force. The relatively poor correspondence between EMG data and individual dynamic finger forces may result from the fact that actual muscle load is higher than reflected in key strike force.

Marklin, R.W. and Simoneau, G.G. (1996). *Upper Extremity Posture of Typists Using Alternative Keyboards*. In Proceedings of the Silicon Valley Ergonomics Conference & Exposition, ErgoCon 96 (pp. 126–132). San Jose, CA: San Jose State University.

Split and tilted keyboards appear to reduce ulnar deviation of the wrist and forearm pronation of the typist, as compared to using a conventional flat keyboard.

Moon, S.D. and Sauter, S.L. (Eds.) (1996). *Beyond Biomechanics. Psychosocial Aspects of Musculoskeletal Disorders in Office Work*. London, UK: Taylor & Francis.

The chapters in this book establish various correlations between musculoskeletal disorders associated with office work, especially on keyboards, and psychosocial factors which are complementary to, and interactive with, physical workplace demands. The psychosocial factors can be individual, or related to the workplace situation and the work organization.

Olacsi, G.S. and Beaton, R.J. (1997). *A New Force-Displacement Metrology for Keyboards*. In Proceedings of the Human Factors and Ergonomics Society 41st Annual Meeting (pp. 637–641). Santa Monica, CA: Human Factors and Ergonomics Society.

A technique for continuous force-displacement measurement was developed that assesses dynamic force applied to a key.

Straker, L., Jones, K.J. and Miller, J. (1997). *A Comparison of the Postures Assumed When Using Laptop Computers and Desktop Computers*. Applied Ergonomics, 28, 263–268.

Significantly greater neck flexion and head tilt occurred during work with laptop compared to desktop computers, but other body angles and keying performance showed no differences.

Swanson, N.G., Galinsky, T.L., Cole, L.L., Pan, C.S. and Sauter, S.L. (1997). *The Impact of Keyboard Design on Comfort and Productivity in a Text-Entry Task*. Applied Ergonomics, 28, 9–16.

After one day of working on a conventional keyboard, and, subsequently, two days of keying on split, tilted, and sloped keyboards, 50 subjects showed no significant differences in discomfort, fatigue or performance.

Rempel, D., Keir, P.J., Smutz, W.P. and Hargens, A. (1997). *Effects of Static Fingertip Loading on Carpal Tunnel Pressure*. J. Orthopedic Research, 15, 422–426.

Increasing forces exerted with the fingertip from 0 to 12 N increase carpal tunnel pressure independently of various wrist flexion and extension angles.

Bernard, B.P. (Ed.) (1997). *Musculoskeletal Disorders and Workplace Factors*. (DHHS NIOSH Publication No. 97-141, second printing). Washington, DC: U.S. Department of Health and Human Services.

This is a critical review of epidemiologic evidence for the work-relatedness of musculoskeletal disorders (MSDs) of the neck, upper extremity, and low back. Regarding keyboarding, the following conclusions can be drawn. For neck and neck/shoulder MSDs, there is reasonable evidence for a causal relationship (page 2–22). For shoulder MSDs, there is evidence for a positive association with highly repetitive work (page 3–25). For carpal tunnel syndrome, there is evidence for a positive association with highly repetitive work (page 5a–28). For hand/wrist tendinitis, there is evidence for an association with any single factor (repetition, force, and posture), and there is strong evidence that job tasks that require a combination of these factors increase the risk for hand/wrist tendinitis (page 5b–17).

Helander, M.G., Landauer, T.K. and Prabhu, P.V. (Eds.) (1997). *Handbook of Human-Computer Interaction* (2nd ed.). Amsterdam, NL: Elsevier.

This is the second edition of the book; the first had been published in 1988 by Helander. It provides a comprehensive overview of the contemporaneous principles and procedures used in designing the interface between the computer and the operator.

Nordin, M., Andersson, G.B.J. and Pope, M.H. (1997). *Musculoskeletal Disorders in the Workplace: Principles and Practices*. St. Louis, MO: Mosby.

This book discusses and evaluates musculoskeletal disorders of the human body in terms of epidemiology, biomechanics, psychophysiology, psychosociology, and treatment and avoidance of the disorders through ergonomic measures.

Lewis, J.R., Potosnak, K.M. and Magyar, R.L. (1997). *Keys and Keyboards*. Chapter 54 in M.G. Helander, T.K. Landauer and P.V. Prabhu (Eds.), *Handbook of Human-Computer Interaction* (2nd ed.) (pp. 1285–1315). Amsterdam, NL: Elsevier.

This is a discussion of the possibilities, advantages, and disadvantages of various kinds of keys and keyboards proposed in the past.

Kroemer, K.H.E. (1997). *Design of the Computer Workstation*. Chapter 57 in M.G. Helander, T.K. Landauer and P.V. Prabhu (Eds.), *Handbook of Human-Computer Interaction* (2nd ed.) (pp. 1395–1414). Amsterdam, NL: Elsevier.

This is a discussion of ergonomic design aspects of the computer workstation, including the keyboard.

Matias, A.C. and Salvendy, G. (1998). *Carpal Tunnel Syndrome Causation Among VDT Operators*. Occupational Ergonomics, 1, 55–66.

The main causes of CTS were found to be long periods of continuous typing, static and bent wrist postures, seating posture and individual wrist size.

Martin, B., Rempel, D., Sudarsan, P., Dennerlein, J., Jacobson, M., Gerard, M. and Armstrong, T. (1998). *Reliability and Sensitivity of Methods to Quantify Muscle Load during Keyboard Work*. In S. Kumar (Ed.), *Advances in Occupational Ergonomics and Safety* (pp. 495–498). Amsterdam, NL: IOS Press.

The reliability and sensitivity of several techniques to assess the muscle loading during keyboarding were investigated. Intramuscular and surface EMGs are in good agreement. The flexor carpi radialis and ulnaris are the prime movers in typing on flat keyboards. Muscle load is double as high in extensors than in flexors. Muscle load increases linearly with typing speed.

Greening, J. and Lynn, B. (1998). *Vibration Sense in the Upper limb in Patients with Repetitive Strain Injury and a Group of At-Risk Office Workers*. International Archive Occupational Environmental Health, 71, 29–34.

RSI patients and office workers had significantly raised thresholds for vibration, demonstrating a work-related exacerbation. Perception of vibration may be useful for early detection of onset of RSI.

National Research Council (Ed.) (1998). *Work-Related Musculoskeletal Disorders: A Review of the Evidence*. Washington, DC: National Academy Press.

Among the conclusions drawn are the following: There is a strong biological plausibility to the relationship between the incidence of musculoskeletal disorders and the causative exposure factors in high-exposure occupational settings. Specific interventions can reduce the reported rate of musculoskeletal disorders for workers who perform high-risk tasks (page 23).

Carayon, P., Smith, M.J. and Haims, M.C. (1999). *Work Organisation, Job Stress, and Work-Related Musculoskeletal Disorders*. Human Factors 41, 644–663.

A review and discussion of previous work indicates that there are possible links between stressful psychosocial factors at work on the one hand, and force, repetition and posture on the other hand.

Marklin, R.W., Simoneau, G.G. and Monroe, J.F. (1999). *Wrist and Forearm Posture from Typing on Split and Vertically Inclined Computer Keyboards*. Human Factors 41, 559–569.

In an experiment with 90 experienced office workers, wrist and forearm posture were compared on conventional and keyboards that were split and slanted, and split, slanted and tilted. The split keyboards reduced the ulnar deviation from that seen with the conventional keyboard.

Tittiranonda, P., Rempel, D., Armstrong, T. and Burastero, S. (1999). *Effect of Four Computer Keyboards in Computer Users With Upper Extremity Musculoskeletal Disorders*. Am. J. Industrial Medicine, 35, 647–661.

Eighty computer users with upper extremity musculoskeletal disorders participated in a 6-month, randomized, placebo-controlled experiment with four different keyboards. Compared to placebo (a conventional flat keyboard), two of the three other keyboards (all split, slanted, and more or less tilted) showed an improving trend in pain severity and hand function, and more individual satisfaction with the keyboards, but no consistent improvement in clinical findings.

National Research Council (Ed.) (1999). *Work-Related Musculoskeletal Disorders: Report, Workshop Summary, and Workshop Papers*. Washington, DC: National Academy Press. The conclusions of the panel can be summarized as follows (page 59): Musculoskeletal disorders (MSDs) are multifactorial. The biomechanical demands of work constitute the most important factors but individual, social, and organizational factors also contribute.

The causes of repetitive overexertion of the key-boarder's musculoskeletal system, and the nature of such occupational diseases, were researched in considerable detail: see, for example, the publications by Low (1990), Burt, Hornung, and Fine (1990), McPhee (1990), Rempel and Gerson (1991), Armstrong et al. (1991), Moore et al. (1991), Armstrong (1991), Stock (1991), Rempel, Harrison and Barnhart (1992), Wells et al. (1994), Armstrong and Lackey (1994), Faucett and Rempel (1994), and Martin et al. (1998). As a result, the biomechanical stressors and their physiological consequences became well understood, as described by Kuorinka and Forcier (1995), Moon and Sauter (1996), Bernard (1997). Partly in response to the intention of the U.S. OSHA (Occupational Safety and Health Agency) to establish ergonomic requirements that specifically target repetitive injuries, the (U.S.) National Research Council was asked to convene a workshop on work-related musculoskeletal disorders. The findings can be summarized by the statement that (National Research Council 1999, page 59) musculoskeletal disorders are multifactorial. The biomechanical demands of work constitute the most important risk factors but individual, social, and organizational factors also contribute.

The market of the 1990s showed a demand for human-engineered input devices to computers, and for ergonomically correct workstations and working conditions that result in safe and efficient performance. Accordingly, novel input devices were described, often based on biomechanical principles, for example reported by Zapp (1990), Faubert and Pritchard (1990), Cushman and Rosenberg (1991), Rempel and Gerson (1991), Kroemer (1991), Grant (1992), Smutz, Serina, and Rempel (1994), Armstrong et al. (1994), McMulkin and Kroemer (1994), Burastero et al. (1994), Marklin and Simoneau (1996), and Tittiranonda et al. (1999). Progress in computer-related technology and better understanding of the capabilities and limitations of the human operator opened the door to a new century during which, one hopes, ergonomic human-computer interfaces will become common.

To summarize the evaluation of the state-of-art during the 1990s, this decade demonstrated the trend already discernable in the preceding decade: investigation of the stressors that can lead to musculoskeletal disorders in keyboarders, and ergonomic means to avoid such disorders. Also, more data input devices were invented that should decrease the physical workload on keyboarders.

7 Summary and conclusions

A listing of the many publications between 1878 and 1999 can be discussed from many perspectives. I will limit myself to the design of keyboards and the activity of keying.

7.1 *Designing and re-designing the keyboard*

We know of no comments by Sholes and his co-inventors on their reasons for shifting from a keyboard with two rows of 21 piano-like keys, proposed in 1868, to a set of keys with four rows and 11 round buttons on each row, in a “QWERTY” arrangement, proposed in 1878. The 1878 layout was, with minor modifications, quickly adopted internationally for typewriters.

The principle of allocating one key to one character, as used in QWERTY, has several ramifications: for example, letters that frequently follow each other in text, such as t and h in English, are spaced apart on the keyboard. This requires complex and time-consuming motions of the fingers. Starting in the early 1900s, many proposals for new key arrangements were published to overcome the disadvantages of the QWERTY layout, to make key operation easier, and to improve typing performance – see the reviews by Kroemer (1964, 1972, 1994, 1997), Michaels (1971), Alden et al. (1972), Seibel (1972), Litterick (1981), Norman and Fisher (1982), Noyes (1983), Gilad and Pollatscheck (1986), Gopher and Raij (1988), and Lewis, Potosnak, and Magyar (1997). Proposed designs include relocating keys within the standard set of keys, or changing the entire keyboard layout.

On the mechanical typewriter, the key top served to receive, from the fingertip, all the energy needed to move an elaborate lever system so that finally a type bar would strike an inked ribbon to generate the imprint of a character on paper. The mechanical workings of the typewriter made changes difficult until new technologies appeared. In the so-called electric typewriters of around 1960, auxiliary energy input into the lever system lessened the physical effort of the typist’s finger, but the old design of mechanical leverages was essentially retained.

From the 1960s on, fingertip-operated switches controlling electric and then electronic circuitry replaced the mechanical keys. This drastically reduced both the key displacement and the force needed to actuate the key, diminishing the dynamic work (force-displacement integral) required from the operator’s finger. This technology opened the way for activating several keys (switches) at the same time, such as in “chording”, so that chunks of inputs could be generated instead of pressing a single key for a single character (Klemmer 1958, Ratz and Ritchie 1961, Seibel and Rochester 1962, Bowen and Guinness 1965, Conrad and Longman 1965, Galitz 1965/6). Chording was and is used by court reporters for on-the-spot recording of verbal discussions, and for mail sorting. Kinkead and Gonzalez published, in 1969, the first “Human Factors Design Recommendations for Touch-Operated Keyboards”.

Using the new switch technology, one could freely relocate the keys and re-design the keyboard. For computer use, keys were added, mostly “function” keys, to the left, behind, and especially to the right of the original QWERTY set. With an additional numeric keypad and a cur-

sor control keypad (both commonly on the right-hand side), in the 1980s the total number of keys was customarily just over one hundred – in some cases about 125 – as compared to the 45 to 50 keys on the old typewriter. So many keys took much keyboard space, and the big keyboard required large finger and hand movements to reach keys on the edges.

On the customary 1980/90s computer keyboards, all keys were arranged side-by-side in straight rows (while fingertips are not). Key columns followed two different design rules: on the QWERTY set, the zigzag columns ran slanted to the left (as seen by the operator), but on the added key sets, the columns were straight. Neither the original QWERTY keyboard nor its derivatives were “human engineered” as proposed in 1915 by Heidner and then by his many successors, and considered by IBM’s Industrial Design Laboratories in Germany in 1962, when switch technology would have made it easily possible.

A major argument for not changing the basic QWERTY design of the keyboard has been, since the 1950s (Strong 1956, Droege and Hill 1961) and repeated into the 1990s (Lewis, Potosnak, and Magyar 1997), that an alteration of the layout would require re-training the operators and hence would slow keying performance, at least initially. This reasoning is probably true for most minor changes, such as exchanging the letter designations of certain keys, prominently proposed by Dvorak in the 1930s and 1940s, or using an alphabetic layout, discussed for example by Hirsch and Michaels in 1970/1. In these cases, the new skills are so close to the old habits that confusion is indeed likely. However, when the design is changed fundamentally, such as when re-locating whole sets of keys, or when chording is used, then the new operation is unique and different from the old ways, and there may be little or no interference – the human is amazingly fast in acquiring new keyboarding skills (Kroemer 1964, Norman and Fisher 1982, McMulkin and Kroemer 1994, Gerard et al. 1994, Marklin, Simoneau, and Monroe 1999).

7.2 *New kinds of keys*

For more than half a century, the mechanical nature of the keys of typewriters did not allow much variation in overall design, in the required energy input, and in the “feel” to the operator. This changed with the use of switch technology in the 1960s. From then on, keys with much reduced travel and resistance could be designed, and those characteristics could be varied rather freely. Seemingly minute differences in key switch resistance during displacement can cause major changes in the “feel” of the key and in the operator’s biomechanical effort (Kinkead and Gonzalez 1969, Clare 1976, Eikelberger 1981, Rempel and Gerson 1991, Akagi 1992, Rempel et al. 1992, Armstrong et al. 1994, Martin et al. 1994). Furthermore, many kinds of key activation are feasible other than tapping a key down from “off” to “on”: for example, keys may be tilted sideways, can achieve several different “on”

positions (Langley 1988, Zapp 1990), or might function in other unconventional ways (Kroemer 1997). Finger movements may be registered (for example by means of instrumented gloves) as inputs to computers without the use of conventional keyboards. Voice recognition provides input to the computer without any finger or switch movement.

7.2.1 Alternative keyboards

As Heidner's 1915 patent shows, it was recognized soon after Sholes' "flat" keyboard became common that its horizontal rows of keys required the typist to rotate the forearms into extreme pronation, a posture difficult and often painful to maintain. Following Heidner, Hoke in 1924, Klockenberg in 1926, and Marloth in 1930 proposed to break the QWERTY keyboard into left and right halves and to tilt them sideways-down to facilitate the work of the key operator. Numerous designs and patents since 1967 included such splitting of the keyboard, angling the sections in slant, tilt, and slope, and relocating the keys from their usual flat rows and columns along curved and dished shapes that follow the natural structure and mobility of the hand and its digits. By the end of the century, conventional keyboards were in competition with more ergonomic "alternative keyboards" with the users hoping to avoid musculoskeletal strain and overuse disorders.

7.3 The effort of keying

In retrospect (with today's emphasis on ergonomics, human well-being, and performance) it is surprising that Sholes and his co-inventors provided no explanation of their keyboard choices. One would believe that they were aware of the then well-known repetitive injuries in telegraphists and musicians, especially pianists. (See the publications by Ayoub and Wittels 1989, Burnette and Ayoub 1989, Hochberg 1983, Burry and Stoke 1985, Fry 1986, and Lockwood 1989 for summaries of contemporary knowledge.)

The underlying biomechanical problem is twofold. One problem lies in the number of key operations: with 5 letters per word, and with 50 words per minute, a key operator performs 90 000 digit flexions, each followed by a digit extension, over a 6-hour working time. A fast enduring typist, typing 100 words per minute over 8 hours, performs 264 000 keystrokes. Even a slow keyboarder makes 24 000 digit flexions and 24 000 digit extensions in just 4 hours while typing only 20 words per minute.

The other aspect is the biomechanical "quality" of the keying effort, i.e., the direction of exertion and the quantity of applied energy, and the physiological work necessary to maintain hands, arms, and trunk in their working positions. In his 1915 patent, Heidner specifically addressed these problems, which he tried to alleviate by re-designing the keyboard. Related proposals followed: in 1920 by Banaji, Nelson, and Wolcott; in 1924 by Hoke;

and, in great detail, by Klockenberg in 1926. None of these proposals was adopted by the dominant manufacturers of that time. The other underlying problem, the large number of keystrokes, remained (and remains until today) unsolved since we still use the technique of "one key for one letter".

7.3.1 Cumulative trauma disorders

Well described since the 1920s, cumulative trauma disorders, usually described as myalgia, tendinitis, and tenosynovitis of the hand and arm, occurred commonly in typists and were recognized as occupational diseases around 1960.

The prominent reasons for such diseases are highly repetitive activities, often together with awkward positions and movements of the body segments involved, and pressure from equipment on the body – all occurring in keyboarding. If the activity or condition takes place only once in a while, then such a single small trauma is not injurious to the body, but the cumulative effects of these "microtrauma" can lead to injury and disease.

A formal definition is: "Cumulative trauma disorders (CTDs) are regional impairments of muscles, tendons, tendon sheaths, ligaments, nerves, and joints associated with activity-related repetitive mechanical trauma."

Other terms used are repetitive trauma injury (or illness, or disorder), repetitive motion injury (or illness, or disorder), repetitive strain injury (RSI), (occupational) overuse disorder (OD), musculoskeletal disorder, and work-related musculoskeletal disorder (WRMD or WRMSD). Currently, the terms cumulative trauma disorder (CTD) and work-related musculoskeletal disorder (WRMD or WRMSD) seem to be most often employed.

Medically, CTDs are usually classified as irritations, sprains, strains, or inflammation. Terms used to describe their nature include tendinitis (tendonitis), tenosynovitis, myalgia, bursitis, peritendinitis, epicondylitis, brachial plexus syndrome, thoracic outlet syndrome, carpal tunnel syndrome (CTS), regional pain syndrome, neurovascular syndrome, cervicobrachial syndrome, compression syndrome, entrapment syndrome, and neuropathy.

7.3.2 Knowledge about CTDs

Early in the 20th century, Schroetter (1925), Klockenberg (1926), Zollinger (1927), Conn (1931), Hammer (1934), and Flowerdew and Bode (1942) provided insights into the pathology of cumulative trauma disorders. By the middle of the century, CTDs were firmly linked to physical strain of the musculoskeletal system, with the stress generated not so much by the amount of energy expended in single muscle activations, but rather by the accumulation of their effects in highly repetitive work. Cumulative trauma disorders were understood to be related to design features and modes of operation of equipment such as hand tools, telegraphs, and the typewriter.

The work required of the typists' hands and arms when pounding on the keyboard of mechanical typewriters was fairly heavy, given the large displacement of the keys and their stiff resistance to energy input by the typist's hands. Furthermore, the arrangement of the keyboard itself, and of the keys on it, still followed Sholes' original design, which was certainly not "ergonomic" because it forced the arms into strong inward twist (pronation) and the hands into lateral bend (ulnar deviation), and required complex motions between ill-located keys.

Around 1960, physiological/biomechanical knowledge clearly related repetitive activities, particularly typing, to strains in the tendons and sheaths of the hand, wrist, and arm. Lundervold (1951, 1958) demonstrated that electromyography was a useful measure of muscle efforts. Pfeiffer et al. stated (in 1988) that, by 1960, carpal tunnel syndrome was the most frequently diagnosed, best understood, and most easily treated entrapment neuropathy.

During the 1980s, CTDs in keyboarders, especially carpal tunnel syndrome, were becoming a widespread phenomenon. "Epidemics" of CTDs had occurred in the 1960s in Japan, then during the early 1980s in Australia, and pockets of the occupational disease appeared repeatedly in certain industries and services in the U.S.A. Apparently, however, keyboarders were supposed to know about the danger: in ruling on a class-action suit, the U.S. Court of Appeals for the Third Circuit in New York decided in 1997 that keyboard manufacturers did not have to provide warnings about possible health risks associated with the use of their keyboards.

During the 1980s and 1990s, the biomechanical strains underlying CTDs, especially carpal tunnel syndrome, were investigated in detail. Their relations to workload on the job, including that associated with keyboarding, became well understood, as demonstrated in the publications by Putz-Anderson (1988), Kuorinka and Forcier (1995), Habes (1996), Moon and Sauter (1996), Nordin, Andersson, and Pope (1997), Bernard (1997), and the National Research Council (1998, 1999).

8 Looking into the future

Present technology and human factors engineering can overcome the ergonomic problems resulting from the human interacting with the computer via keyboarding, as we know it [4]. There are several reasons for being optimistic:

- * The technical means exist to convey information to the computer by means other than conventional push-down binary keys.
- * New technology facilitates suitable designs of the human-computer interface, possibly the abolishment of traditional keyboards.
- * Ergonomic means to interact with the computer can alleviate current biomechanical problems and, at the same time, improve performance.

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